PES University, Bengaluru

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Department of Science and Humanities

Engineering Chemistry

(UE26CY151A)

Question Bank

Self-study compilation from past ESA papers and class material — not an official university document

Unit - 1: Molecular Spectroscopy, Phase Equilibria and Computational Chemistry

How to use this bank. Each chapter opens with the formulas and method you need, then the past ESA questions worked out in full. After that come exam-style practice questions, new question types that have not been asked yet, and quick-fire checks. Attempt them before looking: every answer is in the *Answers* section at the end of the chapter, numbered the same way. Numerical answers are computed from each question's own constants, so small differences from a printed key come from rounding. Chapter 7 has three timed mock papers.

Contents

Chapter 1: Rotational (Microwave) Spectroscopy ..... *91 questions*

Chapter 2: Vibrational (IR) Spectroscopy — Harmonic Oscillator ..... *83 questions*

Chapter 3: Anharmonic Oscillator and Overtones ..... *67 questions*

Chapter 4: Electronic Spectroscopy, Franck–Condon and Raman ..... *97 questions*

Chapter 5: Phase Rule, Water and Pb–Ag Systems ..... *128 questions*

Chapter 6: Computational Chemistry ..... *49 questions*

Chapter 7: Mock Question 1 Papers ..... *15 questions*

530 numbered questions in all, 38 of them from past ESA papers.

What the examiners ask

Question type	Chapter	Times asked	Practice here
Rigid rotor: derive $I = \mu r_0^2$	1	4	1
Rigid rotor: bond length from first line (HCl, HI, HBr)	1	4	6

Question type	**Chapter**	**Times asked**	**Practice here**
Rigid rotor: I and spacing from B; level diagram $J \leq 5/6$ and spectrum	1	4	5
Rigid rotor: J→J+1 line; first line from I or r_0	1	2	5
SHO: $\bar{\nu}$ and zero-point energy from k (HF ×2, NO, CO)	2	5	6
SHO: k from $\bar{\nu}$ (CO ×3, HBr)	2	4	5
SHO: why one line; CO_2 and HI IR activity	2	6	4
Anharmonic: x_e, ω_e, overtone calculations	3	3	12
Anharmonic: selection rule, B–O diagram, SHO failure	3	3	4
Franck–Condon statement, three cases, spectra	4	5	5
Continuum / B–O / Raman / Stokes / Beer–Lambert	4	5	6
Phase rule: derivation	5	3	5
Phase rule: count C, P, F	5	3	12
Water diagram, F values, metastable curve, Clausius–Clapeyron	5	4	9
Reduced phase rule and Pb–Ag diagram, cooling curves	5	4	9
Computational chemistry: definition, applications, methods	6	1	10
Mock Q1 papers (timed, new numbers)	7	—	15

Left out: conductometric titration, colorimetry and α-Al_2O_3 synthesis appear in older Question 1s but belong to other units. Fe–C, the CO_2 diagram and the Beer–Lambert law are marked *[Not on current slides]*; items marked *(Extension)* go beyond the slides.

Chapter 1: Rotational (Microwave) Spectroscopy

Topics T4

The rigid-rotor chapter is the most heavily examined part of Unit 1: a bond-length calculation, the moment-of-inertia derivation and a level/spectrum sketch appear almost every sitting.

Formulas to remember

$$\varepsilon_J = BJ(J+1)\ \text{cm}^{-1}$$

$$B = \frac{h}{8\pi^2 Ic}$$

$$I = \mu r_0^2, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\Delta J = \pm 1, \quad \text{line}(J \to J+1) = 2B(J+1)$$

$$\text{first line} = 2B, \quad \text{spacing} = 2B$$

Method.

1. Read the first line (or the spacing): $B = \text{line}/2$.
2. $I = h/(8\pi^2 cB)$ with c in cm s^{-1}.
3. μ in **kg**: amu $\times\ 1.66 \times 10^{-27}$.
4. $r_0 = \sqrt{I/\mu}$, then convert m $\to$ Å.

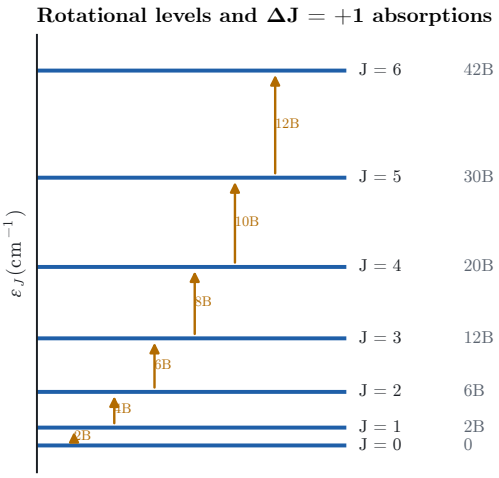


Figure 1.1 (G9): Rigid-rotor energy levels J=0–6 (values J(J+1)B), ΔJ=+1 arrows.

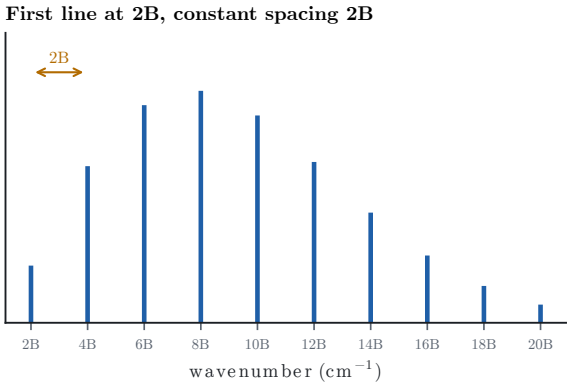


Figure 1.2 (G10): Rotational stick spectrum.

Derivation

Derivation: $I = \mu r_0^2$ (asked $\times 4$)

1. Let masses m_1, m_2 be at distances r_1, r_2 from the centre of mass C, with $r_0 = r_1 + r_2$.
2. Balance about C: $m_1 r_1 = m_2 r_2$. Substitute $r_2 = r_0 - r_1$: $m_1 r_1 = m_2(r_0 - r_1) \Rightarrow r_1 = \dfrac{m_2 r_0}{m_1 + m_2}$ and $r_2 = \dfrac{m_1 r_0}{m_1 + m_2}$.

3. $I = m_1 r_1^2 + m_2 r_2^2 = \dfrac{m_1 m_2^2 r_0^2 + m_2 m_1^2 r_0^2}{(m_1+m_2)^2} = \dfrac{m_1 m_2 (m_1+m_2) r_0^2}{(m_1+m_2)^2}$.
4. $\therefore\ I = \dfrac{m_1 m_2}{m_1+m_2}\, r_0^2 = \mu r_0^2$. (Also the energy: $E_r = \tfrac{1}{2} I\omega^2 = L^2/2I$, giving $E_J = \dfrac{h^2 J(J+1)}{8\pi^2 I}$.)

Past ESA questions with solutions

1. For a rigid-rotor heteronuclear diatomic molecule AX the first line in the pure rotational spectrum is at 21.18 cm^{-1}. Derive an expression for the moment of inertia and calculate the moment of inertia and bond length of AX. (A = 1.008 amu, X = 35.45 amu; $c = 3 \times 10^{10}$ cm s^{-1}, $\pi = 3.14$, $h = 6.627 \times 10^{-34}$ J s, 1 amu = 1.66×10^{-27} kg)

   Jun-24 · UE23 · Set 2 (6 marks)

   Solution.

   i. **Derivation:** see Q1-D above ($I = \mu r_0^2$).

   ii. First line at $2B$, so $B = \dfrac{21.18}{2} = 10.59\ \text{cm}^{-1}$.

   iii. $I = \dfrac{h}{8\pi^2 c B} = \dfrac{6.63 \times 10^{-34}}{8(3.14)^2 (3\times 10^{10})(10.59)} = 2.64 \times 10^{-47}\ \text{kg m}^2$ (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})

   iv. $\mu = \dfrac{m_1 m_2}{m_1+m_2} = \dfrac{1.008 \times 35.45}{36.46} = 0.9801\ \text{amu} = 1.63 \times 10^{-27}\ \text{kg}$.

   v. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{2.64\times10^{-47}}{1.63\times10^{-27}}} = 1.27 \times 10^{-10}\ \text{m} = 1.275\ \text{Å}$.

   Answer: $I = 2.64 \times 10^{-47}$ kg m^2, $r_0 = 1.27$ Å

2. The first line in the pure rotational spectrum of HCl appears at 21.18 cm^{-1}. Calculate the bond length. (H = 1.008, Cl = 35.45 amu; 1 amu = 1.66×10^{-27} kg, $c = 3 \times 10^8$ m s^{-1}, $h = 6.6 \times 10^{-34}$ J s)

   Dec-23 · UE22 (4 marks)

   Solution.

   i. Because c is given in m s^{-1}, convert B to m^{-1} ($\times 100$) or use $c = 3 \times 10^{10}$ cm s^{-1} — same result.

   ii. First line at $2B$, so $B = \dfrac{21.18}{2} = 10.59\ \text{cm}^{-1}$.

   iii. $I = \dfrac{h}{8\pi^2 c B} = \dfrac{6.60 \times 10^{-34}}{8(3.14)^2 (3\times 10^{10})(10.59)} = 2.63 \times 10^{-47}\ \text{kg m}^2$ (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})

   iv. $\mu = \dfrac{m_1 m_2}{m_1+m_2} = \dfrac{1.008 \times 35.45}{36.46} = 0.9801\ \text{amu} = 1.63 \times 10^{-27}\ \text{kg}$.

   v. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{2.63\times10^{-47}}{1.63\times10^{-27}}} = 1.27 \times 10^{-10}\ \text{m} = 1.272\ \text{Å}$.

   Answer: $r_0 \approx 1.27$ Å

3. For HBr, the rotational spectrum shows the first line at 17.19 cm^{-1}. Calculate the bond length (rigid rotor). (H = 1.0, Br = 79.9 amu; $h = 6.627 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Jul-23 · UE22 · A (4 marks)

Solution.

i. First line at $2B$, so $B = \dfrac{17.19}{2} = 8.595\ \text{cm}^{-1}$.

ii. $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.63 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(8.595)} = 3.26 \times 10^{-47}\ \text{kg m}^2$ (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})

iii. $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.0 \times 79.9}{80.9} = 0.9876\ \text{amu} = 1.64 \times 10^{-27}\ \text{kg}$.

iv. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{3.26 \times 10^{-47}}{1.64 \times 10^{-27}}} = 1.41 \times 10^{-10}\ \text{m} = 1.409\ \text{Å}$.

Answer: $r_0 \approx 1.41$ Å

4. The first spectral line in the rotational spectrum of HI appears at 13.022 cm^{-1}. Calculate the bond length. (H = 1.0, I = 126.9 amu; $c = 3 \times 10^8$ m s^{-1}, 1 amu = 1.66×10^{-27} kg, $\pi = 3.14$, $h = 6.627 \times 10^{-34}$ J s)

Jan-25 · UE24 · Set 1 (3 marks)

Solution.

i. First line at $2B$, so $B = \dfrac{13.022}{2} = 6.511\ \text{cm}^{-1}$.

ii. $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.63 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(6.511)} = 4.30 \times 10^{-47}\ \text{kg m}^2$ (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})

iii. $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.0 \times 126.9}{127.9} = 0.9922\ \text{amu} = 1.65 \times 10^{-27}\ \text{kg}$.

iv. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{4.30 \times 10^{-47}}{1.65 \times 10^{-27}}} = 1.62 \times 10^{-10}\ \text{m} = 1.616\ \text{Å}$.

Answer: $I \approx 4.30 \times 10^{-47}$ kg m^2, $r_0 \approx 1.62$ Å

5. For a rigid-rotor diatomic molecule: (i) draw the energy-level diagram up to $J = 5$; (ii) draw the rotational spectrum; (iii) calculate the moment of inertia and the distance between two spectral lines if $B = 10.56$ cm^{-1}. ($c = 3 \times 10^8$ m s^{-1}, $\pi = 3.14$, $h = 6.625 \times 10^{-34}$ J s)

Jun-24 · UE23 · Set 1 (6 marks)

Solution.

i. (i) Levels $\varepsilon_J = BJ(J+1)$: 0, 2B, 6B, 12B, 20B, 30B for J = 0...5 — gaps grow 2B, 4B, 6B, 8B, 10B (**Fig. G9**).

ii. (ii) Allowed lines $J \to J+1$ at 2B, 4B, 6B, 8B, 10B... equally spaced by 2B; intensity rises then falls (**Fig. G10**).

iii. (iii) $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.625 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(10.56)} = 2.65 \times 10^{-47}$ kg m^2.

iv. Spacing between adjacent lines $= 2B = 2(10.56) = 21.12\ \text{cm}^{-1}$.

Answer: $I = 2.65 \times 10^{-47}$ kg m^2, spacing $= 21.12$ cm^{-1}

6. For a rotating diatomic molecule: (i) derive an expression for the moment of inertia I; (ii) if $B = 10$ cm^{-1}, give the rotational energy levels for $J = 0, 1, 2, 3$ in cm^{-1} and draw the energy-level diagram up to $J = 3$.

Jan · UE23A · Set 2 (6 marks)

Solution.

i. (i) See derivation Q1-D.

ii. (ii) $\varepsilon_J = 10\,J(J+1)$: $J=0 \to 0$; $J=1 \to 20$; $J=2 \to 60$; $J=3 \to 120$ cm^{-1}. Draw four horizontal lines at these heights (gaps 20, 40, 60) with $\Delta J = +1$ arrows.

Answer: Levels: 0, 20, 60, 120 cm^{-1}

7. For a rigid-rotor diatomic molecule draw the energy-level diagram up to $J = 6$.

Dec-24 · UE22 (2 marks)

Solution. $\varepsilon_J = BJ(J+1)$: 0, 2B, 6B, 12B, 20B, 30B, 42B — see **Fig. G9** (it is drawn to J = 6). Label J on each level and show gaps increasing 2B, 4B, ... 12B.

Answer: Levels 0, 2B, 6B, 12B, 20B, 30B, 42B

8. (ii) If the rotational transition $J = 0 \to 1$ for HCl occurs at 20.68 cm^{-1}, find the energy required for the $J = 20 \to 21$ transition in cm^{-1}. (iii) If the moment of inertia of HBr is 3.26×10^{-47} kg m^2, at what wavenumber (in m^{-1}) does the first line appear? ($h = 6.62 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m s^{-1})

Jul-23 · UE22 · B (6 marks)

Solution.

i. (ii) $J = 0 \to 1$ gives $2B = 20.68$, so $B = 10.34$ cm^{-1}. $\Delta\varepsilon(J \to J+1) = 2B(J+1)$; for $J = 20$: $2B \times 21 = 21 \times 20.68 = \mathbf{434.28}$ cm^{-1}.

ii. (iii) $B = \dfrac{h}{8\pi^2 I c} = \dfrac{6.62 \times 10^{-34}}{8\pi^2 (3.26 \times 10^{-47})(3 \times 10^8)} = 857.3\ \mathbf{m^{-1}}$ (c in m s^{-1} gives m $^{-1}$ directly).

iii. First line $= 2B = 1715$ m^{-1} $= 17.15$ cm^{-1}.

Answer: (ii) 434.28 cm^{-1} · (iii) ≈ 1715 m^{-1}

9. For HBr, with bond length 1.404 Å, at what energy (in m^{-1}) does the first line appear in the rotational spectrum? (H = 1.008, Br = 79.9 amu; 1 amu $= 1.66 \times 10^{-27}$ kg, $c = 3 \times 10^8$ m s^{-1}, $h = 6.6 \times 10^{-34}$ J s)

Jan-25 · UE24 · Set 2 (6 marks)

Solution.

i. $\mu = \dfrac{1.008 \times 79.9}{80.908}\ \mathbf{amu} = 0.9954$ amu $= 1.65 \times 10^{-27}$ kg.

ii. $I = \mu r_0^2 = 1.65 \times 10^{-27} \times (1.404 \times 10^{-10})^2 = 3.26 \times 10^{-47}$ kg m^2.

iii. $B = \dfrac{h}{8\pi^2 I c} = 855.4$ m^{-1}.

iv. First line $= 2B = 1711$ m^{-1} (≈ 17.11 cm^{-1}).

Answer: First line ≈ 1711 m^{-1}

Practice: exam-style variants

Set 1: Bond length from the first line

Same pattern as the exam. Do all six — the method never changes.

10. The first line in the rotational spectrum of HF is at 41.9 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, F = 18.998 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

11. The first line in the rotational spectrum of CO is at 3.86 cm^{-1}. Calculate B, I and the bond length. (C = 12.0, O = 15.995 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

12. The first line in the rotational spectrum of HBr is at 17.19 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, Br = 79.904 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

13. The first line in the rotational spectrum of HI is at 13.0 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, I = 126.9 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

14. The first line in the rotational spectrum of CS is at 1.632 cm^{-1}. Calculate B, I and the bond length. (C = 12.0, S = 31.97 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

15. The first line in the rotational spectrum of HCl is at 21.18 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, Cl = 35.45 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Set 2: Moment of inertia and line spacing from B

Watch the units: B in cm^{-1} pairs with c in cm s^{-1}.

16. A rigid rotor has $B = 10.56$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

17. A rigid rotor has $B = 5.0$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

18. A rigid rotor has $B = 1.93$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

19. A rigid rotor has $B = 8.5$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

20. A rigid rotor has $B = 0.39$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Set 3: Position of a high-J line

The trick: first line = 2B, so the J→J+1 line is simply (J+1) × first line.

21. For HCl the $J = 0 \to 1$ line is at 20.68 cm^{-1}. Find the line for $J = 20 \to 21$ and the wavenumber gap between it and the previous line.

22. For HCl the $J = 0 \to 1$ line is at 21.18 cm^{-1}. Find the line for $J = 9 \to 10$ and the wavenumber gap between it and the previous line.

23. For HBr the $J = 0 \to 1$ line is at 17.19 cm^{-1}. Find the line for $J = 14 \to 15$ and the wavenumber gap between it and the previous line.

24. For HI the $J = 0 \to 1$ line is at 13.022 cm^{-1}. Find the line for $J = 4 \to 5$ and the wavenumber gap between it and the previous line.

25. For CO the $J = 0 \to 1$ line is at 3.86 cm^{-1}. Find the line for $J = 29 \to 30$ and the wavenumber gap between it and the previous line.

Set 4: The reverse problem: from I or r_0 to the first line

Exam variants (HBr $I = 3.26 \times 10^{-47}$; HBr $r_0 = 1.404$ Å) are in this family.

26. The moment of inertia of HBr is 3.26×10^{-47} kg m^2. At what wavenumber (cm^{-1}) does the first rotational line appear? ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

27. The moment of inertia of HCl is 2.64×10^{-47} kg m^2. At what wavenumber (cm^{-1}) does the first rotational line appear? ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

28. The bond length of HI is 1.61 Å. At what wavenumber (cm^{-1}) does the first rotational line appear? (H = 1.008, I = 126.9 amu) ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

29. The bond length of CO is 1.128 Å. At what wavenumber (cm^{-1}) does the first rotational line appear? (C = 12.0, O = 15.995 amu) ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

30. The bond length of HF is 0.917 Å. At what wavenumber (cm^{-1}) does the first rotational line appear? (H = 1.008, F = 18.998 amu) ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Set 5: Concept twists

Same ideas, asked differently.

31. DCl has the same bond length as HCl (D = 2.014 amu). If HCl's first line is at 21.18 cm^{-1}, predict DCl's first line. Is it higher or lower? Why?

32. A rotational spectrum shows lines at 42.4, 63.6 and 84.8 cm^{-1} with no line at 21.2 cm^{-1} shown. Deduce B and the J values involved.

33. Which of these show a pure rotational spectrum: H_2, HCl, CO_2, CO, N_2, HF, Cl_2, H_2O? Give the rule.

34. Why does the intensity of rotational lines first rise and then fall with J?

35. How many lines appear between 0 and 100 cm^{-1} for $B = 10.56$ cm^{-1}?

New question types (not yet asked)

Centre of mass and moment of inertia from the masses

Tests the derivation numerically: the two routes to I must agree.

36. HCl: $m_1 = 1.008$, $m_2 = 35.45$ amu, bond length 1.275 Å. Find the distances of each atom from the centre of mass, then $I = m_1 r_1^2 + m_2 r_2^2$ and verify $I = \mu r_0^2$.

37. HF: $m_1 = 1.008$, $m_2 = 18.998$ amu, bond length 0.917 Å. Find the distances of each atom from the centre of mass, then $I = m_1 r_1^2 + m_2 r_2^2$ and verify $I = \mu r_0^2$.

38. CO: $m_1 = 12.0$, $m_2 = 15.995$ amu, bond length 1.128 Å. Find the distances of each atom from the centre of mass, then $I = m_1 r_1^2 + m_2 r_2^2$ and verify $I = \mu r_0^2$.

39. NaCl: $m_1 = 22.99$, $m_2 = 35.45$ amu, bond length 2.36 Å. Find the distances of each atom from the centre of mass, then $I = m_1 r_1^2 + m_2 r_2^2$ and verify $I = \mu r_0^2$.

40. LiH: $m_1 = 6.94$, $m_2 = 1.008$ amu, bond length 1.595 Å. Find the distances of each atom from the centre of mass, then $I = m_1 r_1^2 + m_2 r_2^2$ and verify $I = \mu r_0^2$.

41. HBr: $m_1 = 1.008$, $m_2 = 79.9$ amu, bond length 1.414 Å. Find the distances of each atom from the centre of mass, then $I = m_1 r_1^2 + m_2 r_2^2$ and verify $I = \mu r_0^2$.

Frequency, wavelength and energy of rotational lines

Links rotational spectroscopy to the EM-spectrum table: lines are in the microwave region.

42. HCl has $B = 10.59$ cm^{-1}. For the $J = 0 \to 1$ absorption find: the frequency (GHz), wavelength (mm), photon energy (J and kJ mol^{-1}), and verify that it lies in the microwave region (1 cm – 100 µm).

43. CO has $B = 1.93$ cm^{-1}. For the $J = 0 \to 1$ absorption find: the frequency (GHz), wavelength (mm), photon energy (J and kJ mol^{-1}), and verify that it lies in the microwave region (1 cm – 100 µm).

44. HF has $B = 20.9$ cm^{-1}. For the $J = 0 \to 1$ absorption find: the frequency (GHz), wavelength (mm), photon energy (J and kJ mol^{-1}), and verify that it lies in the microwave region (1 cm – 100 µm).

45. HBr has $B = 8.46$ cm^{-1}. For the $J = 0 \to 1$ absorption find: the frequency (GHz), wavelength (mm), photon energy (J and kJ mol^{-1}), and verify that it lies in the microwave region (1 cm – 100 µm).

46. HI has $B = 6.51$ cm^{-1}. For the $J = 0 \to 1$ absorption find: the frequency (GHz), wavelength (mm), photon energy (J and kJ mol^{-1}), and verify that it lies in the microwave region (1 cm – 100 µm).

Identify J and B from two adjacent lines

Reverse engineering a spectrum. The gap is always 2B; the line number is position ÷ gap.

47. Two adjacent lines in the rotational spectrum of HCl are at 82.80 and 103.50 cm^{-1}. Find B, and the $J \to J + 1$ transitions that produced them.

48. Two adjacent lines in the rotational spectrum of CO are at 27.02 and 30.88 cm^{-1}. Find B, and the $J \to J + 1$ transitions that produced them.

49. Two adjacent lines in the rotational spectrum of HF are at 125.40 and 167.20 cm^{-1}. Find B, and the $J \to J + 1$ transitions that produced them.

50. Two adjacent lines in the rotational spectrum of HBr are at 101.52 and 118.44 cm^{-1}. Find B, and the $J \to J + 1$ transitions that produced them.

Isotopic substitution

A classic experimental use: isotopic shifts confirm that a spectrum belongs to a given molecule.

51. ^{12}CO has $B = 1.9313$ cm^{-1}. Predict B and the first rotational line of ^{13}CO (same bond length). Is the line higher or lower, and by how much?

52. H^{35}Cl has $B = 10.59$ cm^{-1}. Predict B and the first rotational line of H^{37}Cl (same bond length). Is the line higher or lower, and by how much?

53. HCl has $B = 10.59$ cm^{-1}. Predict B and the first rotational line of DCl (same bond length). Is the line higher or lower, and by how much?

54. HBr has $B = 8.46$ cm^{-1}. Predict B and the first rotational line of DBr (same bond length). Is the line higher or lower, and by how much?

Find the unknown atom

Run the bond-length method backwards: from I and r_0 to μ and then to a mass.

55. A hydrogen halide HX has its first rotational line at 21.18 cm^{-1} and bond length 1.275 Å. Find I, μ and the atomic mass of X, and identify X. (H = 1.008 amu)

56. A hydrogen halide HX has its first rotational line at 17.0 cm^{-1} and bond length 1.414 Å. Find I, μ and the atomic mass of X, and identify X. (H = 1.008 amu)

57. A hydrogen halide HX has its first rotational line at 41.9 cm^{-1} and bond length 0.917 Å. Find I, μ and the atomic mass of X, and identify X. (H = 1.008 amu)

58. A hydrogen halide HX has its first rotational line at 13.022 cm^{-1} and bond length 1.617 Å. Find I, μ and the atomic mass of X, and identify X. (H = 1.008 amu)

Which line is the strongest? (extension)

Beyond the slides but explains why the spectrum's intensity rises then falls — a favourite 'why' question.

59. (Extension) The most intense rotational line has $J_{max} = \sqrt{kT/2hcB} - \frac{1}{2}$. Find J_{max} for HCl ($B = 10.59$ cm^{-1}) at 300 K ($kT/hc = 0.695\,T$ cm^{-1}) and the wavenumber of the strongest line.

60. (Extension) The most intense rotational line has $J_{max} = \sqrt{kT/2hcB} - \frac{1}{2}$. Find J_{max} for CO ($B = 1.93$ cm^{-1}) at 300 K ($kT/hc = 0.695\,T$ cm^{-1}) and the wavenumber of the strongest line.

61. (Extension) The most intense rotational line has $J_{max} = \sqrt{kT/2hcB} - \frac{1}{2}$. Find J_{max} for HF ($B = 20.9$ cm^{-1}) at 300 K ($kT/hc = 0.695\,T$ cm^{-1}) and the wavenumber of the strongest line.

62. (Extension) The most intense rotational line has $J_{max} = \sqrt{kT/2hcB} - \frac{1}{2}$. Find J_{max} for CO ($B = 1.93$ cm^{-1}) at 500 K ($kT/hc = 0.695\,T$ cm^{-1}) and the wavenumber of the strongest line.

63. (Extension) The most intense rotational line has $J_{max} = \sqrt{kT/2hcB} - \frac{1}{2}$. Find J_{max} for HI ($B = 6.51$ cm^{-1}) at 298 K ($kT/hc = 0.695\,T$ cm^{-1}) and the wavenumber of the strongest line.

Level tables and thermal energy

64. For HCl ($B = 10.59$ cm^{-1}) tabulate ε_J for $J = 0$ to 4 and the allowed absorption lines $J \to J + 1$. Which gaps are equal?

65. For CO ($B = 1.93$ cm^{-1}) tabulate ε_J for $J = 0$ to 4 and the allowed absorption lines $J \to J + 1$. Which gaps are equal?

66. For HF ($B = 20.9$ cm^{-1}) tabulate ε_J for $J = 0$ to 4 and the allowed absorption lines $J \to J + 1$. Which gaps are equal?

67. At 298 K, $kT/hc \approx 207$ cm^{-1}. For HCl, how does the first rotational line (21.18 cm^{-1}) compare with kT? What does this say about populations?

Selection rules and active molecules

More variants of 'which is microwave active, and why?'

68. Classify as microwave active/inactive and explain: N_2, NO, HBr, CS_2, OCS, SO_2, CH_4, H_2O, HD.

69. Why can a molecule with no permanent dipole still be IR-active but never microwave-active?

70. If B is halved, what happens to I, to the first line and to the line spacing?

71. If the bond length is doubled, by what factor does B change?

72. Why is the rigid-rotor model an approximation? What really happens at high J?

73. How many rotational lines lie between 0 and 200 cm^{-1} for HI ($B = 6.51\ \text{cm}^{-1}$)?

Quick-fire checks and spot the error

True or false

Answer first; the key explains why.

74. The first line in a rigid-rotor spectrum appears at B.

75. Adjacent lines in the rotational spectrum are separated by $2B$.

76. A molecule must have a permanent dipole to absorb microwave radiation.

77. J can start at 1.

78. CO_2 shows a pure rotational spectrum.

79. $I = \mu r_0^2$ where μ is reduced mass.

80. Increasing the reduced mass shifts rotational lines to higher wavenumber.

81. The energy gap between levels J and $J+1$ is $2B(J+1)$.

82. $\Delta J = \pm 2$ is the selection rule for the rigid rotor.

83. The most intense line is always the first line.

84. H_2 and HD both show microwave spectra.

85. Rotational energy levels get closer together as J increases.

Spot the error

Each worked 'solution' contains one mistake.

86. HCl first line = 21.18 cm^{-1}, so B = 21.18 cm^{-1} and I = $h/8\pi^2 cB$.

87. I = $h/(8\pi^2 cB) = 6.63\times10^{-34}/(8\pi^2\cdot3\times10^8\cdot10.59) = 2.64\times10^{-47}$ (c in m s^{-1}, B in cm^{-1}).

88. μ of HCl = $(1.008\times35.45)/(1.008+35.45) = 0.98$ kg, so $r_0 = \sqrt{(I/\mu)}$.

89. Rotational levels J = 0 to 3 for B = 10: 10, 20, 30, 40 cm^{-1}.

90. The line J = 20→21 is at 20 × 2B.

91. Bond length of HCl is 1.27 m.

Answers

Set 1: Bond length from the first line

10. ***Working.*** $B = 41.9/2 = 20.95$ cm^{-1}; $I = h/8\pi^2 cB = 1.34 \times 10^{-47}$ kg m^2; $\mu = 0.9572$ amu.
    Answer: $B = 20.95$ cm^{-1}, $I = 1.34 \times 10^{-47}$ kg m^2, $r_0 = 0.917$ Å

11. ***Working.*** $B = 3.86/2 = 1.93$ cm^{-1}; $I = h/8\pi^2 cB = 1.45 \times 10^{-46}$ kg m^2; $\mu = 6.856$ amu.
    Answer: $B = 1.93$ cm^{-1}, $I = 1.45 \times 10^{-46}$ kg m^2, $r_0 = 1.128$ Å

12. ***Working.*** $B = 17.19/2 = 8.595$ cm^{-1}; $I = h/8\pi^2 cB = 3.25 \times 10^{-47}$ kg m^2; $\mu = 0.9954$ amu.
    Answer: $B = 8.595$ cm^{-1}, $I = 3.25 \times 10^{-47}$ kg m^2, $r_0 = 1.403$ Å

13. ***Working.*** $B = 13.0/2 = 6.5$ cm^{-1}; $I = h/8\pi^2 cB = 4.30 \times 10^{-47}$ kg m^2; $\mu = 1$ amu.
    Answer: $B = 6.5$ cm^{-1}, $I = 4.30 \times 10^{-47}$ kg m^2, $r_0 = 1.610$ Å

14. ***Working.*** $B = 1.632/2 = 0.816$ cm^{-1}; $I = h/8\pi^2 cB = 3.43 \times 10^{-46}$ kg m^2; $\mu = 8.725$ amu.
    Answer: $B = 0.816$ cm^{-1}, $I = 3.43 \times 10^{-46}$ kg m^2, $r_0 = 1.538$ Å

15. ***Working.*** $B = 21.18/2 = 10.59$ cm^{-1}; $I = h/8\pi^2 cB = 2.64 \times 10^{-47}$ kg m^2; $\mu = 0.9801$ amu.
    Answer: $B = 10.59$ cm^{-1}, $I = 2.64 \times 10^{-47}$ kg m^2, $r_0 = 1.274$ Å

Set 2: Moment of inertia and line spacing from B

16. ***Working.*** $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.
    Answer: $I = 2.65 \times 10^{-47}$ kg m^2; spacing 21.12 cm^{-1}; lines 21.12, 42.24, 63.36 cm^{-1}

17. ***Working.*** $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.
    Answer: $I = 5.59 \times 10^{-47}$ kg m^2; spacing 10 cm^{-1}; lines 10, 20, 30 cm^{-1}

18. ***Working.*** $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.
    Answer: $I = 1.45 \times 10^{-46}$ kg m^2; spacing 3.86 cm^{-1}; lines 3.86, 7.72, 11.58 cm^{-1}

19. ***Working.*** $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.
    Answer: $I = 3.29 \times 10^{-47}$ kg m^2; spacing 17 cm^{-1}; lines 17, 34, 51 cm^{-1}

20. ***Working.*** $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.
    Answer: $I = 7.17 \times 10^{-46}$ kg m^2; spacing 0.78 cm^{-1}; lines 0.78, 1.56, 2.34 cm^{-1}

Set 3: Position of a high-J line

21. ***Working.*** $\Delta\varepsilon = 2B(J+1) = (J+1) \times (\text{first line})$; adjacent-line gap $= 2B$.
    Answer: 434.28 cm^{-1}; gap $= 20.68$ cm^{-1}

22. ***Working.*** $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.
Answer: 211.80 cm^{-1}; gap $= 21.18$ cm^{-1}

23. ***Working.*** $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.
Answer: 257.85 cm^{-1}; gap $= 17.19$ cm^{-1}

24. ***Working.*** $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.
Answer: 65.11 cm^{-1}; gap $= 13.022$ cm^{-1}

25. ***Working.*** $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.
Answer: 115.80 cm^{-1}; gap $= 3.86$ cm^{-1}

Set 4: The reverse problem: from I or r_0 to the first line

26. ***Working.*** If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.
Answer: $B = 8.581$ cm^{-1}, first line $= 17.16$ cm^{-1}

27. ***Working.*** If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.
Answer: $B = 10.596$ cm^{-1}, first line $= 21.19$ cm^{-1}

28. ***Working.*** If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.
Answer: $B = 6.501$ cm^{-1}, first line $= 13.00$ cm^{-1}

29. ***Working.*** If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.
Answer: $B = 1.932$ cm^{-1}, first line $= 3.86$ cm^{-1}

30. ***Working.*** If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.
Answer: $B = 20.936$ cm^{-1}, first line $= 41.87$ cm^{-1}

Set 5: Concept twists

31. ***Working.***
    i. r_0 is unchanged (electronic structure identical) so $I = \mu r_0^2$ and $B \propto 1/\mu$.
    ii. $\mu_{HCl} = 0.9801$, $\mu_{DCl} = 1.9057$ amu, so $B_{DCl} = B_{HCl}\,\mu_{HCl}/\mu_{DCl}$.
    Answer: First line ≈ 10.89 cm^{-1} — lower, because heavier μ gives larger I and smaller B.

32. ***Working.*** Spacing $= 2B = 21.2$ so $B = 10.6$ cm^{-1}. Lines $= 2B(J+1)$: $42.4 \to J+1 = 2$ ($J = 1 \to 2$); $63.6 \to 3$; $84.8 \to 4$.
Answer: $B = 10.6$ cm^{-1}; transitions $1 \to 2$, $2 \to 3$, $3 \to 4$

33. ***Working.*** Gross rule: permanent dipole moment. Specific rule: $\Delta J = \pm 1$.
Answer: Active: HCl, CO, HF, H_2O. Inactive: H_2, CO_2 (symmetric, zero net dipole), N_2, Cl_2.

34. ***Working.*** Level J has $(2J+1)$ states (degeneracy) but its population also falls exponentially as energy rises (Boltzmann factor).
Answer: Degeneracy $(2J+1)$ grows; Boltzmann population $e^{-E_J/kT}$ decays; the product peaks at an intermediate J.

35. ***Working.*** Lines at $21.12\,(J+1)$; $21.12n < 100 \Rightarrow n \le 4$.

    Answer: 4 lines (21.12, 42.24, 63.36, 84.48 cm^{-1})

Centre of mass and moment of inertia from the masses

36. ***Working.*** $r_1 = m_2 r_0/(m_1+m_2)$, $r_2 = m_1 r_0/(m_1+m_2)$ — the lighter atom is further from C.

    Answer: $r_1 = 1.2397$ Å, $r_2 = 0.0353$ Å; $I = 2.64 \times 10^{-47}$ kg m^2; $\mu r_0^2 = 2.64 \times 10^{-47}$ kg m^2 ✓ (same)

37. ***Working.*** $r_1 = m_2 r_0/(m_1+m_2)$, $r_2 = m_1 r_0/(m_1+m_2)$ — the lighter atom is further from C.

    Answer: $r_1 = 0.8708$ Å, $r_2 = 0.0462$ Å; $I = 1.34 \times 10^{-47}$ kg m^2; $\mu r_0^2 = 1.34 \times 10^{-47}$ kg m^2 ✓ (same)

38. ***Working.*** $r_1 = m_2 r_0/(m_1+m_2)$, $r_2 = m_1 r_0/(m_1+m_2)$ — the lighter atom is further from C.

    Answer: $r_1 = 0.6445$ Å, $r_2 = 0.4835$ Å; $I = 1.45 \times 10^{-46}$ kg m^2; $\mu r_0^2 = 1.45 \times 10^{-46}$ kg m^2 ✓ (same)

39. ***Working.*** $r_1 = m_2 r_0/(m_1+m_2)$, $r_2 = m_1 r_0/(m_1+m_2)$ — the lighter atom is further from C.

    Answer: $r_1 = 1.4316$ Å, $r_2 = 0.9284$ Å; $I = 1.29 \times 10^{-45}$ kg m^2; $\mu r_0^2 = 1.29 \times 10^{-45}$ kg m^2 ✓ (same)

40. ***Working.*** $r_1 = m_2 r_0/(m_1+m_2)$, $r_2 = m_1 r_0/(m_1+m_2)$ — the lighter atom is further from C.

    Answer: $r_1 = 0.2023$ Å, $r_2 = 1.3927$ Å; $I = 3.72 \times 10^{-47}$ kg m^2; $\mu r_0^2 = 3.72 \times 10^{-47}$ kg m^2 ✓ (same)

41. ***Working.*** $r_1 = m_2 r_0/(m_1+m_2)$, $r_2 = m_1 r_0/(m_1+m_2)$ — the lighter atom is further from C.

    Answer: $r_1 = 1.3964$ Å, $r_2 = 0.0176$ Å; $I = 3.30 \times 10^{-47}$ kg m^2; $\mu r_0^2 = 3.30 \times 10^{-47}$ kg m^2 ✓ (same)

Frequency, wavelength and energy of rotational lines

42. ***Working.*** Line $= 2B$ cm^{-1}; $\nu = c\bar{\nu}$; $\lambda = 1/\bar{\nu}$; $E = hc\bar{\nu}$; $E \times N_A$.

    Answer: Line 21.18 cm^{-1}; ν = 635 GHz; λ = 0.472 mm (microwave ✓); E = 4.21×10^{-22} J = 0.254 kJ mol^{-1}

43. ***Working.*** Line $= 2B$ cm^{-1}; $\nu = c\bar{\nu}$; $\lambda = 1/\bar{\nu}$; $E = hc\bar{\nu}$; $E \times N_A$.

    Answer: Line 3.86 cm^{-1}; ν = 116 GHz; λ = 2.591 mm (microwave ✓); E = 7.67×10^{-23} J = 0.046 kJ mol^{-1}

44. ***Working.*** Line $= 2B$ cm^{-1}; $\nu = c\bar{\nu}$; $\lambda = 1/\bar{\nu}$; $E = hc\bar{\nu}$; $E \times N_A$.

    Answer: Line 41.80 cm^{-1}; ν = 1254 GHz; λ = 0.239 mm (microwave ✓); E = 8.31×10^{-22} J = 0.500 kJ mol^{-1}

45. ***Working.*** Line $= 2B$ cm^{-1}; $\nu = c\bar{\nu}$; $\lambda = 1/\bar{\nu}$; $E = hc\bar{\nu}$; $E \times N_A$.

    Answer: Line 16.92 cm^{-1}; ν = 508 GHz; λ = 0.591 mm (microwave ✓); E = 3.36×10^{-22} J = 0.203 kJ mol^{-1}

46. ***Working.*** Line $= 2B$ cm^{-1}; $\nu = c\bar{\nu}$; $\lambda = 1/\bar{\nu}$; $E = hc\bar{\nu}$; $E \times N_A$.

    Answer: Line 13.02 cm^{-1}; ν = 391 GHz; λ = 0.768 mm (microwave ✓); E = 2.59×10^{-22} J = 0.156 kJ mol^{-1}

Identify J and B from two adjacent lines

47. ***Working.*** Adjacent gap $= 2B$; then line $= 2B(J+1)$ gives $J+1$.

    Answer: $2B = 20.70 \Rightarrow B = 10.35$ cm^{-1}; lines are $J = 3 \to 4$ and $J = 4 \to 5$

48. ***Working.*** Adjacent gap $= 2B$; then line $= 2B(J+1)$ gives $J+1$.

    Answer: $2B = 3.86 \Rightarrow B = 1.93$ cm^{-1}; lines are $J = 6 \to 7$ and $J = 7 \to 8$

49. ***Working.*** Adjacent gap $= 2B$; then line $= 2B(J+1)$ gives $J+1$.

    Answer: $2B = 41.80 \Rightarrow B = 20.9$ cm^{-1}; lines are $J = 2 \to 3$ and $J = 3 \to 4$

50. ***Working.*** Adjacent gap $= 2B$; then line $= 2B(J+1)$ gives $J+1$.

    Answer: $2B = 16.92 \Rightarrow B = 8.46$ cm^{-1}; lines are $J = 5 \to 6$ and $J = 6 \to 7$

Isotopic substitution

51. ***Working.*** Same electronic structure ⇒ same $r_0 \Rightarrow I = \mu r_0^2$, so $B \propto 1/\mu$: $B_2 = B_1\,\mu_1/\mu_2$.

    Answer: μ_1 = 6.8562, μ_2 = 7.1723 amu; B_2 = 1.8462 cm^{-1}; first line 3.692 cm^{-1} — lower by 0.170 cm^{-1}

52. ***Working.*** Same electronic structure ⇒ same $r_0 \Rightarrow I = \mu r_0^2$, so $B \propto 1/\mu$: $B_2 = B_1\,\mu_1/\mu_2$.

    Answer: μ_1 = 0.9798, μ_2 = 0.9812 amu; B_2 = 10.5740 cm^{-1}; first line 21.148 cm^{-1} — lower by 0.032 cm^{-1}

53. ***Working.*** Same electronic structure ⇒ same $r_0 \Rightarrow I = \mu r_0^2$, so $B \propto 1/\mu$: $B_2 = B_1\,\mu_1/\mu_2$.

    Answer: μ_1 = 0.9801, μ_2 = 1.9057 amu; B_2 = 5.4465 cm^{-1}; first line 10.893 cm^{-1} — lower by 10.287 cm^{-1}

54. ***Working.*** Same electronic structure ⇒ same $r_0 \Rightarrow I = \mu r_0^2$, so $B \propto 1/\mu$: $B_2 = B_1\,\mu_1/\mu_2$.

    Answer: μ_1 = 0.9954, μ_2 = 1.9645 amu; B_2 = 4.2868 cm^{-1}; first line 8.574 cm^{-1} — lower by 8.346 cm^{-1}

Find the unknown atom

55. ***Working.*** $B = \text{line}/2$; $I = h/8\pi^2 cB$; $\mu = I/r_0^2$; then $\mu = \dfrac{m_H m_X}{m_H + m_X} \Rightarrow m_X = \dfrac{\mu m_H}{m_H - \mu}$.
Answer: $I = 2.64 \times 10^{-47}$ kg m^2; μ = 0.9788 amu; $m_X \approx 33.8$ amu → X = Cl

56. ***Working.*** $B = \text{line}/2$; $I = h/8\pi^2 cB$; $\mu = I/r_0^2$; then $\mu = \dfrac{m_H m_X}{m_H + m_X} \Rightarrow m_X = \dfrac{\mu m_H}{m_H - \mu}$.
Answer: $I = 3.29 \times 10^{-47}$ kg m^2; μ = 0.9915 amu; $m_X \approx 60.8$ amu → X = Br

57. ***Working.*** $B = \text{line}/2$; $I = h/8\pi^2 cB$; $\mu = I/r_0^2$; then $\mu = \dfrac{m_H m_X}{m_H + m_X} \Rightarrow m_X = \dfrac{\mu m_H}{m_H - \mu}$.
Answer: $I = 1.34 \times 10^{-47}$ kg m^2; μ = 0.9566 amu; $m_X \approx 18.7$ amu → X = F

58. ***Working.*** $B = \text{line}/2$; $I = h/8\pi^2 cB$; $\mu = I/r_0^2$; then $\mu = \dfrac{m_H m_X}{m_H + m_X} \Rightarrow m_X = \dfrac{\mu m_H}{m_H - \mu}$.
Answer: $I = 4.30 \times 10^{-47}$ kg m^2; μ = 0.9898 amu; $m_X \approx 54.9$ amu → X = I

Which line is the strongest? (extension)

59. ***Working.*** Population $\propto (2J+1)e^{-\varepsilon_J/kT}$; set its derivative to zero.
Answer: $J_{max} = 2.64$ → nearest level 3; strongest line $\approx 2B(J+1) = 84.7$ cm^{-1}

60. ***Working.*** Population $\propto (2J+1)e^{-\varepsilon_J/kT}$; set its derivative to zero.
Answer: $J_{max} = 6.85$ → nearest level 7; strongest line $\approx 2B(J+1) = 30.9$ cm^{-1}

61. ***Working.*** Population $\propto (2J+1)e^{-\varepsilon_J/kT}$; set its derivative to zero.
Answer: $J_{max} = 1.73$ → nearest level 2; strongest line $\approx 2B(J+1) = 125.4$ cm^{-1}

62. ***Working.*** Population $\propto (2J+1)e^{-\varepsilon_J/kT}$; set its derivative to zero.
Answer: $J_{max} = 8.99$ → nearest level 9; strongest line $\approx 2B(J+1) = 38.6$ cm^{-1}

63. ***Working.*** Population $\propto (2J+1)e^{-\varepsilon_J/kT}$; set its derivative to zero.
Answer: $J_{max} = 3.49$ → nearest level 3; strongest line $\approx 2B(J+1) = 52.1$ cm^{-1}

Level tables and thermal energy

64. ***Working.*** $\varepsilon_J = BJ(J+1)$; line $= \varepsilon_{J+1} - \varepsilon_J = 2B(J+1)$.
Answer: ε: J=0: 0.00, J=1: 21.18, J=2: 63.54, J=3: 127.08, J=4: 211.80 cm^{-1}. Lines: 0→1: 21.18, 1→2: 42.36, 2→3: 63.54, 3→4: 84.72. Adjacent lines are equally spaced (2B); adjacent levels are not.

65. ***Working.*** $\varepsilon_J = BJ(J+1)$; line $= \varepsilon_{J+1} - \varepsilon_J = 2B(J+1)$.
Answer: ε: J=0: 0.00, J=1: 3.86, J=2: 11.58, J=3: 23.16, J=4: 38.60 cm^{-1}. Lines: 0→1: 3.86, 1→2: 7.72, 2→3: 11.58, 3→4: 15.44. Adjacent lines are equally spaced (2B); adjacent levels are not.

66. ***Working.*** $\varepsilon_J = BJ(J+1)$; line $= \varepsilon_{J+1} - \varepsilon_J = 2B(J+1)$.

Answer: ε: J=0: 0.00, J=1: 41.80, J=2: 125.40, J=3: 250.80, J=4: 418.00 cm^{-1}. Lines: 0→1: 41.80, 1→2: 83.60, 2→3: 125.40, 3→4: 167.20. Adjacent lines are equally spaced (2B); adjacent levels are not.

67. **Answer:** $21.18 \ll 207$: many rotational levels are thermally populated at room temperature, so the spectrum shows many lines starting from J > 0 (as the slide notes).

Selection rules and active molecules

68. **Answer:** Active (permanent dipole): NO, HBr, OCS, SO_2, H_2O, HD (tiny dipole, weak). Inactive: N_2, CS_2 (linear symmetric), CH_4 (tetrahedral, zero dipole).

69. **Answer:** IR needs a dipole change during vibration, which symmetric molecules can still show; rotation of a non-polar molecule produces no oscillating dipole.

70. **Answer:** $I \propto 1/B$ doubles; the first line and the spacing both halve ($= 2B$).

71. ***Working.*** $B \propto 1/I \propto 1/r_0^2$.

Answer: B falls by a factor of 4; the spectrum compresses by 4.

72. **Answer:** Bonds stretch under centrifugal force at high J, so I increases and levels lie slightly below $BJ(J+1)$ (non-rigid rotor). Lines then converge slightly.

73. ***Working.*** Lines at $13.02\,n$; $n \leq 200/13.02 = 15.4$.

Answer: 15 lines.

True or false

74. **Answer: False** — it appears at $2B$ ($J = 0 \to 1$).

75. **Answer: True** — $2B(J+2) - 2B(J+1) = 2B$.

76. **Answer: True** — gross selection rule.

77. **Answer: False** — $J = 0, 1, 2\ldots$; $\varepsilon_0 = 0$.

78. **Answer: False** — linear and symmetric: no permanent dipole.

79. **Answer: True** — result of the derivation.

80. **Answer: False** — larger μ ⇒ larger I ⇒ smaller B ⇒ lines move lower.

81. **Answer: True** — $BJ(J+1) \to B(J+1)(J+2)$ gives $2B(J+1)$.

82. **Answer: False** — ΔJ = ±1 (±2 is for rotational Raman).

83. **Answer: False** — intensity peaks at J_{max} (thermal population × degeneracy).

84. **Answer: False** — H_2 has no dipole; HD has a very small one, so only HD is (weakly) active.

85. **Answer: False** — gaps $2B(J+1)$ increase with J.

Spot the error

86. **Answer:** B = 21.18/2 = 10.59 cm^{-1}; the first line is 2B.

87. **Answer:** c and B must use matching length units: use c = 3×10^{10} cm s^{-1} with B in cm^{-1} (or c in m s^{-1} with B in m^{-1}).

88. **Answer:** 0.98 is in amu; multiply by 1.66×10^{-27} to get kg (1.63×10^{-27} kg).

89. **Answer:** ε_J = BJ(J+1): 0, 20, 60, 120 cm^{-1}.

90. **Answer:** It is 2B(J+1) = 21 × 2B = 21 × first line.

91. **Answer:** $r_0 = 1.27\times10^{-10}$ m = 1.27 Å — check units at the end.

Chapter 2: Vibrational (IR) Spectroscopy — Harmonic Oscillator

Topics T5

Every paper has a k $\leftrightarrow$ $\bar{\nu}$calculation, usually with zero-point energy, plus the 'why only one line' reason. Learn the one-line formula and practise the unit handling.

Formulas to remember

$$\bar{\nu}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}\ \text{cm}^{-1}$$

$$k = 4\pi^2 c^2 \bar{\nu}^2 \mu$$

$$\varepsilon_v = \left(v + \tfrac{1}{2}\right)\bar{\nu}_{osc}$$

$$\text{ZPE} = \varepsilon_0 = \tfrac{1}{2}\bar{\nu}_{osc}\ \ (E_0 = \tfrac{1}{2}hc\bar{\nu})$$

$$\Delta v = \pm 1 \ \Rightarrow\ \text{one line at } \bar{\nu}_{osc}$$

Method.

1. Find μ in kg.
2. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{k/\mu}$ (c in cm s^{-1} $\rightarrow$ cm^{-1}).
3. ZPE $= \tfrac{1}{2}\bar{\nu}$ (multiply by hc for joules).
4. For k: $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$.

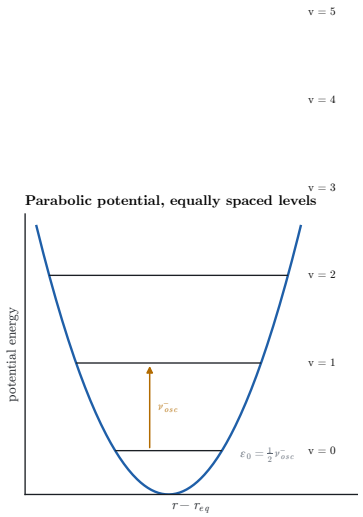


Figure 2.1 (G13): SHO parabola with equally spaced levels.

Past ESA questions with solutions

1. (i) The force constant of HF (SHO) is 970 N m^{-1}. Calculate the oscillation frequency (cm^{-1}) and the zero-point energy. (ii) Why does the SHO spectrum show only one line? (c = 3×10^8 m s^{-1}, π = 3.14, 1 amu = 1.66×10^{-27} kg; H = 1.008, F = 18.99 amu)

   Jun-24 · UE23 · Set 1 (6 marks)

   Solution.

   i. $\mu = \dfrac{1.008 \times 18.99}{19.998} = 0.9572$ amu $= 1.59 \times 10^{-27}$ kg.

   ii. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3\times10^{10})}\sqrt{\dfrac{970}{1.59\times10^{-27}}} = 4147$ cm^{-1}.

   iii. ZPE $= \frac{1}{2}\bar{\nu} = 2074$ cm^{-1} $= 4.12 \times 10^{-20}$ J ($\times hc$).

   iv. (ii) Levels $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$ are equally spaced and the selection rule is $\Delta v = \pm 1$, so every allowed transition needs the same energy $\bar{\nu}$ — one line.

   Answer: $\bar{\nu} \approx 4147$ cm^{-1}, ZPE ≈ 2074 cm^{-1} $\approx 4.12 \times 10^{-20}$ J

2. Same HF question with H = 1.004 amu, F = 18.998 amu (k = 970 N m^{-1}): (i) oscillation frequency in cm^{-1}; (ii) zero-point energy; (iii) why one line?

   Dec-24 · UE22 (6 marks)

   Solution.

   i. $\mu = \dfrac{1.004 \times 18.998}{20.002} = 0.9536$ amu $= 1.58 \times 10^{-27}$ kg.

   ii. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3\times10^{10})}\sqrt{\dfrac{970}{1.58\times10^{-27}}} = 4155$ cm^{-1}.

   iii. ZPE $= \frac{1}{2}\bar{\nu} = 2077$ cm^{-1} $= 4.13 \times 10^{-20}$ J ($\times hc$).

   iv. (iii) Equal spacing + $\Delta v = \pm 1$.

   Answer: $\bar{\nu} \approx 4155$ cm^{-1}; ZPE ≈ 2077 cm^{-1}

3. (i) CO absorbs at 2168 cm^{-1} in a vibrational transition. Calculate the ground-state vibrational energy and force constant (SHO). ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.625 \times 10^{-34}$ J s, C = 12, O = 15.999 amu). (ii) Write a note on the failure of the SHO model.

   Jun-24 · UE23 · Set 2 (6 marks)

   Solution.

   i. Ground state ($v = 0$): $\varepsilon_0 = \frac{1}{2}(2168) = 1084$ cm^{-1} $= 1084 \times hc = 2.15 \times 10^{-20}$ J.

   ii. $\mu = 6.8570$ amu; $k = 4\pi^2 c^2 \bar{\nu}^2 \mu = 1.90 \times 10^3 \approx 1899$ N m^{-1}.

   iii. (ii) See Q3 and the note after this chapter: real bonds dissociate, levels converge, overtones/hot bands occur — SHO predicts none.

   Answer: $\varepsilon_0 = 1084$ cm^{-1}; $k \approx 1899$ N m^{-1}

4. (i) Radiation absorbed by CO in a vibrational transition is 2168 cm^{-1}. Calculate the ground-state vibrational energy and the force constant (SHO; π = 3.14, h = 6.627×10^{-34} J s, c = 3×10^{10} cm s^{-1}, C = 12, O = 15.999). (ii) Only one peak is observed in the SHO model. Substantiate.

Jan · UE23A · Set 2 (6 marks)

Solution.

i. Same as the previous CO question: $\varepsilon_0 = 1084$ cm^{-1}; $k \approx 1.90 \times 10^3$ N m^{-1}.

ii. (ii) Equal level spacing and $\Delta v = \pm 1$.

Answer: ZPE 1084 cm^{-1}; k ≈ 1.90×10^3 N m^{-1}

5. If the IR spectrum of CO shows only one line at 2050 cm^{-1}, calculate the force constant of CO. (C = 12, O = 15.9994 amu; 1 amu = 1.66×10^{-27} kg, $c = 3 \times 10^8$ m s^{-1})

Jul-23 · UE22 · B (4 marks)

Solution.

i. $\mu = 6.8570$ amu $= 1.14 \times 10^{-26}$ kg; $\bar{\nu} = 2050$ $cm^{-1} = 205000$ m^{-1}.

ii. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu = 1.70 \times 10^3$ N m^{-1}.

Answer: $k \approx 1700$ N m^{-1}

6. For HBr (SHO) the vibrational spectrum shows a fundamental at 2649 cm^{-1}. Determine the force constant. (Reduced mass of HBr = 1.639×10^{-27} kg, $c = 3 \times 10^8$ m s^{-1})

Jan–May-24 · UE22 (6 marks)

Solution.

i. μ is given — no need to compute it.

ii. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$ with $\bar{\nu} = 264900$ m^{-1}: $k = 4.09 \times 10^2 \approx 409$ N m^{-1}.

Answer: $k \approx 409$ N m^{-1}

7. The force constant of CO is 1840 N m^{-1}. Calculate the oscillation frequency and wavenumber. (1 amu = 1.66×10^{-27} kg, $c = 3 \times 10^8$ m s^{-1}; C = 12.000, O = 15.9994)

Dec-24 · UE23 (4 marks)

Solution.

i. $\mu = \dfrac{12.0 \times 15.9994}{27.999} = 6.8570$ amu $= 1.14 \times 10^{-26}$ kg.

ii. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3\times10^{10})}\sqrt{\dfrac{1840}{1.14\times10^{-26}}} = 2133$ cm^{-1}.

iii. Frequency $\nu = c\bar{\nu} = (3\times10^{10})(2133) = 6.40\times10^{13}$ Hz.

Answer: $\bar{\nu} \approx 2133$ cm^{-1}; $\nu \approx 6.40 \times 10^{13}$ Hz

8. (i) Using the harmonic oscillator model calculate the zero-point vibrational energy of NO (k = 1530 N m^{-1}; N = 14, O = 16 amu). (ii) Draw the vibrational level diagram and the spectrum. (iii) Calculate the energies of $v = 3$ and $v = 4$. ($c = 3 \times 10^8$ m s^{-1}, $\pi = 3.14$, $h = 6.627 \times 10^{-34}$ J s)

Jan-25 · UE24 · Set 1 (7 marks)

Solution.

i. $\mu = \dfrac{14 \times 16}{30} = 7.4667$ amu $= 1.24 \times 10^{-26}$ kg.

ii. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3 \times 10^{10})}\sqrt{\dfrac{1530}{1.24 \times 10^{-26}}} = 1865$ cm^{-1}.

iii. ZPE $= \frac{1}{2}\bar{\nu} = 932$ cm^{-1} $= 1.85 \times 10^{-20}$ J (using $h = 6.627 \times 10^{-34}$).

iv. (ii) Equally spaced levels (gap $\bar{\nu}$) on a parabola; one line at $\bar{\nu}$ (**Fig. G13**).

v. (iii) $\varepsilon_3 = 3.5\bar{\nu} = 6527$ cm^{-1}; $\varepsilon_4 = 4.5\bar{\nu} = 8392$ cm^{-1}.

Answer: ZPE $\approx$ 932 cm^{-1}; $\varepsilon_3 \approx 6527$, $\varepsilon_4 \approx 8392$ cm^{-1}

Theory questions with model answers

9. **Why does the SHO spectrum show only one line? (asked ×5)**

Answer: Levels $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$ are **equally spaced**; the selection rule $\Delta v = \pm 1$ allows only neighbouring-level jumps. Every jump has the same energy $\bar{\nu}$, so all coincide in one line.

10. **Why is CO_2 IR-active though it has no permanent dipole? (asked ×3)**

Answer: IR activity needs a change in dipole during vibration, not a permanent dipole. CO_2's symmetric stretch gives no change (inactive), but the asymmetric stretch and the bending mode do (active). It has no permanent dipole, so it is microwave-inactive.

11. **Why is HI IR-active but H_2 and I_2 inactive?**

Answer: HI is heteronuclear: stretching changes its dipole moment. H_2 and I_2 are homonuclear: zero dipole at every bond length, so the vibration never changes it.

12. **Write a note on the failure of the SHO model. (asked ×1, relevant ×4)**

Answer: The SHO assumes a parabolic potential and equal spacing, so it predicts a single line and never dissociation. Real bonds break when stretched (anharmonic, Morse), levels converge, $\Delta v = \pm 2, \pm 3$ overtones and hot bands appear, and the molecule can dissociate. The SHO therefore fails at high v and cannot explain overtones.

Practice: exam-style variants

Set 1: k → wavenumber and zero-point energy

Exam pattern: HF, NO, CO k-values. Practise with other diatomics.

13. The force constant of HCl is 516 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 35.45 amu)

14. The force constant of HBr is 412 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 79.9 amu)

15. The force constant of HI is 314 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 126.9 amu)

16. The force constant of CO is 1857 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 12.0, 15.995 amu)

17. The force constant of HF is 966 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 18.998 amu)

18. The force constant of NO is 1595 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 14.0, 16.0 amu)

Set 2: $\bar{\nu} \rightarrow$ force constant

The reverse calculation. A stiffer bond (larger k) gives a higher $\bar{\nu}$.

19. The fundamental IR band of HCl (treated as an SHO) is at 2886 cm^{-1}. Calculate the force constant. (1.008 and 35.45 amu; $c = 3 \times 10^{10}$ cm s^{-1})

20. The fundamental IR band of CO (treated as an SHO) is at 2143 cm^{-1}. Calculate the force constant. (12.0 and 15.995 amu; $c = 3 \times 10^{10}$ cm s^{-1})

21. The fundamental IR band of HF (treated as an SHO) is at 4138 cm^{-1}. Calculate the force constant. (1.008 and 18.998 amu; $c = 3 \times 10^{10}$ cm s^{-1})

22. The fundamental IR band of NO (treated as an SHO) is at 1904 cm^{-1}. Calculate the force constant. (14.0 and 16.0 amu; $c = 3 \times 10^{10}$ cm s^{-1})

23. The fundamental IR band of HI (treated as an SHO) is at 2309 cm^{-1}. Calculate the force constant. (1.008 and 126.9 amu; $c = 3 \times 10^{10}$ cm s^{-1})

Set 3: Energy of the v-th level

Draw the ladder yourself first; the exam also asks for the level diagram and the single-line spectrum.

24. For NO ($k = 1530$ N m^{-1}) calculate the energies of levels $v = 0, 1, 2, 3, 4$ in cm^{-1}, and the spacing between neighbouring levels.

25. For HCl ($k = 516$ N m^{-1}) calculate the energies of levels $v = 0, 1, 2, 5$ in cm^{-1}, and the spacing between neighbouring levels.

26. For CO ($k = 1857$ N m^{-1}) calculate the energies of levels $v = 0, 2, 3, 6$ in cm^{-1}, and the spacing between neighbouring levels.

27. For HF ($k = 966$ N m^{-1}) calculate the energies of levels $v = 1, 2, 4$ in cm^{-1}, and the spacing between neighbouring levels.

Set 4: Concept twists

28. HCl and DCl have the same force constant (516 N m^{-1}). Compare their fundamental wavenumbers and ZPE.

29. Which of the following are IR active: N_2, HCl, CO_2 (symmetric stretch), CO_2 (bend), H_2O (bend), O_2, CH_4 (all stretches)?

30. If k is doubled, by what factor does the SHO wavenumber change? If μ is doubled?

31. State the gross and specific selection rules for IR absorption of a diatomic.

New question types (not yet asked)

Convert IR absorptions: λ, ν, E

The conversions that link the table of regions to calculations.

32. A fundamental IR band of HCl is at 2886 cm^{-1}. Find the wavelength (μm), frequency (Hz), photon energy (J, eV and kJ mol^{-1}) and confirm the region.

33. A fundamental IR band of CO is at 2143 cm^{-1}. Find the wavelength (μm), frequency (Hz), photon energy (J, eV and kJ mol^{-1}) and confirm the region.

34. A fundamental IR band of HF is at 3959 cm^{-1}. Find the wavelength (μm), frequency (Hz), photon energy (J, eV and kJ mol^{-1}) and confirm the region.

35. A fundamental IR band of NO is at 1876 cm^{-1}. Find the wavelength (μm), frequency (Hz), photon energy (J, eV and kJ mol^{-1}) and confirm the region.

36. A fundamental IR band of HI is at 2230 cm^{-1}. Find the wavelength (μm), frequency (Hz), photon energy (J, eV and kJ mol^{-1}) and confirm the region.

37. A fundamental IR band of O–H stretch is at 3400 cm^{-1}. Find the wavelength (μm), frequency (Hz), photon energy (J, eV and kJ mol^{-1}) and confirm the region.

Zero-point energy in J, eV and kJ/mol

Zero-point energy is the most-asked sub-part in Unit 1; this set drills every unit it can appear in.

38. HCl ($k = 516$ N m^{-1}): express the zero-point energy in joules, eV and kJ mol^{-1}, and find the energy needed to excite $v = 0 \to v = 1$ in each unit.

39. HF ($k = 966$ N m^{-1}): express the zero-point energy in joules, eV and kJ mol^{-1}, and find the energy needed to excite $v = 0 \to v = 1$ in each unit.

40. CO ($k = 1857$ N m^{-1}): express the zero-point energy in joules, eV and kJ mol^{-1}, and find the energy needed to excite $v = 0 \to v = 1$ in each unit.

41. NO ($k = 1595$ N m^{-1}): express the zero-point energy in joules, eV and kJ mol^{-1}, and find the energy needed to excite $v = 0 \to v = 1$ in each unit.

42. HBr ($k = 412$ N m^{-1}): express the zero-point energy in joules, eV and kJ mol^{-1}, and find the energy needed to excite $v = 0 \to v = 1$ in each unit.

Isotope effect on the IR band

Same force constant, heavier atom, lower frequency.

43. HCl $\to$ DCl: the original band is at 2886 cm^{-1}. Assuming the same force constant, predict the new band and the ZPE of both.

44. ^{12}CO $\to$ ^{13}CO: the original band is at 2143 cm^{-1}. Assuming the same force constant, predict the new band and the ZPE of both.

45. H^{79}Br $\to$ D^{79}Br: the original band is at 2559 cm^{-1}. Assuming the same force constant, predict the new band and the ZPE of both.

46. ^{14}N^{16}O $\to$ ^{15}N^{16}O: the original band is at 1876 cm^{-1}. Assuming the same force constant, predict the new band and the ZPE of both.

47. HF $\to$ DF: the original band is at 3959 cm^{-1}. Assuming the same force constant, predict the new band and the ZPE of both.

Number of vibrational modes and IR activity (extension)

Beyond diatomics: the same dipole-change test applied to polyatomics. Added for clarity — the slides treat CO_2 only.

48. (Extension) How many fundamental vibrational modes does HCl have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

49. (Extension) How many fundamental vibrational modes does N_2 have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

50. (Extension) How many fundamental vibrational modes does CO_2 have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

51. (Extension) How many fundamental vibrational modes does H_2O have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

52. (Extension) How many fundamental vibrational modes does SO_2 have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

53. (Extension) How many fundamental vibrational modes does NH_3 have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

54. (Extension) How many fundamental vibrational modes does CH_4 have ($3N - 5$ for linear, $3N - 6$ for non-linear) and which are IR-active?

Compare bond strengths from two spectra

55. HF absorbs at 3959 cm^{-1} and HCl at 2886 cm^{-1}. Which has the stronger bond? Calculate both force constants and their ratio.

56. CO absorbs at 2143 cm^{-1} and NO at 1876 cm^{-1}. Which has the stronger bond? Calculate both force constants and their ratio.

57. HCl absorbs at 2886 cm^{-1} and HBr at 2559 cm^{-1}. Which has the stronger bond? Calculate both force constants and their ratio.

58. HBr absorbs at 2559 cm^{-1} and HI at 2230 cm^{-1}. Which has the stronger bond? Calculate both force constants and their ratio.

Hooke's law numerically

59. HCl has $k = 516$ N m^{-1}. Using Hooke's law find the restoring force and potential energy when the bond is stretched by 0.1 Å, and compare with the zero-point energy.

60. HF has $k = 966$ N m^{-1}. Using Hooke's law find the restoring force and potential energy when the bond is stretched by 0.05 Å, and compare with the zero-point energy.

61. CO has $k = 1857$ N m^{-1}. Using Hooke's law find the restoring force and potential energy when the bond is stretched by 0.05 Å, and compare with the zero-point energy.

Find the unknown atom from IR data

62. A diatomic HX has a fundamental at 2886 cm^{-1} and $k = 516$ N m^{-1}. Find μ and the atomic mass of X (H = 1.008 amu).

63. A diatomic HX has a fundamental at 2559 cm^{-1} and $k = 412$ N m^{-1}. Find μ and the atomic mass of X (H = 1.008 amu).

64. A diatomic HX has a fundamental at 4138 cm^{-1} and $k = 966$ N m^{-1}. Find μ and the atomic mass of X (H = 1.008 amu).

65. The ratio of $\bar{\nu}$ for HCl to $\bar{\nu}$ for HBr (same k) equals...?

Quick-fire checks and spot the error

True or false

66. The SHO zero-point energy is zero.

67. IR absorption requires a permanent dipole.

68. Doubling the force constant doubles the wavenumber.

69. The energy levels of the SHO are equally spaced.

70. For SHO, $\Delta v = \pm 2$ transitions are allowed.

71. Replacing H by D lowers the vibrational frequency.

72. Homonuclear diatomics are IR-active.

73. CO_2's symmetric stretch is IR-inactive.

74. A stiffer bond has a larger force constant and a higher stretching frequency (same μ).

75. ZPE can be removed by cooling to 0 K.

76. The vibrational spectrum of the SHO has one line for all molecules of a given type.

77. In an exam, μ must be converted to kg before using $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$ with k in N m $^{-1}$.

Spot the error

78. ZPE of HF = $\bar{\nu}$ = 4145 cm^{-1}.

79. $\mu = m_1 m_2/(m_1+m_2)$ = 0.95 kg, used directly in $\sqrt{(k/\mu)}$.

80. $\bar{\nu} = \sqrt{(k/\mu)}/(2\pi c)$ with c = 3×10^8 cm s^{-1}.

81. The v = 3 level of the SHO is at $3\bar{\nu}$.

82. Only one line appears because $\Delta v = \pm2$ is forbidden and overtones are weak.

83. CO_2 is IR-inactive because it has no permanent dipole.

Answers

Set 1: k → wavenumber and zero-point energy

13. ***Working.*** $\mu = 0.9801$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \tfrac{1}{2}\bar{\nu}$.
Answer: $\bar{\nu} = 2988$ cm^{-1}; $\nu = 8.96 \times 10^{13}$ Hz; ZPE $= 1494$ cm^{-1} $= 2.97 \times 10^{-20}$ J

14. ***Working.*** $\mu = 0.9954$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \tfrac{1}{2}\bar{\nu}$.
Answer: $\bar{\nu} = 2649$ cm^{-1}; $\nu = 7.95 \times 10^{13}$ Hz; ZPE $= 1325$ cm^{-1} $= 2.63 \times 10^{-20}$ J

15. ***Working.*** $\mu = 1.0001$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \tfrac{1}{2}\bar{\nu}$.
Answer: $\bar{\nu} = 2307$ cm^{-1}; $\nu = 6.92 \times 10^{13}$ Hz; ZPE $= 1154$ cm^{-1} $= 2.29 \times 10^{-20}$ J

16. ***Working.*** $\mu = 6.8562$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \tfrac{1}{2}\bar{\nu}$.
Answer: $\bar{\nu} = 2143$ cm^{-1}; $\nu = 6.43 \times 10^{13}$ Hz; ZPE $= 1071$ cm^{-1} $= 2.13 \times 10^{-20}$ J

17. ***Working.*** $\mu = 0.9572$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \tfrac{1}{2}\bar{\nu}$.
Answer: $\bar{\nu} = 4136$ cm^{-1}; $\nu = 1.24 \times 10^{14}$ Hz; ZPE $= 2068$ cm^{-1} $= 4.11 \times 10^{-20}$ J

18. ***Working.*** $\mu = 7.4667$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \tfrac{1}{2}\bar{\nu}$.
Answer: $\bar{\nu} = 1903$ cm^{-1}; $\nu = 5.71 \times 10^{13}$ Hz; ZPE $= 952$ cm^{-1} $= 1.89 \times 10^{-20}$ J

Set 2: ν̄ → force constant

19. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.
Answer: $k = 4.81 \times 10^2 \approx 481$ N m^{-1}

20. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.
Answer: $k = 1.86 \times 10^3 \approx 1857$ N m^{-1}

21. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.
Answer: $k = 9.67 \times 10^2 \approx 967$ N m^{-1}

22. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.
Answer: $k = 1.60 \times 10^3 \approx 1597$ N m^{-1}

23. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.
Answer: $k = 3.14 \times 10^2 \approx 314$ N m^{-1}

Set 3: Energy of the v-th level

24. ***Working.*** $\varepsilon_v = (v + \tfrac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.
Answer: $\bar{\nu} = 1864$: $\varepsilon_0 = 932$, $\varepsilon_1 = 2796$, $\varepsilon_2 = 4660$, $\varepsilon_3 = 6524$, $\varepsilon_4 = 8388$ cm^{-1}; spacing 1864 cm^{-1}

25. ***Working.*** $\varepsilon_v = (v + \tfrac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.
Answer: $\bar{\nu} = 2988$: $\varepsilon_0 = 1494$, $\varepsilon_1 = 4481$, $\varepsilon_2 = 7469$, $\varepsilon_5 = 16432$ cm^{-1}; spacing 2988 cm^{-1}

26. ***Working.*** $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.

    Answer: $\bar{\nu} = 2143$: $\varepsilon_0 = 1071$, $\varepsilon_2 = 5357$, $\varepsilon_3 = 7500$, $\varepsilon_6 = 13929$ cm^{-1}; spacing 2143 cm^{-1}

27. ***Working.*** $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.

    Answer: $\bar{\nu} = 4136$: $\varepsilon_1 = 6205$, $\varepsilon_2 = 10341$, $\varepsilon_4 = 18614$ cm^{-1}; spacing 4136 cm^{-1}

Set 4: Concept twists

28. ***Working.*** $\bar{\nu} \propto 1/\sqrt{\mu}$: $\mu_{HCl} = 0.9801$, $\mu_{DCl} = 1.9057$ amu.

    Answer: HCl 2988 cm^{-1} (ZPE 1494); DCl 2143 cm^{-1} (ZPE 1071). Ratio 1.394. Heavier isotope → lower ν̄ and ZPE.

29. ***Working.*** Test: does the dipole moment change during the vibration?

    Answer: Active: HCl, CO_2 (bend), H_2O (bend), CH_4 (some stretches). Inactive: N_2, O_2, CO_2 symmetric stretch, CH_4 totally symmetric stretch.

30. ***Working.*** $\bar{\nu} \propto \sqrt{k/\mu}$.

    Answer: $\times\sqrt{2} = 1.414$ when k doubles; $\times 1/\sqrt{2} = 0.707$ when μ doubles.

31. ***Working.*** Gross: dipole moment must change on vibration. Specific (SHO): $\Delta v = \pm 1$.

    Answer: Gross: change in dipole; Specific: Δv = ±1 (SHO), Δv = ±1, ±2, ±3... (anharmonic).

Convert IR absorptions: λ, ν, E

32. ***Working.*** $\lambda(\mu m) = 10^4/\bar{\nu}$; $\nu = c\bar{\nu}$; $E = h\nu$.

    Answer: λ = 3.465 μm (IR: 1–100 μm ✓); ν = 8.66×10^{13} Hz; E = 5.74×10^{-20} J = 0.358 eV = 34.5 kJ mol^{-1}

33. ***Working.*** $\lambda(\mu m) = 10^4/\bar{\nu}$; $\nu = c\bar{\nu}$; $E = h\nu$.

    Answer: λ = 4.666 μm (IR: 1–100 μm ✓); ν = 6.43×10^{13} Hz; E = 4.26×10^{-20} J = 0.266 eV = 25.7 kJ mol^{-1}

34. ***Working.*** $\lambda(\mu m) = 10^4/\bar{\nu}$; $\nu = c\bar{\nu}$; $E = h\nu$.

    Answer: λ = 2.526 μm (IR: 1–100 μm ✓); ν = 1.19×10^{14} Hz; E = 7.87×10^{-20} J = 0.491 eV = 47.4 kJ mol^{-1}

35. ***Working.*** $\lambda(\mu m) = 10^4/\bar{\nu}$; $\nu = c\bar{\nu}$; $E = h\nu$.

    Answer: λ = 5.330 μm (IR: 1–100 μm ✓); ν = 5.63×10^{13} Hz; E = 3.73×10^{-20} J = 0.233 eV = 22.5 kJ mol^{-1}

36. ***Working.*** $\lambda(\mu m) = 10^4/\bar{\nu}$; $\nu = c\bar{\nu}$; $E = h\nu$.

    Answer: λ = 4.484 μm (IR: 1–100 μm ✓); ν = 6.69×10^{13} Hz; E = 4.43×10^{-20} J = 0.276 eV = 26.7 kJ mol^{-1}

37. ***Working.*** $\lambda(\mu m) = 10^4/\bar{\nu}$; $\nu = c\bar{\nu}$; $E = h\nu$.

Answer: λ = 2.941 µm (IR: 1–100 µm ✓); ν = 1.02×10^{14} Hz; E = 6.76×10^{-20} J = 0.422 eV = 40.7 kJ mol^{-1}

Zero-point energy in J, eV and kJ/mol

38. ***Working.*** First $\bar{\nu}$; ZPE = $\frac{1}{2}\bar{\nu}$; $E = hc\bar{\nu}$; 1 eV = 8065.5 cm^{-1}.

Answer: $\bar{\nu}$ = 2988 cm^{-1}; ZPE = 2.97×10^{-20} J = 0.185 eV = 17.88 kJ mol^{-1}; 0→1: 5.94×10^{-20} J = 0.370 eV = 35.76 kJ mol^{-1}

39. ***Working.*** First $\bar{\nu}$; ZPE = $\frac{1}{2}\bar{\nu}$; $E = hc\bar{\nu}$; 1 eV = 8065.5 cm^{-1}.

Answer: $\bar{\nu}$ = 4136 cm^{-1}; ZPE = 4.11×10^{-20} J = 0.256 eV = 24.76 kJ mol^{-1}; 0→1: 8.22×10^{-20} J = 0.513 eV = 49.52 kJ mol^{-1}

40. ***Working.*** First $\bar{\nu}$; ZPE = $\frac{1}{2}\bar{\nu}$; $E = hc\bar{\nu}$; 1 eV = 8065.5 cm^{-1}.

Answer: $\bar{\nu}$ = 2143 cm^{-1}; ZPE = 2.13×10^{-20} J = 0.133 eV = 12.83 kJ mol^{-1}; 0→1: 4.26×10^{-20} J = 0.266 eV = 25.65 kJ mol^{-1}

41. ***Working.*** First $\bar{\nu}$; ZPE = $\frac{1}{2}\bar{\nu}$; $E = hc\bar{\nu}$; 1 eV = 8065.5 cm^{-1}.

Answer: $\bar{\nu}$ = 1903 cm^{-1}; ZPE = 1.89×10^{-20} J = 0.118 eV = 11.39 kJ mol^{-1}; 0→1: 3.78×10^{-20} J = 0.236 eV = 22.78 kJ mol^{-1}

42. ***Working.*** First $\bar{\nu}$; ZPE = $\frac{1}{2}\bar{\nu}$; $E = hc\bar{\nu}$; 1 eV = 8065.5 cm^{-1}.

Answer: $\bar{\nu}$ = 2649 cm^{-1}; ZPE = 2.63×10^{-20} J = 0.164 eV = 15.86 kJ mol^{-1}; 0→1: 5.27×10^{-20} J = 0.328 eV = 31.71 kJ mol^{-1}

Isotope effect on the IR band

43. ***Working.*** $\bar{\nu} \propto 1/\sqrt{\mu}$, so $\bar{\nu}_2 = \bar{\nu}_1\sqrt{\mu_1/\mu_2}$.

Answer: μ_1 = 0.9801, μ_2 = 1.9057 amu; $\bar{\nu}_2$ = 2070 cm^{-1} (shift 816 lower); ZPE 1443 → 1035 cm^{-1}

44. ***Working.*** $\bar{\nu} \propto 1/\sqrt{\mu}$, so $\bar{\nu}_2 = \bar{\nu}_1\sqrt{\mu_1/\mu_2}$.

Answer: μ_1 = 6.8562, μ_2 = 7.1723 amu; $\bar{\nu}_2$ = 2095 cm^{-1} (shift 48 lower); ZPE 1072 → 1048 cm^{-1}

45. ***Working.*** $\bar{\nu} \propto 1/\sqrt{\mu}$, so $\bar{\nu}_2 = \bar{\nu}_1\sqrt{\mu_1/\mu_2}$.

Answer: μ_1 = 0.9953, μ_2 = 1.9639 amu; $\bar{\nu}_2$ = 1822 cm^{-1} (shift 737 lower); ZPE 1280 → 911 cm^{-1}

46. ***Working.*** $\bar{\nu} \propto 1/\sqrt{\mu}$, so $\bar{\nu}_2 = \bar{\nu}_1\sqrt{\mu_1/\mu_2}$.

Answer: μ_1 = 7.4664, μ_2 = 7.7408 amu; $\bar{\nu}_2$ = 1842 cm^{-1} (shift 34 lower); ZPE 938 → 921 cm^{-1}

47. ***Working.*** $\bar{\nu} \propto 1/\sqrt{\mu}$, so $\bar{\nu}_2 = \bar{\nu}_1\sqrt{\mu_1/\mu_2}$.

Answer: μ_1 = 0.9572, μ_2 = 1.8210 amu; $\bar{\nu}_2$ = 2870 cm^{-1} (shift 1089 lower); ZPE 1980 → 1435 cm^{-1}

Number of vibrational modes and IR activity (extension)

48. ***Working.*** $N = 2$, linear: $3N - 5 = 1$.

    Answer: 1 modes (1 (stretch)); IR: active.

49. ***Working.*** $N = 2$, linear: $3N - 5 = 1$.

    Answer: 1 modes (1 (stretch)); IR: inactive: no dipole change.

50. ***Working.*** $N = 3$, linear: $3N - 5 = 4$.

    Answer: 4 modes (4 (sym. stretch, asym. stretch, 2 bends)); IR: asym. stretch and bend active; sym. stretch inactive.

51. ***Working.*** $N = 3$, non-linear: $3N - 6 = 3$.

    Answer: 3 modes (3 (sym. stretch, asym. stretch, bend)); IR: all three active.

52. ***Working.*** $N = 3$, non-linear: $3N - 6 = 3$.

    Answer: 3 modes (3); IR: all three active.

53. ***Working.*** $N = 4$, non-linear: $3N - 6 = 6$.

    Answer: 6 modes (6); IR: all active.

54. ***Working.*** $N = 5$, non-linear: $3N - 6 = 9$.

    Answer: 9 modes (9); IR: only the asymmetric stretch and bend are IR-active.

Compare bond strengths from two spectra

55. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$ — note a higher $\bar{\nu}$ does not automatically mean a higher k (μ matters).

    Answer: k(HF) = 885 N m^{-1}; k(HCl) = 481 N m^{-1}; ratio = 1.84 → HF has the stronger bond.

56. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$ — note a higher $\bar{\nu}$ does not automatically mean a higher k (μ matters).

    Answer: k(CO) = 1857 N m^{-1}; k(NO) = 1550 N m^{-1}; ratio = 1.20 → CO has the stronger bond.

57. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$ — note a higher $\bar{\nu}$ does not automatically mean a higher k (μ matters).

    Answer: k(HCl) = 481 N m^{-1}; k(HBr) = 384 N m^{-1}; ratio = 1.25 → HCl has the stronger bond.

58. ***Working.*** $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$ — note a higher $\bar{\nu}$ does not automatically mean a higher k (μ matters).

    Answer: k(HBr) = 384 N m^{-1}; k(HI) = 293 N m^{-1}; ratio = 1.31 → HBr has the stronger bond.

Hooke's law numerically

59. ***Working.*** $F = -kx$; $PE = \frac{1}{2}kx^2$.
Answer: |F| = 5.16×10^{-9} N; PE = 2.58×10^{-20} J; ZPE = 2.97×10^{-20} J (stretch energy is 0.87 × ZPE)

60. ***Working.*** $F = -kx$; $PE = \frac{1}{2}kx^2$.
Answer: |F| = 4.83×10^{-9} N; PE = 1.21×10^{-20} J; ZPE = 4.11×10^{-20} J (stretch energy is 0.29 × ZPE)

61. ***Working.*** $F = -kx$; $PE = \frac{1}{2}kx^2$.
Answer: |F| = 9.29×10^{-9} N; PE = 2.32×10^{-20} J; ZPE = 2.13×10^{-20} J (stretch energy is 1.09 × ZPE)

Find the unknown atom from IR data

62. ***Working.*** $\mu = k/(4\pi^2c^2\bar{\nu}^2)$ then $m_X = \mu m_H/(m_H - \mu)$.
Answer: μ = 1.0504 amu; m_X ≈ -25.0 amu

63. ***Working.*** $\mu = k/(4\pi^2c^2\bar{\nu}^2)$ then $m_X = \mu m_H/(m_H - \mu)$.
Answer: μ = 1.0667 amu; m_X ≈ -18.3 amu

64. ***Working.*** $\mu = k/(4\pi^2c^2\bar{\nu}^2)$ then $m_X = \mu m_H/(m_H - \mu)$.
Answer: μ = 0.9565 amu; m_X ≈ 18.7 amu

65. ***Working.*** $\sqrt{\mu_{HBr}/\mu_{HCl}}$.
Answer: 1.0078 (≈ 1.0 — both μ ≈ 1 amu because H is light).

True or false

66. **Answer: False** — $\varepsilon_0 = \frac{1}{2}\bar{\nu}_{osc}$.

67. **Answer: False** — it requires a change in dipole during vibration.

68. **Answer: False** — $\bar{\nu} \propto \sqrt{k}$, so it rises by √2.

69. **Answer: True** — spacing = $\bar{\nu}_{osc}$.

70. **Answer: False** — only $\Delta v = \pm 1$.

71. **Answer: True** — μ increases ⇒ ν̄ decreases by about √2.

72. **Answer: False** — no dipole change on stretching.

73. **Answer: True** — no change in dipole moment.

74. **Answer: True** — $\bar{\nu} \propto \sqrt{k/\mu}$.

75. **Answer: False** — it is the energy of the lowest quantum state itself.

76. **Answer: True** — equal spacing and Δv = ±1.

77. **Answer: True** — N = kg m s^{-2}; units must be SI.

Spot the error

78. **Answer:** ZPE $= \frac{1}{2}\bar{\nu} \approx 2073$ cm^{-1}.

79. **Answer:** 0.95 is in amu; convert to kg (1.58×10^{-27} kg).

80. **Answer:** c $= 3\times10^{10}$ cm s^{-1} (or 3×10^{8} m s^{-1} and then convert m$^{-1} \to$ cm^{-1} by $\div 100$).

81. **Answer:** $\varepsilon_3 = (3 + \frac{1}{2})\bar{\nu} = 3.5\bar{\nu}$

82. **Answer:** The reason is equal level spacing with $\Delta v = \pm 1$; overtones are absent in SHO (they appear only in the anharmonic model).

83. **Answer:** IR needs a dipole change; asymmetric stretch and bending change it, so CO_2 is IR-active.

Chapter 3: Anharmonic Oscillator and Overtones

Topics T6

Anharmonic problems are two-equation puzzles: you are given two of fundamental / overtone / ω_e / x_e and must find the others. Memorise the three transition expressions.

Formulas to remember

$$\varepsilon_v = (v+\tfrac{1}{2})\bar{\omega}_e - (v+\tfrac{1}{2})^2\bar{\omega}_e x_e$$

$$0 \to 1 : \ \bar{\omega}_e(1-2x_e)$$

$$0 \to 2 : \ 2\bar{\omega}_e(1-3x_e)$$

$$0 \to 3 : \ 3\bar{\omega}_e(1-4x_e)$$

$$1 \to 2\ (\text{hot}) : \ \bar{\omega}_e(1-4x_e)$$

Method.

1. Write the two given transitions as two equations in $\bar{\omega}_e$ and x_e.
2. Subtract: $\bar{\omega}_e x_e = f - o/2$; then $\bar{\omega}_e = 3f - o$ and $x_e = (f - o/2)/\bar{\omega}_e$.
3. If x_e is given instead: $\bar{\omega}_e = f/(1-2x_e)$.

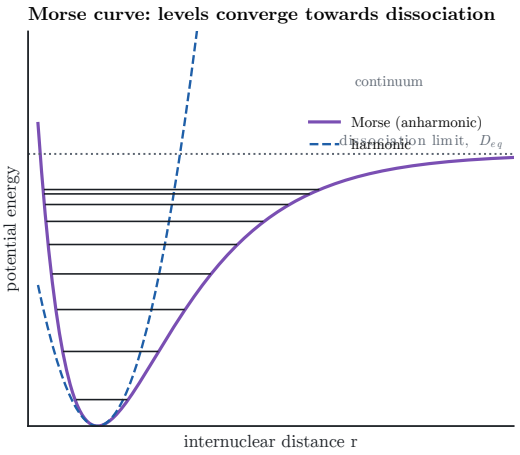


Figure 3.1 (G15): Morse curve vs harmonic parabola.

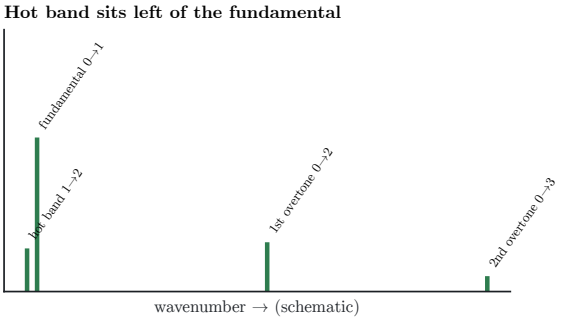


Figure 3.2 (G17): Anharmonic IR spectrum (schematic).

Past ESA questions with solutions

1. For HBr the vibrational spectrum shows the fundamental at 2559.08 cm^{-1} and the first overtone at 5027.54 cm^{-1}. Calculate the anharmonicity constant (anharmonic model).

   Jul-23 · UE22 · A (4 marks)

   Solution.

   i. Fundamental: $\bar{\omega}_e(1-2x_e) = 2559.08$. First overtone: $2\bar{\omega}_e(1-3x_e) = 5027.54 \Rightarrow \bar{\omega}_e(1-3x_e) = 2513.77$.

   ii. Subtract: $\bar{\omega}_e x_e = 2559.08 - 2513.77 = 45.31$ cm^{-1}. Then $\bar{\omega}_e = 2559.08 + 2(45.31) = 3(2559.08) - 5027.54 = 2649.70$ cm^{-1}.

   iii. $x_e = 45.31/2649.70 = 0.0171$.

   Answer: $x_e \approx 0.0171$ ($\bar{\omega}_e \approx 2649.7$ cm^{-1})

2. The fundamental stretching frequency of CO is 2143 cm^{-1} and its anharmonicity constant is 0.01. Calculate its oscillational (equilibrium) frequency.

   Jan-25 · UE24 · Set 2 (2 marks)

   Solution.

   i. $\bar{\omega}_e(1-2x_e) = 2143$.

   ii. $\bar{\omega}_e = \dfrac{2143}{1-2(0.01)} = \dfrac{2143}{0.98} = 2186.7$ cm^{-1}.

   Answer: $\bar{\omega}_e \approx 2186.7$ cm^{-1}

3. Assuming CO shows anharmonic oscillation, calculate the wavenumber of the first overtone if $x_e = 0.017$ and the equilibrium vibrational frequency is 2050 cm^{-1}.

   Jul-23 · UE22 · B (2 marks)

   Solution. First overtone $0 \to 2$: $2\bar{\omega}_e(1-3x_e) = 2(2050)(1-3\times 0.017) = 2(2050)(0.949)$.

   Answer: 3890.9 cm^{-1}

4. (i) What is the selection rule for the anharmonic oscillator? What is the first overtone transition? Give its expression and explain the terms.

   Jan-25 · UE24 · Set 2 (4 marks)

   Solution.

   i. Selection rule: $\Delta v = \pm 1, \pm 2, \pm 3 \ldots$ (gross rule: dipole must change).

   ii. First overtone $v = 0 \to 2$: $\Delta\bar{\nu} = 2\bar{\omega}_e(1-3x_e)$, where $\bar{\omega}_e$ is the equilibrium vibrational wavenumber (cm^{-1}) and x_e is the small positive anharmonicity constant. It is a little less than twice the fundamental.

   Answer: $2\bar{\omega}_e(1-3x_e)$

5. Briefly describe the Born–Oppenheimer approximation and draw the ground and excited electronic states of a diatomic molecule showing rotational and vibrational energy levels (anharmonic model).

   Dec-24 · UE23 (5 marks)

   Solution.

   i. B–O: electronic, vibrational and rotational energies are independent; $E = E_{rot} + E_{vib} + E_{elec}$ (translation negligible), with $E_{rot} \ll E_{vib} \ll E_{elec}$.

   ii. Diagram: two Morse curves (ground below, excited above and displaced to longer r_e); levels converge towards each dissociation limit; each vibrational level carries closely spaced rotational sub-levels (**Fig. G15** for the Morse shape).

   Answer: See statement + diagram

Practice: exam-style variants

Set 1: $\mathbf{x_e}$ and $\boldsymbol{\omega_e}$ from fundamental + first overtone

The standard exam puzzle.

6. HCl: fundamental 2885.3 cm^{-1}, first overtone 5665.0 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

7. CO: fundamental 2143.2 cm^{-1}, first overtone 4259.9 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

8. HF: fundamental 3958.5 cm^{-1}, first overtone 7737.2 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

9. NO: fundamental 1876.1 cm^{-1}, first overtone 3724.0 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

10. HBr: fundamental 2559.08 cm^{-1}, first overtone 5027.54 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

Set 2: $\boldsymbol{\omega_e}$ from the fundamental and $\mathbf{x_e}$; overtones

Exam: CO 2143, $x_e = 0.01$. Same method, other molecules.

11. CO: fundamental 2143 cm^{-1} and $x_e = 0.01$. Find $\bar{\omega}_e$ and the first and second overtones.

12. HCl: fundamental 2886 cm^{-1} and $x_e = 0.0177$. Find $\bar{\omega}_e$ and the first and second overtones.

13. HF: fundamental 3959 cm^{-1} and $x_e = 0.0217$. Find $\bar{\omega}_e$ and the first and second overtones.

14. NO: fundamental 1876 cm^{-1} and $x_e = 0.0074$. Find $\bar{\omega}_e$ and the first and second overtones.

15. HI: fundamental 2230 cm^{-1} and $x_e = 0.0183$. Find $\bar{\omega}_e$ and the first and second overtones.

Set 3: All transitions from $\boldsymbol{\omega_e}$ and $\mathbf{x_e}$

Exam: CO $\omega_e = 2050$, $x_e = 0.017$, first overtone.

16. CO: $\bar{\omega}_e = 2050$ cm^{-1}, $x_e = 0.017$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

17. HCl: $\bar{\omega}_e = 2990$ cm^{-1}, $x_e = 0.0177$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

18. NO: $\bar{\omega}_e = 1904$ cm^{-1}, $x_e = 0.0074$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

19. HF: $\bar{\omega}_e = 4138$ cm^{-1}, $x_e = 0.0217$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

Set 4: Concept twists

20. Why do the lines of an anharmonic spectrum get closer together at higher energy?

21. Give the transition energy of the hot band relative to the fundamental for $\bar{\omega}_e = 2000$, $x_e = 0.02$.

22. Compare SHO and anharmonic oscillator on: potential, level spacing, selection rule, number of lines.

23. Calculate the anharmonic zero-point energy of CO ($\bar{\omega}_e = 2169.8$, $x_e = 0.00612$) and compare with $\frac{1}{2}\bar{\omega}_e$.

New question types (not yet asked)

x_e and ω_e from other pairs of bands

Past papers used fundamental + first overtone. Any two bands work — set up two equations.

24. CO: the fundamental is at 2143.2 cm^{-1} and the **second overtone** at 6350.0 cm^{-1}. Find x_e and $\bar{\omega}_e$.

25. HCl: the fundamental is at 2885.3 cm^{-1} and the **second overtone** at 8339.2 cm^{-1}. Find x_e and $\bar{\omega}_e$.

26. HF: the fundamental is at 3958.5 cm^{-1} and the **second overtone** at 11336.3 cm^{-1}. Find x_e and $\bar{\omega}_e$.

27. NO: the fundamental is at 1876.1 cm^{-1} and the **second overtone** at 5543.7 cm^{-1}. Find x_e and $\bar{\omega}_e$.

28. CO: first overtone 4259.9 cm^{-1} and second overtone 6350.0 cm^{-1}. Find x_e and $\bar{\omega}_e$.

29. HCl: first overtone 5665.1 cm^{-1} and second overtone 8339.2 cm^{-1}. Find x_e and $\bar{\omega}_e$.

Anharmonic ladder vs SHO ladder

See levels converge: the spacing falls by $2\bar{\omega}_e x_e$ at each step.

30. CO: $\bar{\omega}_e = 2169.8$ cm^{-1}, $x_e = 0.00612$. Tabulate ε_v for $v = 0$–5 (anharmonic) and compare with the SHO values $(v + \frac{1}{2})\bar{\omega}_e$. Give the spacing $\varepsilon_{v+1} - \varepsilon_v$.

31. HCl: $\bar{\omega}_e = 2990.9$ cm^{-1}, $x_e = 0.01765$. Tabulate ε_v for $v = 0$–5 (anharmonic) and compare with the SHO values $(v + \frac{1}{2})\bar{\omega}_e$. Give the spacing $\varepsilon_{v+1} - \varepsilon_v$.

32. HF: $\bar{\omega}_e = 4138.3$ cm^{-1}, $x_e = 0.02172$. Tabulate ε_v for $v = 0$–5 (anharmonic) and compare with the SHO values $(v + \frac{1}{2})\bar{\omega}_e$. Give the spacing $\varepsilon_{v+1} - \varepsilon_v$.

Dissociation energy from ω_e and x_e (extension)

Shows what 'the molecule dissociates when stretched' means quantitatively. Added for clarity — not on the slides.

33. (Extension) For the Morse oscillator $D_e \approx \bar{\omega}_e / 4x_e$. CO: $\bar{\omega}_e = 2169.8$ cm^{-1}, $x_e = 0.00612$. Find D_e (cm^{-1}, eV, kJ mol^{-1}), the spectroscopic dissociation energy $D_0 = D_e - \varepsilon_0$ and the highest bound level $v_{max} \approx 1/2x_e - \frac{1}{2}$.

34. (Extension) For the Morse oscillator $D_e \approx \bar{\omega}_e / 4x_e$. HCl: $\bar{\omega}_e = 2990.9$ cm^{-1}, $x_e = 0.01765$. Find D_e (cm^{-1}, eV, kJ mol^{-1}), the spectroscopic dissociation energy $D_0 = D_e - \varepsilon_0$ and the highest bound level $v_{max} \approx 1/2x_e - \frac{1}{2}$.

35. (Extension) For the Morse oscillator $D_e \approx \bar{\omega}_e / 4x_e$. HF: $\bar{\omega}_e = 4138.3$ cm^{-1}, $x_e = 0.02172$. Find D_e (cm^{-1}, eV, kJ mol^{-1}), the spectroscopic dissociation energy $D_0 = D_e - \varepsilon_0$ and the highest bound level $v_{max} \approx 1/2x_e - \frac{1}{2}$.

36. (Extension) For the Morse oscillator $D_e \approx \bar{\omega}_e / 4x_e$. NO: $\bar{\omega}_e = 1904.2$ cm^{-1}, $x_e = 0.00739$. Find D_e (cm^{-1}, eV, kJ mol^{-1}), the spectroscopic dissociation energy $D_0 = D_e - \varepsilon_0$ and the highest bound level $v_{max} \approx 1/2x_e - \frac{1}{2}$.

Hot bands

37. CO: fundamental 2143 cm^{-1}, $x_e = 0.0061$. Find the hot-band $(1 \to 2)$ position, its shift relative to the fundamental, and state on which side of the fundamental it lies.

38. HCl: fundamental 2886 cm^{-1}, $x_e = 0.0177$. Find the hot-band $(1 \to 2)$ position, its shift relative to the fundamental, and state on which side of the fundamental it lies.

39. HF: fundamental 3959 cm^{-1}, $x_e = 0.0217$. Find the hot-band $(1 \to 2)$ position, its shift relative to the fundamental, and state on which side of the fundamental it lies.

40. NO: fundamental 1876 cm^{-1}, $x_e = 0.0074$. Find the hot-band $(1 \to 2)$ position, its shift relative to the fundamental, and state on which side of the fundamental it lies.

Overtone/fundamental ratio and SHO error

41. The ratio (first overtone)/(fundamental) for a diatomic is 1.95. Find x_e. Explain why the ratio is less than 2.

42. The ratio (first overtone)/(fundamental) for a diatomic is 1.9. Find x_e. Explain why the ratio is less than 2.

43. The ratio (first overtone)/(fundamental) for a diatomic is 1.98. Find x_e. Explain why the ratio is less than 2.

44. The ratio (first overtone)/(fundamental) for a diatomic is 1.92. Find x_e. Explain why the ratio is less than 2.

45. CO: by how much does the SHO overestimate the energy of level $v = 3$ when $\bar{\omega}_e = 2169.8$, $x_e = 0.00612$?

46. HCl: by how much does the SHO overestimate the energy of level $v = 3$ when $\bar{\omega}_e = 2990.9$, $x_e = 0.01765$?

47. HF: by how much does the SHO overestimate the energy of level $v = 4$ when $\bar{\omega}_e = 4138.3$, $x_e = 0.02172$?

Why do hot bands appear only for low-frequency modes? (extension)

Gives a quantitative reason behind 'at higher temperature, hot bands appear'.

48. (Extension) At 298 K ($kT/hc = 207$ cm^{-1}) estimate $N_{v=1}/N_{v=0} = e^{-\bar{\nu}/(kT/hc)}$ for HCl ($\bar{\nu} = 2886$ cm^{-1}). Will hot bands be visible?

49. (Extension) At 298 K ($kT/hc = 207$ cm^{-1}) estimate $N_{v=1}/N_{v=0} = e^{-\bar{\nu}/(kT/hc)}$ for CO ($\bar{\nu} = 2143$ cm^{-1}). Will hot bands be visible?

50. (Extension) At 298 K ($kT/hc = 207$ cm^{-1}) estimate $N_{v=1}/N_{v=0} = e^{-\bar{\nu}/(kT/hc)}$ for I_2 (stretch) ($\bar{\nu} = 214$ cm^{-1}). Will hot bands be visible?

51. (Extension) At 298 K ($kT/hc = 207$ cm^{-1}) estimate $N_{v=1}/N_{v=0} = e^{-\bar{\nu}/(kT/hc)}$ for Br_2 (stretch) ($\bar{\nu} = 323$ cm^{-1}). Will hot bands be visible?

52. (Extension) At 298 K ($kT/hc = 207$ cm^{-1}) estimate $N_{v=1}/N_{v=0} = e^{-\bar{\nu}/(kT/hc)}$ for HF ($\bar{\nu} = 3959$ cm^{-1}). Will hot bands be visible?

Quick-fire checks and spot the error

True or false

53. In the Morse model the energy levels converge as v increases.

54. The first overtone is exactly twice the fundamental.

55. x_e is small and positive for bond stretching.

56. The hot band lies to the right (higher wavenumber) of the fundamental.

57. The anharmonic selection rule allows Δv = $\pm$1, $\pm$2, $\pm$3....

58. Most absorptions start from $v = 0$ because it is the most populated level.

59. A harmonic oscillator can dissociate.

60. $\bar{\omega}_e = 3f - o$ where f is the fundamental and o is the first overtone.

61. The Morse curve is wider than the harmonic parabola at high energy.

62. A larger x_e means faster convergence of levels.

Spot the error

63. Fundamental = $\bar{\omega}_e(1 - x_e)$.

64. First overtone = $3\bar{\omega}_e(1 - 4x_e)$.

65. Hot band is at $\bar{\omega}_e(1 - 2x_e) + 2\bar{\omega}_e x_e$.

66. $x_e = 2559.08/2649.7$ for HBr.

67. The spacing between all anharmonic levels is $\bar{\omega}_e$.

Answers

Set 1: ω_e and $\omega_e x_e$ from fundamental + first overtone

6. ***Working.*** $\bar{\omega}_e x_e = f - o/2 = 52.80$; $\bar{\omega}_e = 3f - o = 2990.9$.
   Answer: $\bar{\omega}_e = 2990.9$ cm^{-1}, $x_e = 0.0177$

7. ***Working.*** $\bar{\omega}_e x_e = f - o/2 = 13.25$; $\bar{\omega}_e = 3f - o = 2169.7$.
   Answer: $\bar{\omega}_e = 2169.7$ cm^{-1}, $x_e = 0.0061$

8. ***Working.*** $\bar{\omega}_e x_e = f - o/2 = 89.90$; $\bar{\omega}_e = 3f - o = 4138.3$.
   Answer: $\bar{\omega}_e = 4138.3$ cm^{-1}, $x_e = 0.0217$

9. ***Working.*** $\bar{\omega}_e x_e = f - o/2 = 14.10$; $\bar{\omega}_e = 3f - o = 1904.3$.
   Answer: $\bar{\omega}_e = 1904.3$ cm^{-1}, $x_e = 0.0074$

10. ***Working.*** $\bar{\omega}_e x_e = f - o/2 = 45.31$; $\bar{\omega}_e = 3f - o = 2649.7$.
    Answer: $\bar{\omega}_e = 2649.7$ cm^{-1}, $x_e = 0.0171$

Set 2: ω_e from the fundamental and x_e; overtones

11. ***Working.*** $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.
    Answer: $\bar{\omega}_e = 2186.7$; 1st overtone 4242.3; 2nd overtone 6297.8 cm^{-1}

12. ***Working.*** $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.
    Answer: $\bar{\omega}_e = 2991.9$; 1st overtone 5666.1; 2nd overtone 8340.3 cm^{-1}

13. ***Working.*** $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.
    Answer: $\bar{\omega}_e = 4138.6$; 1st overtone 7738.4; 2nd overtone 11338.2 cm^{-1}

14. ***Working.*** $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.
    Answer: $\bar{\omega}_e = 1904.2$; 1st overtone 3723.8; 2nd overtone 5543.5 cm^{-1}

15. ***Working.*** $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.
    Answer: $\bar{\omega}_e = 2314.7$; 1st overtone 4375.3; 2nd overtone 6435.8 cm^{-1}

Set 3: All transitions from ω_e and x_e

16. ***Working.*** Use the four expressions in the recap.
    Answer: Fundamental 1980.3; 1st overtone 3890.9; 2nd overtone 5731.8; hot band 1910.6 cm^{-1} (hot band lies below the fundamental)

17. ***Working.*** Use the four expressions in the recap.
    Answer: Fundamental 2884.2; 1st overtone 5662.5; 2nd overtone 8334.9; hot band 2778.3 cm^{-1} (hot band lies below the fundamental)

18. ***Working.*** Use the four expressions in the recap.
    Answer: Fundamental 1875.8; 1st overtone 3723.5; 2nd overtone 5542.9; hot band 1847.6 cm^{-1} (hot band lies below the fundamental)

19. ***Working.*** Use the four expressions in the recap.

    Answer: Fundamental 3958.4; 1st overtone 7737.2; 2nd overtone 11336.5; hot band 3778.8 cm^{-1} (hot band lies below the fundamental)

Set 4: Concept twists

20. ***Working.*** Levels converge towards the dissociation limit: spacing $= \bar{\omega}_e[1 - 2x_e(v + 1)]$ shrinks with v.

    Answer: Because the Morse levels converge; the gap between v and $v + 1$ decreases as v increases.

21. ***Working.*** Fundamental $2000(0.96) = 1920$; hot band $2000(0.92) = 1840$.

    Answer: Hot band 1840 cm^{-1} — 80 cm^{-1} below the fundamental (the left side of the spectrum).

22. **Answer:** SHO: parabola, equal, Δv = $\pm$1, one line. Anharmonic: Morse curve, converging, Δv = $\pm$1, $\pm$2, $\pm$3..., fundamental + overtones (+ hot bands).

23. ***Working.*** $\varepsilon_0 = \frac{1}{2}\bar{\omega}_e(1 - \frac{1}{2}x_e)$ (added for clarity).

    Answer: $\varepsilon_0 = 1081.6$ cm^{-1} vs 1084.9 — slightly lower.

$\mathbf{x_e}$ and $\boldsymbol{\omega_e}$ from other pairs of bands

24. ***Working.*** Fundamental $\bar{\omega}_e(1 - 2x_e)$; second overtone $3\bar{\omega}_e(1 - 4x_e) \Rightarrow t/3 = \bar{\omega}_e(1 - 4x_e)$. Subtract: $2\bar{\omega}_e x_e = f - t/3$.

    Answer: $\bar{\omega}_e x_e = 13.28$; $\bar{\omega}_e = 2169.8$ cm^{-1}; $x_e = 0.00612$

25. ***Working.*** Fundamental $\bar{\omega}_e(1 - 2x_e)$; second overtone $3\bar{\omega}_e(1 - 4x_e) \Rightarrow t/3 = \bar{\omega}_e(1 - 4x_e)$. Subtract: $2\bar{\omega}_e x_e = f - t/3$.

    Answer: $\bar{\omega}_e x_e = 52.79$; $\bar{\omega}_e = 2990.9$ cm^{-1}; $x_e = 0.01765$

26. ***Working.*** Fundamental $\bar{\omega}_e(1 - 2x_e)$; second overtone $3\bar{\omega}_e(1 - 4x_e) \Rightarrow t/3 = \bar{\omega}_e(1 - 4x_e)$. Subtract: $2\bar{\omega}_e x_e = f - t/3$.

    Answer: $\bar{\omega}_e x_e = 89.88$; $\bar{\omega}_e = 4138.3$ cm^{-1}; $x_e = 0.02172$

27. ***Working.*** Fundamental $\bar{\omega}_e(1 - 2x_e)$; second overtone $3\bar{\omega}_e(1 - 4x_e) \Rightarrow t/3 = \bar{\omega}_e(1 - 4x_e)$. Subtract: $2\bar{\omega}_e x_e = f - t/3$.

    Answer: $\bar{\omega}_e x_e = 14.07$; $\bar{\omega}_e = 1904.2$ cm^{-1}; $x_e = 0.00739$

28. ***Working.*** $o_1/2 = \bar{\omega}_e(1 - 3x_e)$, $o_2/3 = \bar{\omega}_e(1 - 4x_e)$; subtract.

    Answer: $\bar{\omega}_e x_e = 13.28$; $\bar{\omega}_e = 2169.8$; $x_e = 0.00612$

29. ***Working.*** $o_1/2 = \bar{\omega}_e(1 - 3x_e)$, $o_2/3 = \bar{\omega}_e(1 - 4x_e)$; subtract.

    Answer: $\bar{\omega}_e x_e = 52.79$; $\bar{\omega}_e = 2990.9$; $x_e = 0.01765$

Anharmonic ladder vs SHO ladder

30. ***Working.*** $\varepsilon_v = (v+\frac{1}{2})\bar{\omega}_e - (v+\frac{1}{2})^2\bar{\omega}_e x_e$.

    Answer: v=0: 1081.6 (SHO 1084.9); v=1: 3224.8 (SHO 3254.7); v=2: 5341.5 (SHO 5424.5); v=3: 7431.6 (SHO 7594.3); v=4: 9495.2 (SHO 9764.1); v=5: 11532.2 (SHO 11933.9). Spacings: 2143.2, 2116.7, 2090.1, 2063.6, 2037.0 cm^{-1} — decreasing.

31. ***Working.*** $\varepsilon_v = (v+\frac{1}{2})\bar{\omega}_e - (v+\frac{1}{2})^2\bar{\omega}_e x_e$.

    Answer: v=0: 1482.3 (SHO 1495.5); v=1: 4367.6 (SHO 4486.4); v=2: 7147.3 (SHO 7477.2); v=3: 9821.5 (SHO 10468.1); v=4: 12390.1 (SHO 13459.1); v=5: 14853.1 (SHO 16450.0). Spacings: 2885.3, 2779.7, 2674.2, 2568.6, 2463.0 cm^{-1} — decreasing.

32. ***Working.*** $\varepsilon_v = (v+\frac{1}{2})\bar{\omega}_e - (v+\frac{1}{2})^2\bar{\omega}_e x_e$.

    Answer: v=0: 2046.7 (SHO 2069.2); v=1: 6005.2 (SHO 6207.5); v=2: 9784.0 (SHO 10345.8); v=3: 13383.0 (SHO 14484.1); v=4: 16802.2 (SHO 18622.4); v=5: 20041.7 (SHO 22760.7). Spacings: 3958.5, 3778.8, 3599.0, 3419.2, 3239.5 cm^{-1} — decreasing.

Dissociation energy from ω_e and x_e (extension)

33. ***Working.*** $\varepsilon_0 = \frac{1}{2}\bar{\omega}_e(1-\frac{1}{2}x_e)$; 1 cm^{-1} = 0.01196 kJ mol^{-1}.

    Answer: D_e ≈ 88636 cm^{-1} = 10.99 eV = 1060 kJ mol^{-1}; ε_0 = 1082; D_0 ≈ 87554 cm^{-1} (1047 kJ mol^{-1}); v_max ≈ 81

34. ***Working.*** $\varepsilon_0 = \frac{1}{2}\bar{\omega}_e(1-\frac{1}{2}x_e)$; 1 cm^{-1} = 0.01196 kJ mol^{-1}.

    Answer: D_e ≈ 42364 cm^{-1} = 5.25 eV = 507 kJ mol^{-1}; ε_0 = 1482; D_0 ≈ 40882 cm^{-1} (489 kJ mol^{-1}); v_max ≈ 28

35. ***Working.*** $\varepsilon_0 = \frac{1}{2}\bar{\omega}_e(1-\frac{1}{2}x_e)$; 1 cm^{-1} = 0.01196 kJ mol^{-1}.

    Answer: D_e ≈ 47632 cm^{-1} = 5.91 eV = 570 kJ mol^{-1}; ε_0 = 2047; D_0 ≈ 45586 cm^{-1} (545 kJ mol^{-1}); v_max ≈ 23

36. ***Working.*** $\varepsilon_0 = \frac{1}{2}\bar{\omega}_e(1-\frac{1}{2}x_e)$; 1 cm^{-1} = 0.01196 kJ mol^{-1}.

    Answer: D_e ≈ 64418 cm^{-1} = 7.99 eV = 771 kJ mol^{-1}; ε_0 = 949; D_0 ≈ 63470 cm^{-1} (759 kJ mol^{-1}); v_max ≈ 67

Hot bands

37. ***Working.*** $\bar{\omega}_e = f/(1-2x_e)$; hot = $\bar{\omega}_e(1-4x_e)$; shift = $2\bar{\omega}_e x_e$.

    Answer: $\bar{\omega}_e$ = 2169.5; hot band = 2116.5 cm^{-1}, i.e. 26.5 cm^{-1} below the fundamental (left side).

38. ***Working.*** $\bar{\omega}_e = f/(1-2x_e)$; hot = $\bar{\omega}_e(1-4x_e)$; shift = $2\bar{\omega}_e x_e$.

    Answer: $\bar{\omega}_e$ = 2991.9; hot band = 2780.1 cm^{-1}, i.e. 105.9 cm^{-1} below the fundamental (left side).

39. ***Working.*** $\bar{\omega}_e = f/(1-2x_e)$; hot = $\bar{\omega}_e(1-4x_e)$; shift = $2\bar{\omega}_e x_e$.

    Answer: $\bar{\omega}_e$ = 4138.6; hot band = 3779.4 cm^{-1}, i.e. 179.6 cm^{-1} below the fundamental (left side).

40. ***Working.*** $\bar{\omega}_e = f/(1-2x_e)$; hot $= \bar{\omega}_e(1-4x_e)$; shift $= 2\bar{\omega}_e x_e$.

    Answer: $\bar{\omega}_e$ = 1904.2; hot band = 1847.8 cm^{-1}, i.e. 28.2 cm^{-1} below the fundamental (left side).

Overtone/fundamental ratio and SHO error

41. ***Working.*** $r = \dfrac{2(1-3x_e)}{(1-2x_e)} \Rightarrow x_e = \dfrac{2-r}{6-2r}$.

    Answer: x_e = 0.02381. The ratio is < 2 because the levels converge (anharmonicity): the 0→2 gap is smaller than twice the 0→1 gap.

42. ***Working.*** $r = \dfrac{2(1-3x_e)}{(1-2x_e)} \Rightarrow x_e = \dfrac{2-r}{6-2r}$.

    Answer: x_e = 0.04545. The ratio is < 2 because the levels converge (anharmonicity): the 0→2 gap is smaller than twice the 0→1 gap.

43. ***Working.*** $r = \dfrac{2(1-3x_e)}{(1-2x_e)} \Rightarrow x_e = \dfrac{2-r}{6-2r}$.

    Answer: x_e = 0.00980. The ratio is < 2 because the levels converge (anharmonicity): the 0→2 gap is smaller than twice the 0→1 gap.

44. ***Working.*** $r = \dfrac{2(1-3x_e)}{(1-2x_e)} \Rightarrow x_e = \dfrac{2-r}{6-2r}$.

    Answer: x_e = 0.03704. The ratio is < 2 because the levels converge (anharmonicity): the 0→2 gap is smaller than twice the 0→1 gap.

45. ***Working.*** Difference $= (v+\tfrac{1}{2})^2 \bar{\omega}_e x_e$.

    Answer: anharmonic 7431.6 vs SHO 7594.3: error 162.7 cm^{-1} (2.1 %).

46. ***Working.*** Difference $= (v+\tfrac{1}{2})^2 \bar{\omega}_e x_e$.

    Answer: anharmonic 9821.5 vs SHO 10468.1: error 646.7 cm^{-1} (6.2 %).

47. ***Working.*** Difference $= (v+\tfrac{1}{2})^2 \bar{\omega}_e x_e$.

    Answer: anharmonic 16802.2 vs SHO 18622.4: error 1820.1 cm^{-1} (9.8 %).

Why do hot bands appear only for low-frequency modes? (extension)

48. ***Working.*** Boltzmann factor.

    Answer: $N_1/N_0 = 8.88\times10^{-7}$ — hot bands practically invisible.

49. ***Working.*** Boltzmann factor.

    Answer: $N_1/N_0 = 3.21\times10^{-5}$ — hot bands practically invisible.

50. ***Working.*** Boltzmann factor.

    Answer: $N_1/N_0 = 3.56\times10^{-1}$ — hot bands are significant.

51. ***Working.*** Boltzmann factor.

    Answer: $N_1/N_0 = 2.10\times10^{-1}$ — hot bands are significant.

52. ***Working.*** Boltzmann factor.

    Answer: $N_1/N_0 = 4.99\times10^{-9}$ — hot bands practically invisible.

True or false

53. **Answer: True** — spacing $= \bar{\omega}_e[1 - 2x_e(v+1)]$ shrinks.

54. **Answer: False** — it is $2\bar{\omega}_e(1 - 3x_e)$, a little less than $2\bar{\omega}_e(1 - 2x_e)$.

55. **Answer: True** — typically 0.005–0.03.

56. **Answer: False** — it lies to the left (lower), by $2\bar{\omega}_e x_e$.

57. **Answer: True** — overtones become weakly allowed.

58. **Answer: True** — Boltzmann population at room temperature.

59. **Answer: False** — the parabola rises forever; only the Morse potential has a dissociation limit.

60. **Answer: True** — derived from the two equations.

61. **Answer: True** — it flattens towards D_e.

62. **Answer: True** — spacing falls by $2\bar{\omega}_e x_e$ per level.

Spot the error

63. **Answer:** It is $\bar{\omega}_e(1 - 2x_e)$.

64. **Answer:** That is the second overtone; the first overtone (0→2) is $2\bar{\omega}_e(1 - 3x_e)$.

65. **Answer:** Hot band is below the fundamental: $\bar{\omega}_e(1 - 4x_e)$ = fundamental $- 2\bar{\omega}_e x_e$.

66. **Answer:** $x_e = \bar{\omega}_e x_e / \bar{\omega}_e = 45.31/2649.7 = 0.0171$; $\bar{\omega}_e x_e$ = f − o/2.

67. **Answer:** The spacing decreases: $\bar{\omega}_e[1 - 2x_e(v+1)]$.

Chapter 4: Electronic Spectroscopy, Franck–Condon and Raman

Topics T1–T3, T7

Largely theory: statements, diagrams and 'what can you conclude?' questions. Franck–Condon diagrams appear in more than half of all papers.

Formulas to remember

$$E = E_{rot} + E_{vib} + E_{elec}, \quad E_{rot} \ll E_{vib} \ll E_{elec}$$

$$E = h\nu = hc\bar{\nu}, \quad \bar{\nu} = 1/\lambda$$

$$\text{electronic lines labelled } (v', v'')$$

$$A = \log \frac{I_0}{I} = \varepsilon c l \quad \text{(Beer–Lambert; older syllabus)}$$

Method.

1. Franck–Condon: always draw the vertical arrow from the centre of $v'' = 0$.
2. Compare r_e' with r_e'': equal → (0,0) strongest; larger → (v′,0) with v′>0 strongest; much larger → continuum.

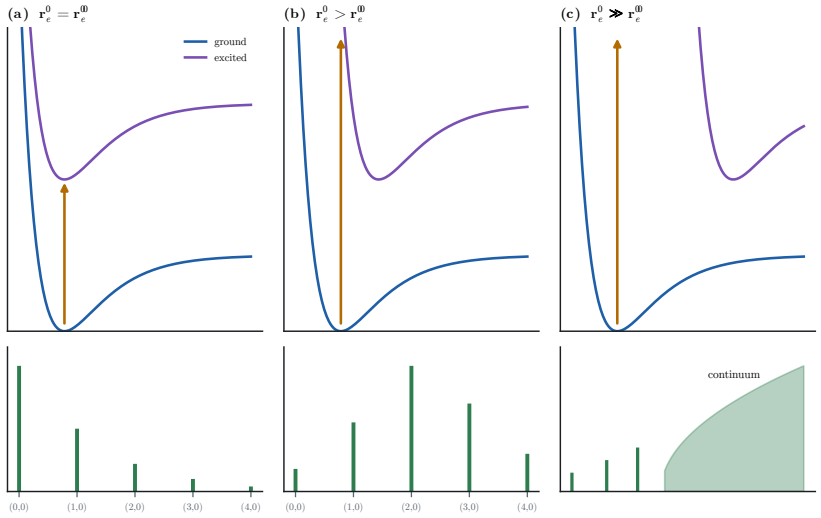


Figure 4.1 (G21): Franck–Condon, three cases (a, b, c) with spectra.

Past ESA questions with solutions

1. State the Franck–Condon principle. With a diagram show the electronic transitions and spectra when (i) $v'' = 0 \to v' = 0$ is strongest; (ii) $v'' = 0 \to v' = 2$ is strongest.

   Dec-24 · UE22 (7 marks)

   Solution. Statement as in the theory box. (i) is case (a) of **Fig. G21** ($r_e' = r_e''$): (0,0) strongest. (ii) is case (b) ($r_e' > r_e''$): the vertical arrow ends on $v' = 2$, so the (2,0) line is strongest and the progression peaks there.

   Answer: Cases (a) and (b)

2. Justify the Franck–Condon principle by drawing the transitions and spectra when (1) the internuclear distances in ground and excited states are the same; (2) the excited-state distance is much larger.

   Jun-24 · UE23 · Set 2 (4 marks)

   Solution. (1) case (a): (0,0) most intense. (2) case (c): vertical arrow above the dissociation limit → continuum (**Fig. G21**).

   Answer: Cases (a) and (c)

3. (i) What is the Franck–Condon principle? If a vibrational-electronic spectrum shows a continuum, what can one infer? (ii) Why is bulk aluminium stable whereas nano aluminium is combustible?

   Jul-23 · UE22 · B (6 marks)

   Solution. (i) Statement + dissociation (theory box). (ii) is a nano-materials question (Unit 4) — not covered here.

   Answer: Dissociation

4. (i) State the Born–Oppenheimer approximation. (ii) What does the appearance of a continuum in an electronic spectrum let us conclude?

   Jan–May-24 · UE22 (6 marks)

   Solution. Statement and dissociation reasoning — see theory box.

   Answer: See theory box

5. What do you mean by Rayleigh scattering and Raman scattering? Mention the difference between Stokes and anti-Stokes lines.

   Dec-23 · UE22 (4 marks)

   Solution. See theory box.

   Answer: See theory box

6. How is it that carbon dioxide is IR active even though it has no permanent dipole? (also Jan-24 UE22 'Why is CO_2 IR active but microwave inactive?')

   Jul-23 · UE22 · B (2 marks)

   Solution. See Q2 theory box: asymmetric stretch and bend change the dipole; symmetric stretch does not.

   Answer: Dipole change, not permanent dipole

Theory questions with model answers

7. **State the Franck–Condon principle. Draw the three cases and comment on each spectrum. (asked ×5)**

   Answer: Statement: "An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition," so transitions are vertical.
   (a) $r'_e = r''_e$: vertical arrow lands on $v' = 0$; the (0,0) line is strongest and intensity falls for (1,0), (2,0)...
   (b) $r'_e > r''_e$: arrow lands on a higher v' (e.g. 2); the (v′,0) line with v′>0 is strongest, so the maximum is displaced along the progression.
   (c) $r'_e \gg r''_e$: arrow lands above the dissociation limit; a few discrete lines then a **continuum** (**Fig. G21**).

8. **The vibrational–electronic spectrum of a molecule shows a continuum. What can be inferred? (asked ×2)**

   Answer: The molecule has been excited above the dissociation limit of the upper state, so it **dissociates** into atoms. The atoms carry away the excess energy as unquantised kinetic energy, so no discrete lines appear — a continuum. (It also implies $r'_e \gg r''_e$.)

9. **State the Born–Oppenheimer approximation. (asked ×3)**

   Answer: The total energy of a molecule is the sum of independent translational, rotational, vibrational and electronic parts; translation is negligible, so $E = E_{rot} + E_{vib} + E_{elec}$ with $E_{rot} \ll E_{vib} \ll E_{elec}$ (electronic levels are far apart, rotational levels very close).

10. **What are Rayleigh scattering and Raman scattering? Differentiate Stokes and anti-Stokes lines. (asked ×2)**

    Answer: Rayleigh: elastic scattering — scattered light has the same frequency as the incident light. **Raman:** inelastic scattering — frequency shifts because the molecule exchanges vibrational/rotational energy with the photon. **Stokes:** scattered frequency lower than incident (molecule gains energy, goes up). **Anti-Stokes:** higher than incident (molecule loses energy, comes down); weaker because fewer molecules start in an excited level. (Transcript-based; not on the 2026 slides.)

11. **State the Beer–Lambert law and explain the terms. (older syllabus)**

    Answer: $A = \log(I_0/I) = \varepsilon c l$: absorbance is proportional to concentration c (mol L^{-1}) and path length l (cm); ε is the molar absorptivity (L mol^{-1} cm^{-1}); I_0 and I are incident and transmitted intensities. **Flag: not on current slides.**

12. **Why does the excited electronic state usually have a larger equilibrium bond length?**

    Answer: An electron is promoted to an antibonding orbital (e.g. π→π*), weakening the bond, so the minimum of the upper potential curve shifts to larger r.

13. **Why do homonuclear molecules (H_2, N_2) show vibration/rotation only in electronic spectra?**

    Answer: They have no dipole change for IR or rotation, but all molecules can undergo electronic transitions, which carry vibrational (coarse) and rotational (fine) structure.

Practice: exam-style variants

Set 1: Wavelength, wavenumber, frequency, energy

Basic conversions that underpin every spectroscopy question.

14. Light of wavelength 500 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

15. Light of wavelength 250 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

16. Light of wavelength 700 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

17. Light of wavelength 10000 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

18. Light of wavelength 3000 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Set 2: Franck–Condon outcomes

Decide the case, then describe the spectrum.

19. A diatomic's ground and excited states have almost identical bond lengths. Which line is the most intense?

20. The spectrum shows the strongest line as (3,0). Compare r_e' and r_e'' and sketch the transition.

21. A spectrum has a few discrete lines that merge into a featureless band at high energy. What happened to the molecule?

22. Why can't a vibrational spectrum of an electronic transition be explained without the Franck–Condon principle even though there is no vibrational selection rule?

23. State two differences between the case where $r_e' = r_e''$ and $r_e' > r_e''$.

Set 3: Selecting the right region and transition

24. Which transition type and region: (a) spin reversal of nuclei; (b) rotation; (c) vibration; (d) valence-electron redistribution; (e) inner-electron; (f) nuclear rearrangement?

25. Order rotation, vibration and electronic transitions by increasing energy and say which spectroscopy each belongs to.

26. Name the three electronic transitions common in organic molecules and the orbitals involved.

27. Draw/describe Stokes and anti-Stokes lines relative to the Rayleigh line.

New question types (not yet asked)

HOMO–LUMO gap ↔ absorption wavelength

Electronic spectroscopy by number: smaller gap → longer wavelength → colour.

28. The HOMO→LUMO absorption maximum of ethene (π→π*) is at 165 nm. Find the wavenumber (cm^{-1}), the energy gap (J, eV, kJ mol^{-1}) and the spectral region.

29. The HOMO→LUMO absorption maximum of 1,3-butadiene (π→π*) is at 217 nm. Find the wavenumber (cm^{-1}), the energy gap (J, eV, kJ mol^{-1}) and the spectral region.

30. The HOMO→LUMO absorption maximum of benzene is at 255 nm. Find the wavenumber (cm^{-1}), the energy gap (J, eV, kJ mol^{-1}) and the spectral region.

31. The HOMO→LUMO absorption maximum of β-carotene is at 450 nm. Find the wavenumber (cm^{-1}), the energy gap (J, eV, kJ mol^{-1}) and the spectral region.

32. The HOMO→LUMO absorption maximum of methylene blue is at 664 nm. Find the wavenumber (cm^{-1}), the energy gap (J, eV, kJ mol^{-1}) and the spectral region.

33. A molecule has a HOMO–LUMO gap of 6.2 eV. At what wavelength does it absorb? ($hc = 1239.84$ eV nm)

34. A molecule has a HOMO–LUMO gap of 4.4 eV. At what wavelength does it absorb? ($hc = 1239.84$ eV nm)

35. A molecule has a HOMO–LUMO gap of 3.1 eV. At what wavelength does it absorb? ($hc = 1239.84$ eV nm)

36. A molecule has a HOMO–LUMO gap of 2.0 eV. At what wavelength does it absorb? ($hc = 1239.84$ eV nm)

Beer–Lambert law (flagged: older syllabus)

Asked once (Dec 2024 / Jun 2024 Q4). Short, numeric and easy marks.

37. (Beer–Lambert; older syllabus) A solution transmits light with $I_0 = 100$, $I = 25$. Find absorbance and % transmittance.

38. (Beer–Lambert; older syllabus) A solution transmits light with $I_0 = 80$, $I = 20$. Find absorbance and % transmittance.

39. (Beer–Lambert; older syllabus) A solution transmits light with $I_0 = 100$, $I = 10$. Find absorbance and % transmittance.

40. (Beer–Lambert; older syllabus) A solution transmits light with $I_0 = 5.0$, $I = 3.5$. Find absorbance and % transmittance.

41. A solution of concentration 3.0×10^{-5} mol L^{-1} in a 1.0 cm cell has absorbance 0.45. Find ε.

42. A solution of concentration 1.6×10^{-4} mol L^{-1} in a 1.0 cm cell has absorbance 0.8. Find ε.

43. A solution of concentration 5.0×10^{-5} mol L^{-1} in a 2.0 cm cell has absorbance 0.3. Find ε.

44. A solution of concentration 2.0×10^{-4} mol L^{-1} in a 0.5 cm cell has absorbance 1.2. Find ε.

45. $\varepsilon = 15000$ L mol^{-1} cm^{-1}, $A = 0.6$, path 1.0 cm. Find the concentration.

46. $\varepsilon = 8500$ L mol^{-1} cm^{-1}, $A = 0.25$, path 1.0 cm. Find the concentration.

47. $\varepsilon = 22000$ L mol^{-1} cm^{-1}, $A = 0.9$, path 2.0 cm. Find the concentration.

48. Why does Beer's law fail at high concentration?

Raman shifts

Rayleigh, Stokes and anti-Stokes as numbers (concept is examined; arithmetic is an extension).

49. (Transcript-based) A laser of 514.5 nm excites a sample showing a Raman shift of 1000 cm^{-1}. Find the wavelengths of the Rayleigh, Stokes and anti-Stokes lines.

50. (Transcript-based) A laser of 532.0 nm excites a sample showing a Raman shift of 1332 cm^{-1}. Find the wavelengths of the Rayleigh, Stokes and anti-Stokes lines.

51. (Transcript-based) A laser of 488.0 nm excites a sample showing a Raman shift of 520 cm^{-1}. Find the wavelengths of the Rayleigh, Stokes and anti-Stokes lines.

52. (Transcript-based) A laser of 632.8 nm excites a sample showing a Raman shift of 2331 cm^{-1}. Find the wavelengths of the Rayleigh, Stokes and anti-Stokes lines.

53. (Transcript-based) A laser of 514.5 nm excites a sample showing a Raman shift of 218 cm^{-1}. Find the wavelengths of the Rayleigh, Stokes and anti-Stokes lines.

Identify the region and the transition

A quick-fire table drill — the whole EM-spectrum slide in ten questions.

54. Radiation of wavelength 5 m: name the region, the type of transition it can excite, and its wavenumber.

55. Radiation of wavelength 3 cm: name the region, the type of transition it can excite, and its wavenumber.

56. Radiation of wavelength 200 µm: name the region, the type of transition it can excite, and its wavenumber.

57. Radiation of wavelength 10 µm: name the region, the type of transition it can excite, and its wavenumber.

58. Radiation of wavelength 500 nm: name the region, the type of transition it can excite, and its wavenumber.

59. Radiation of wavelength 2.5 µm: name the region, the type of transition it can excite, and its wavenumber.

60. Radiation of wavelength 1 nm: name the region, the type of transition it can excite, and its wavenumber.

61. Radiation of wavelength 10 pm: name the region, the type of transition it can excite, and its wavenumber.

62. Radiation of wavelength 0.5 mm: name the region, the type of transition it can excite, and its wavenumber.

63. Radiation of wavelength 350 nm: name the region, the type of transition it can excite, and its wavenumber.

Classify the Franck–Condon case from bond lengths

64. Ground-state bond length $r_e'' = \mathbf{1.21}$ Å, excited-state $r_e' = \mathbf{1.21}$ Å. (Take $\Delta r < 0.02$ Å as 'same' and $\Delta r > 0.6$ Å as 'much larger'.) Identify the Franck–Condon case and describe the spectrum.

65. Ground-state bond length $r_e'' = \mathbf{1.21}$ Å, excited-state $r_e' = \mathbf{1.35}$ Å. (Take $\Delta r < 0.02$ Å as 'same' and $\Delta r > 0.6$ Å as 'much larger'.) Identify the Franck–Condon case and describe the spectrum.

66. Ground-state bond length $r_e'' = \mathbf{1.21}$ Å, excited-state $r_e' = \mathbf{2.3}$ Å. (Take $\Delta r < 0.02$ Å as 'same' and $\Delta r > 0.6$ Å as 'much larger'.) Identify the Franck–Condon case and describe the spectrum.

67. Ground-state bond length $r_e'' = \mathbf{1.5}$ Å, excited-state $r_e' = \mathbf{1.5}$ Å. (Take $\Delta r < 0.02$ Å as 'same' and $\Delta r > 0.6$ Å as 'much larger'.) Identify the Franck–Condon case and describe the spectrum.

68. Ground-state bond length $r_e'' = \mathbf{1.4}$ Å, excited-state $r_e' = \mathbf{1.6}$ Å. (Take $\Delta r < 0.02$ Å as 'same' and $\Delta r > 0.6$ Å as 'much larger'.) Identify the Franck–Condon case and describe the spectrum.

69. Ground-state bond length $r_e'' = \mathbf{1.1}$ Å, excited-state $r_e' = \mathbf{3.0}$ Å. (Take $\Delta r < 0.02$ Å as 'same' and $\Delta r > 0.6$ Å as 'much larger'.) Identify the Franck–Condon case and describe the spectrum.

70. If the excited state had a shorter bond than the ground state ($r_e' < r_e''$) what would the vertical transition land on?

Born–Oppenheimer and energy scales

Order-of-magnitude reasoning (added for clarity): $E_elec \sim 10^4$–10^5 cm^{-1}, $E_vib \sim 10^2$–10^4 cm^{-1}, $E_rot \sim 1$–10^2 cm^{-1}.

71. For HCl: rotation ≈ 21 cm^{-1} (first line), vibration ≈ 2886 cm^{-1}, electronic ≈ 80000 cm^{-1}. Find the ratios E_{vib}/E_{rot} and E_{elec}/E_{vib} and the region of each.

72. A single electronic transition may show vibrational and rotational structure. Explain using B–O.

73. Why do we neglect translational energy in the B–O treatment?

74. Why is the electronic energy gap much larger than the vibrational gap?

75. Which spectroscopy gives structural information about N_2?

Franck–Condon in words (long answer practice)

Write each answer in 4–5 lines, then compare.

76. Explain why 'vertical transition' is used. What is 'vertical' about it?

77. Why do the transitions begin at the centre of $v'' = 0$?

78. Explain 'progression' and why the lines converge.

79. Differentiate 'coarse structure' from 'fine structure'.

80. Why is the intensity unequal even though there is no vibrational selection rule?

Quick-fire checks and spot the error

True or false

81. Franck–Condon transitions are vertical.

82. If $r_e' = r_e''$ the (0,0) line is the strongest.

83. A continuum implies the molecule remained intact.

84. Stokes lines appear at higher frequency than the Rayleigh line.

85. Rayleigh scattering is inelastic.

86. The excited state usually has a longer bond than the ground state.

87. All molecules can show electronic transitions.

88. There is no selection rule for vibrational changes during an electronic transition.

89. $E_{rot} \gg E_{vib}$.

90. UV–Vis spectroscopy gives rise to rotational spectra.

91. $A = \log(I_0/I)$ is the Beer–Lambert absorbance.

92. The Raman effect needs the molecule to have a permanent dipole.

Spot the error

93. Franck–Condon: the molecule rapidly changes bond length during excitation, so the arrow is slanted.

94. Case (c) gives (0,0) as the strongest line.

95. Stokes line $= \bar{\nu}_0 + \Delta\bar{\nu}$.

96. Wavenumber $= \lambda$ in nm $\times 10^7$.

97. The first overtone appears in the electronic spectrum as the (0,2) line.

Answers

Set 1: Wavelength, wavenumber, frequency, energy

14. ***Working.*** $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.
Answer: $\bar{\nu} = 2.00 \times 10^{4}$ cm^{-1}; $\nu = 6.00 \times 10^{14}$ Hz; $E = 3.98 \times 10^{-19}$ J; region: visible

15. ***Working.*** $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.
Answer: $\bar{\nu} = 4.00 \times 10^{4}$ cm^{-1}; $\nu = 1.20 \times 10^{15}$ Hz; $E = 7.95 \times 10^{-19}$ J; region: UV

16. ***Working.*** $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.
Answer: $\bar{\nu} = 1.43 \times 10^{4}$ cm^{-1}; $\nu = 4.29 \times 10^{14}$ Hz; $E = 2.84 \times 10^{-19}$ J; region: visible

17. ***Working.*** $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.
Answer: $\bar{\nu} = 1.00 \times 10^{3}$ cm^{-1}; $\nu = 3.00 \times 10^{13}$ Hz; $E = 1.99 \times 10^{-20}$ J; region: IR

18. ***Working.*** $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.
Answer: $\bar{\nu} = 3.33 \times 10^{3}$ cm^{-1}; $\nu = 1.00 \times 10^{14}$ Hz; $E = 6.63 \times 10^{-20}$ J; region: IR

Set 2: Franck–Condon outcomes

19. **Answer:** (0,0): case (a); intensity decreases for (1,0), (2,0)...

20. **Answer:** $r_e' > r_e''$ (case b): the vertical arrow from $v'' = 0$ ends on $v' = 3$.

21. **Answer:** It dissociated (case c); the continuum is unquantised kinetic energy of the atoms.

22. **Answer:** With no selection rule all (v′,v″) lines are allowed, yet they differ greatly in intensity. Franck–Condon (vertical transitions + nuclear probability) explains which are strong.

23. **Answer:** (a) one dominant (0,0) line, intensity falls quickly; (b) intensity maximum shifts to (v′,0) with v′>0 and the progression is longer.

Set 3: Selecting the right region and transition

24. **Answer:** (a) radio/NMR 10 m–1 cm; (b) microwave 1 cm–100 µm; (c) IR 100 µm–1 µm; (d) UV-Vis 1 µm–10 nm; (e) X-ray 10 nm–100 pm; (f) γ-ray 100–1 pm.

25. **Answer:** $E_{rot} < E_{vib} < E_{elec}$: microwave < IR < UV-Vis.

26. **Answer:** σ→σ*, π→π*, n→π* (HOMO → LUMO).

27. **Answer:** Rayleigh at the incident frequency; Stokes at lower frequency (red side); anti-Stokes at higher frequency (blue side), weaker.

HOMO–LUMO gap ↔ absorption wavelength

28. ***Working.*** $\bar{\nu} = 10^7/\lambda(\text{nm})$; $\Delta E = hc\bar{\nu}$.
    Answer: ν̄ = 60606 cm^{-1}; ΔE = 1.20×10^{-18} J = 7.51 eV = 725 kJ mol^{-1}; region: far UV

29. ***Working.*** $\bar{\nu} = 10^7/\lambda(\text{nm})$; $\Delta E = hc\bar{\nu}$.
    Answer: ν̄ = 46083 cm^{-1}; ΔE = 9.16×10^{-19} J = 5.71 eV = 552 kJ mol^{-1}; region: UV

30. ***Working.*** $\bar{\nu} = 10^7/\lambda(\text{nm})$; $\Delta E = hc\bar{\nu}$.
    Answer: ν̄ = 39216 cm^{-1}; ΔE = 7.80×10^{-19} J = 4.86 eV = 469 kJ mol^{-1}; region: UV

31. ***Working.*** $\bar{\nu} = 10^7/\lambda(\text{nm})$; $\Delta E = hc\bar{\nu}$.
    Answer: ν̄ = 22222 cm^{-1}; ΔE = 4.42×10^{-19} J = 2.76 eV = 266 kJ mol^{-1}; region: visible (blue absorbed → orange)

32. ***Working.*** $\bar{\nu} = 10^7/\lambda(\text{nm})$; $\Delta E = hc\bar{\nu}$.
    Answer: ν̄ = 15060 cm^{-1}; ΔE = 2.99×10^{-19} J = 1.87 eV = 180 kJ mol^{-1}; region: visible (red absorbed → blue)

33. ***Working.*** $\lambda = hc/\Delta E$.
    Answer: λ = 200 nm (UV)

34. ***Working.*** $\lambda = hc/\Delta E$.
    Answer: λ = 282 nm (UV)

35. ***Working.*** $\lambda = hc/\Delta E$.
    Answer: λ = 400 nm (UV)

36. ***Working.*** $\lambda = hc/\Delta E$.
    Answer: λ = 620 nm (visible)

Beer–Lambert law (flagged: older syllabus)

37. ***Working.*** $A = \log_{10}(I_0/I)$; $\%T = 100\, I/I_0$.
    Answer: A = 0.602; %T = 25.0 %

38. ***Working.*** $A = \log_{10}(I_0/I)$; $\%T = 100\, I/I_0$.
    Answer: A = 0.602; %T = 25.0 %

39. ***Working.*** $A = \log_{10}(I_0/I)$; $\%T = 100\, I/I_0$.
    Answer: A = 1.000; %T = 10.0 %

40. ***Working.*** $A = \log_{10}(I_0/I)$; $\%T = 100\, I/I_0$.
    Answer: A = 0.155; %T = 70.0 %

41. ***Working.*** $\varepsilon = A/(cl)$.
    Answer: ε = 1.5×10^4 L mol^{-1} cm^{-1}

42. ***Working.*** $\varepsilon = A/(cl)$.
    Answer: ε = 5.0×10^3 L mol^{-1} cm^{-1}

43. ***Working.*** $\varepsilon = A/(cl)$.

Answer: $\varepsilon = 3.0\times10^{3}$ L mol^{-1} cm^{-1}

44. ***Working.*** $\varepsilon = A/(cl)$.

Answer: $\varepsilon = 1.2\times10^{4}$ L mol^{-1} cm^{-1}

45. ***Working.*** $c = A/(\varepsilon l)$.

Answer: $c = 4.00\times10^{-5}$ mol L^{-1}

46. ***Working.*** $c = A/(\varepsilon l)$.

Answer: $c = 2.94\times10^{-5}$ mol L^{-1}

47. ***Working.*** $c = A/(\varepsilon l)$.

Answer: $c = 2.05\times10^{-5}$ mol L^{-1}

48. **Answer:** Interactions between absorbing species, changes in refractive index and stray light cause deviations; use dilute solutions.

Raman shifts

49. ***Working.*** $\bar{\nu}_0 = 10^7/\lambda$; Stokes $= \bar{\nu}_0 - \Delta$; anti-Stokes $= \bar{\nu}_0 + \Delta$.

Answer: Rayleigh 514.5 nm (19436 cm^{-1}); Stokes 542.4 nm (18436 cm^{-1}); anti-Stokes 489.3 nm (20436 cm^{-1}). Stokes is on the red side, anti-Stokes on the blue side.

50. ***Working.*** $\bar{\nu}_0 = 10^7/\lambda$; Stokes $= \bar{\nu}_0 - \Delta$; anti-Stokes $= \bar{\nu}_0 + \Delta$.

Answer: Rayleigh 532.0 nm (18797 cm^{-1}); Stokes 572.6 nm (17465 cm^{-1}); anti-Stokes 496.8 nm (20129 cm^{-1}). Stokes is on the red side, anti-Stokes on the blue side.

51. ***Working.*** $\bar{\nu}_0 = 10^7/\lambda$; Stokes $= \bar{\nu}_0 - \Delta$; anti-Stokes $= \bar{\nu}_0 + \Delta$.

Answer: Rayleigh 488.0 nm (20492 cm^{-1}); Stokes 500.7 nm (19972 cm^{-1}); anti-Stokes 475.9 nm (21012 cm^{-1}). Stokes is on the red side, anti-Stokes on the blue side.

52. ***Working.*** $\bar{\nu}_0 = 10^7/\lambda$; Stokes $= \bar{\nu}_0 - \Delta$; anti-Stokes $= \bar{\nu}_0 + \Delta$.

Answer: Rayleigh 632.8 nm (15803 cm^{-1}); Stokes 742.3 nm (13472 cm^{-1}); anti-Stokes 551.5 nm (18134 cm^{-1}). Stokes is on the red side, anti-Stokes on the blue side.

53. ***Working.*** $\bar{\nu}_0 = 10^7/\lambda$; Stokes $= \bar{\nu}_0 - \Delta$; anti-Stokes $= \bar{\nu}_0 + \Delta$.

Answer: Rayleigh 514.5 nm (19436 cm^{-1}); Stokes 520.3 nm (19218 cm^{-1}); anti-Stokes 508.8 nm (19654 cm^{-1}). Stokes is on the red side, anti-Stokes on the blue side.

Identify the region and the transition

54. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

Answer: Radio-frequency (10 m – 1 cm); nuclear/electron spin reversal (NMR, ESR); $\bar{\nu} = 2.0\times10^{-3}$ cm^{-1}

55. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: Radio-frequency (10 m – 1 cm); nuclear/electron spin reversal (NMR, ESR); $\bar{\nu} = 3.3\times10^{-1}$ cm^{-1}

56. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: Microwave (1 cm – 100 µm); rotational; $\bar{\nu} = 5.0\times10^{1}$ cm^{-1}

57. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: Infrared (100 µm – 1 µm); vibrational; $\bar{\nu} = 1.0\times10^{3}$ cm^{-1}

58. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: UV–Visible (1 µm – 10 nm); electronic (valence electrons); $\bar{\nu} = 2.0\times10^{4}$ cm^{-1}

59. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: Infrared (100 µm – 1 µm); vibrational; $\bar{\nu} = 4.0\times10^{3}$ cm^{-1}

60. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: X-ray (10 nm – 100 pm); inner-electron redistribution; $\bar{\nu} = 1.0\times10^{7}$ cm^{-1}

61. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: γ-ray (100 pm – 1 pm); nuclear rearrangement; $\bar{\nu} = 1.0\times10^{9}$ cm^{-1}

62. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: Microwave (1 cm – 100 µm); rotational; $\bar{\nu} = 2.0\times10^{1}$ cm^{-1}

63. ***Working.*** Compare λ with the region table (10 m–1 cm, 1 cm–100 µm, 100 µm–1 µm, 1 µm–10 nm, 10 nm–100 pm, 100–1 pm).

    Answer: UV–Visible (1 µm – 10 nm); electronic (valence electrons); $\bar{\nu} = 2.9\times10^{4}$ cm^{-1}

Classify the Franck–Condon case from bond lengths

64. ***Working.*** Compare r_e' with r_e''; draw the vertical arrow from $v'' = 0$.

    Answer: $\Delta r = +0.00$ Å → case (a): (0,0) strongest, intensity falls steadily.

65. ***Working.*** Compare r_e' with r_e''; draw the vertical arrow from $v'' = 0$.

    Answer: $\Delta r = +0.14$ Å → case (b): strongest line (v',0) with $v'>0$; progression longer.

66. ***Working.*** Compare r_e' with r_e''; draw the vertical arrow from $v'' = 0$.
Answer: Δr = +1.09 Å → case (c): a few lines then a continuum (dissociation).

67. ***Working.*** Compare r_e' with r_e''; draw the vertical arrow from $v'' = 0$.
Answer: Δr = +0.00 Å → case (a): (0,0) strongest, intensity falls steadily.

68. ***Working.*** Compare r_e' with r_e''; draw the vertical arrow from $v'' = 0$.
Answer: Δr = +0.20 Å → case (b): strongest line (v′,0) with v′>0; progression longer.

69. ***Working.*** Compare r_e' with r_e''; draw the vertical arrow from $v'' = 0$.
Answer: Δr = +1.90 Å → case (c): a few lines then a continuum (dissociation).

70. **Answer:** A vibrational level $v' > 0$ on the inner (left) wall — again the strongest line is (v′,0) with v′>0; the spectrum looks like case (b).

Born–Oppenheimer and energy scales

71. **Answer:** E_vib/E_rot ≈ 136; E_elec/E_vib ≈ 28; microwave, IR, UV respectively. Hence E_rot ≪ E_vib ≪ E_elec.

72. **Answer:** Since the three energies are independent and additive, $\Delta E = \Delta E_{elec} + \Delta E_{vib} + \Delta E_{rot}$: each electronic band splits into vibrational coarse structure, and each vibrational line into rotational fine structure.

73. **Answer:** Translational levels are so closely spaced and so small that they are effectively continuous and do not appear in the spectrum.

74. **Answer:** Electrons are far lighter than nuclei, so electronic motion is much faster and electronic levels are much farther apart; nuclear motion is slow.

75. **Answer:** Only electronic spectroscopy (and Raman); N_2 is inactive in microwave and IR.

Franck–Condon in words (long answer practice)

76. **Answer:** Electronic excitation takes ~10^{-15} s but nuclei vibrate in ~10^{-13} s, so r stays the same during excitation: on a PE-vs-r plot the arrow is vertical.

77. **Answer:** For $v = 0$ the probability of finding the nuclei is maximum at the equilibrium position (centre), while for higher v it is at the turning points.

78. **Answer:** A series of lines $(v', 0)$ from $v'' = 0$ to successive v'; they converge because the excited-state levels converge (anharmonic), reaching the dissociation limit.

79. **Answer:** Coarse = vibrational changes accompanying the electronic transition; fine = rotational changes.

80. **Answer:** Franck–Condon overlap of nuclear wave functions governs relative intensity.

True or false

81. **Answer: True** — internuclear distance is unchanged.

82. **Answer: True** — case (a).

83. **Answer: False** — it implies dissociation.

84. **Answer: False** — Stokes: lower; anti-Stokes: higher.

85. **Answer: False** — it is elastic; Raman is inelastic.

86. **Answer: True** — antibonding character.

87. **Answer: True** — unlike microwave or IR, no dipole condition.

88. **Answer: True** — intensity is set by Franck–Condon factors.

89. **Answer: False** — $E_{rot} \ll E_{vib} \ll E_{elec}$.

90. **Answer: False** — microwave; though rotational *fine structure* rides on electronic bands.

91. **Answer: True** — flagged older syllabus.

92. **Answer: False** — it depends on polarisability, not permanent dipole.

Spot the error

93. **Answer:** The bond length does not change appreciably; the arrow is vertical.

94. **Answer:** In case (c) the arrow lands above the dissociation limit; a continuum results.

95. **Answer:** Stokes = $\bar{\nu}_0 - \Delta\bar{\nu}$(lower frequency); anti-Stokes is $\bar{\nu}_0 + \Delta\bar{\nu}$.

96. **Answer:** $\bar{\nu}(\mathrm{cm}^{-1}) = 10^7 \,/\, \lambda(\mathrm{nm})$.

97. **Answer:** Electronic lines are labelled (v′, v″); 'overtone' belongs to IR vibrational spectra.

Chapter 5: Phase Rule, Water and Pb–Ag Systems

Topics T8–T12

Phase equilibria questions come in three flavours: derive the rule, count C/P/F for a system, and draw-and-label a diagram. Practise counting until it is automatic.

Formulas to remember

$$F = C - P + 2$$

$$\text{variables} = P(C-1) + 2, \quad \text{constraints} = C(P-1)$$

$$\text{reduced: } F = C - P + 1$$

$$\text{water: } F = 3 - P$$

$$\frac{dP}{dT} = \frac{\Delta H}{T \Delta V}$$

Method.

1. Count phases P (each solid, each immiscible liquid, all gases = 1).
2. Count components C = species − independent reactions − composition restrictions.
3. $F = C - P + 2$ (or $C - P + 1$ if pressure is fixed / condensed).
4. $F = 2$ bivariant, 1 univariant, 0 invariant.

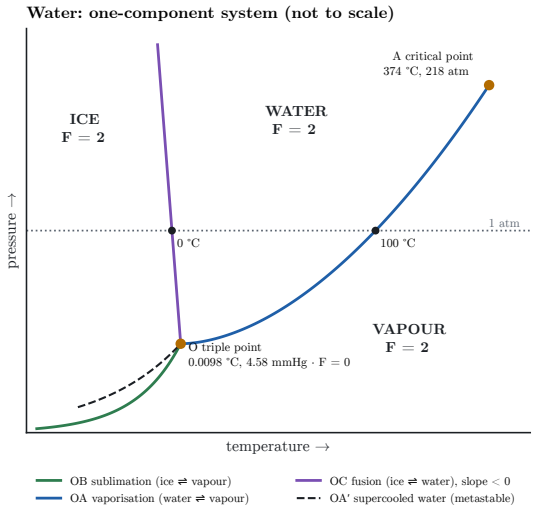


Figure 5.1 (G22): Water phase diagram (schematic, not to scale).

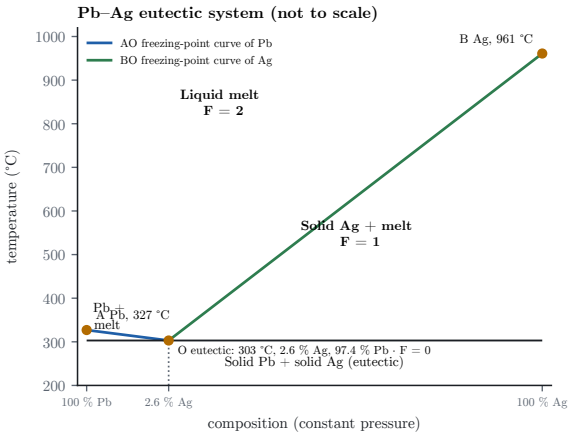


Figure 5.2 (G23): Pb–Ag eutectic diagram (schematic; Pb left, Ag right).

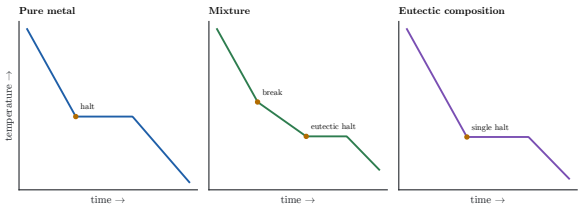


Figure 5.3 (G24): Cooling curves: pure, mixture, eutectic.

Derivation

Derivation of the phase rule (asked ×3)

1. System of C components in P phases at equilibrium. Variables: temperature, pressure and the composition of each phase.
2. Composition of one phase needs $C-1$ mole fractions (they sum to 1). For P phases: $P(C-1)$. Add T and P: **total variables** $= P(C-1)+2$.
3. At equilibrium the chemical potential of each component is equal in every phase: $\mu_i^{\alpha} = \mu_i^{\beta} = \mu_i^{\gamma} = \cdots$ — that is $(P-1)$ equations per component, $C(P-1)$ **constraints** in all.
4. F = variables − constraints $= P(C-1)+2-C(P-1) = PC - P + 2 - CP + C =$ $\mathbf{C - P + 2}$.

Past ESA questions with solutions

1. Derive the phase rule for a system of C components and P phases in equilibrium.
   Jun-24 · UE23 · Set 1 (4 marks)

   Solution. See derivation Q5-D.

   Answer: $F = C - P + 2$

2. Derive the phase rule for a system of C components and P phases in equilibrium.
   Jan-25 · UE24 · Set 1 (4 marks)

   Solution. See derivation Q5-D.

   Answer: $F = C - P + 2$

3. (i) Define component. Give the number of phases and components for the closed system $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$. (ii) Define eutectoid and peritectic points in the Fe–C system.
   Jan · UE23A · Set 2 (4 marks)

   Solution.

   i. (i) **Component:** the smallest number of independently variable constituents by which the composition of every phase can be expressed. Phases: $MgCO_3(s)$, $MgO(s)$, $CO_2(g)$: $P = 3$. Species $3 - 1$ reaction $= C = 2$ (e.g. choose MgO and CO_2: $MgCO_3 = MgO + CO_2$).

   ii. (ii) Eutectoid: one solid phase transforms isothermally into two other solids (austenite → ferrite + cementite, 727 °C, 0.76 wt% C). Peritectic: liquid + one solid → a new solid (liquid + δ-ferrite → austenite, 1493 °C, 0.17 wt% C). **Flag: Fe–C is not in the current material; values are standard textbook figures.**

   Answer: $P = 3$, $C = 2$

4. (i) Why is the phase rule $F = C - P + 2$ for water but $F = C - P + 1$ for the Pb–Ag system? (ii) For $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ in a closed container give the number of components and phases.

   Jan-25 · UE24 · Set 2 (6 marks)

   Solution.

   i. (i) Water is a one-component system with T and P both variable. For Pb–Ag, $C = 2$ so $F = 4 - P$, giving $F = 3$ for one phase — a 3-D diagram. Pressure is therefore fixed ($\approx$1 atm) and, since solid–liquid equilibria (condensed systems, no gas) hardly depend on pressure, the rule reduces to $F = C - P + 1$.

   ii. (ii) $P = 3$ ($CaCO_3$, CaO, CO_2); species $3 - 1$ reaction $\Rightarrow C = 2$.

   Answer: (ii) $P = 3$, $C = 2$

5. (i) Draw a neat labelled phase diagram of water (areas, lines, triple point, critical point, pressure, temperature) and calculate the degree of freedom along the freezing-point curve. (ii) What is the condensed phase rule and why is it applied to the two-component Pb–Ag system?

   Jan-25 · UE24 · Set 1 (6 marks)

   Solution.

   i. (i) Diagram: **Fig. G22**. Triple point O: 0.0098 °C, 4.58 mmHg; critical point A: 374 °C, 218 atm. Along a curve $C = 1, P = 2$: $F = 1 - 2 + 2 = \mathbf{1}$ (univariant).

   ii. (ii) $F = C - P + 1$: used when pressure is constant and the gas phase is absent (condensed systems), as in Pb–Ag where P has little effect.

   Answer: $F = 1$ on the freezing curve

6. (i) Draw a neat labelled phase diagram of the water system with areas, lines and T, P values at the triple and critical points. (ii) Which phases are in equilibrium on the metastable curve? (iii) Calculate F at the triple point.

   Jun-24 · UE23 · Set 2 (6 marks)

   Solution.

   i. (i) **Fig. G22**.

   ii. (ii) Supercooled liquid water $\rightleftharpoons$ vapour (curve OA$\prime$, an extension of OA below the triple point).

   iii. (iii) $C = 1, P = 3$: $F = 1 - 3 + 2 = \mathbf{0}$ — invariant.

   Answer: (ii) liquid + vapour; (iii) F = 0

7. (i) Draw the phase diagram of water, label areas and boundaries, and calculate F in the areas, along phase boundaries and at the triple point. (ii) Using the Clausius–Clapeyron equation, discuss why the melting-point curve has a negative slope. (iii) What is the reduced phase rule and when is it used?

Dec-24 · UE23 (8 marks)

Solution.

i. (i) **Fig. G22**: areas $F = 3 - 1 = 2$; lines $F = 3 - 2 = 1$; triple point $F = 3 - 3 = 0$.

ii. (ii) $\dfrac{dP}{dT} = \dfrac{\Delta H_{fus}}{T\Delta V}$. Fusion: $\Delta H > 0$; ice → water **decreases** volume ($\Delta V < 0$), so $dP/dT < 0$: raising pressure lowers the melting point. The slope is steep because $|\Delta V|$ is small. (OA and OB have $\Delta V > 0$ and large, so gentle positive slopes.)

iii. (iii) $F = C - P + 1$ — for condensed systems at constant pressure.

Answer: F = 2, 1, 0; dP/dT < 0 because ΔV < 0

8. (i) Draw a neat labelled phase diagram of the lead–silver system indicating areas and lines. Give the temperature and composition at the eutectic point. (ii) For the Fe–C system mention the phases in equilibrium, temperature and wt% C at the peritectic point.

Jun-24 · UE23 · Set 1 (6 marks)

Solution.

i. (i) **Fig. G23**: Pb melts at 327 °C, Ag at 961 °C; curves AO and BO fall to the eutectic point O at **303 °C, 2.6 % Ag (97.4 % Pb)**; areas: liquid melt; liquid + solid Ag; liquid + solid Pb; solid Pb + solid Ag. At O: $C = 2, P = 3, F = 0$.

ii. (ii) Peritectic: liquid + δ-ferrite ⇌ austenite at **1493 °C, 0.17 wt% C** (flag: Fe–C not in current material).

Answer: Eutectic: 303 °C, 2.6 % Ag

9. (i) Draw a neat labelled phase diagram of the CO_2 system with areas, lines, triple and critical points (T, P). (ii) What is the reduced phase rule? Why is the general Gibbs rule not applied to a two-component system? *[Not on current slides]*

Jan · UE23A · Set 2 (6 marks)

Solution.

i. (i) Like water but the solid–liquid line leans **right** (positive slope, since solid CO_2 is denser). Triple point **−56.6 °C, 5.11 atm**; critical point **31.1 °C, 73 atm**; at 1 atm CO_2 sublimes (−78.5 °C). **Flag: not in current material; standard values.**

ii. (ii) For $C = 2$, $F = 4 - P$ gives $F = 3$ when $P = 1$, needing three axes (T, P, composition). So pressure is fixed: $F = C - P + 1$.

Answer: Triple −56.6 °C / 5.11 atm; critical 31.1 °C / 73 atm

Practice: exam-style variants

Set 1: Count P, C and F

The exam asks this for $CaCO_3$ and $MgCO_3$. The twist is usually NH_4Cl or an added reactant.

10. For the system: Ice ⇌ water ⇌ water vapour at the triple point, find P, C and F using $F = C - P + 2$.

11. For the system: Liquid water in equilibrium with its vapour, find P, C and F using $F = C - P + 2$.

12. For the system: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ (closed), find P, C and F using $F = C - P + 2$.

13. For the system: $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$ (closed), find P, C and F using $F = C - P + 2$.

14. For the system: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$, starting from NH_4Cl only, find P, C and F using $F = C - P + 2$.

15. For the system: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$ with excess NH_3 added, find P, C and F using $F = C - P + 2$.

16. For the system: Saturated NaCl solution + excess solid NaCl + vapour, find P, C and F using $F = C - P + 2$.

17. For the system: Two immiscible liquids (oil and water) and their vapour, find P, C and F using $F = C - P + 2$.

18. For the system: Rhombic sulphur ⇌ monoclinic sulphur ⇌ sulphur vapour, find P, C and F using $F = C - P + 2$.

19. For the system: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ in a closed vessel, starting from PCl_5 only, find P, C and F using $F = C - P + 2$.

20. For the system: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, all three initially present, find P, C and F using $F = C - P + 2$.

21. For the system: $Na_2SO_4{\cdot}10H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10\ H_2O(g)$, find P, C and F using $F = C - P + 2$.

Set 2: Number of variables and constraints (derivation check)

Using the derivation to confirm $F = C - P + 2$.

22. For a system of $C = 2$ components in $P = 1$ phases: how many composition variables, total variables, constraints and what is F?

23. For a system of $C = 2$ components in $P = 2$ phases: how many composition variables, total variables, constraints and what is F?

24. For a system of $C = 2$ components in $P = 3$ phases: how many composition variables, total variables, constraints and what is F?

25. For a system of $C = 3$ components in $P = 2$ phases: how many composition variables, total variables, constraints and what is F?

26. For a system of $C = 3$ components in $P = 4$ phases: how many composition variables, total variables, constraints and what is F?

Set 3: Water diagram

Draw Fig. G22 from memory, then check.

27. What are F in an area, on a line, and at the triple point of water? Name the systems.

28. Name the curves OA, OB, OC and OA′ and the phases in equilibrium on each.

29. Give the temperature and pressure at the triple point and critical point.

30. Why does water stop existing as a liquid above the critical point?

31. What happens to the melting point of ice if pressure increases, and why?

32. State the F-value and meaning of: ice + water at 1 atm. What is the temperature?

Set 4: Clausius–Clapeyron slopes (extension)

Not asked numerically in past papers — included to make the sign argument concrete (Added for clarity).

33. Ice melting at 273.15 K: $\Delta H_{fus} = 6010$ J mol^{-1}, $\Delta V = -1.63 \times 10^{-6}$ m^3 mol^{-1}. Find dP/dT.

34. Water boiling at 373.15 K: $\Delta H_{vap} = 40700$ J mol^{-1}, $\Delta V \approx 30.6 \times 10^{-3}$ m^3 mol^{-1}. Find dP/dT and compare with Set 4 Q1.

35. Predict the sign of the slope for a substance that expands on melting.

Set 5: Reduced phase rule and the Pb–Ag system

36. Why do we use $F = C - P + 1$ for Pb–Ag?

37. Find F at the eutectic point of Pb–Ag and name the three phases present.

38. Find F along the freezing-point curves AO and BO and in the one-phase liquid region.

39. A Pb–Ag melt contains 10 % Ag. It is cooled slowly from 700 °C. Describe the cooling path.

40. What is a eutectic mixture? Give two properties.

41. How does the phase rule explain why the cooling curve of the eutectic composition shows a single halt?

Set 6: Cooling curves (thermal analysis)

42. Sketch the cooling curve of a pure metal and explain the flat portion.

43. Sketch the cooling curve of a binary mixture (not eutectic) and explain the break and the halt.

44. How is a phase diagram constructed from cooling curves?

New question types (not yet asked)

Counting phases, components and degrees of freedom — 18 systems

Exam: $CaCO_3$, $MgCO_3$, NH_4Cl. Here is the full range of systems an examiner could use.

45. Find P, C and F for: An unsaturated NaCl solution (no solid, no vapour considered). (Use $F = C - P + 2$.)

46. Find P, C and F for: A saturated NaCl solution with no excess solid, and its vapour. (Use $F = C - P + 2$.)

47. Find P, C and F for: A mixture of N_2 and O_2 gases. (Use $F = C - P + 2$.)

48. Find P, C and F for: Ice + saturated salt solution + vapour. (Use $F = C - P + 2$.)

49. Find P, C and F for: $Fe(s) + H_2O(g) \rightleftharpoons FeO(s) + H_2(g)$ (closed vessel). (Use $F = C - P + 2$.)

50. Find P, C and F for: Graphite $\rightleftharpoons$ diamond (solid phases only). (Use $F = C - P + 2$.)

51. Find P, C and F for: Benzene + toluene (miscible liquids) + vapour. (Use $F = C - P + 2$.)

52. Find P, C and F for: Ethanol + water solution + vapour. (Use $F = C - P + 2$.)

53. Find P, C and F for: $CuSO_4{\cdot}5H_2O(s) \rightleftharpoons CuSO_4{\cdot}3H_2O(s) + 2H_2O(g)$. (Use $F = C - P + 2$.)

54. Find P, C and F for: NaCl(aq) + ice. (Use $F = C - P + 2$.)

55. Find P, C and F for: Pure water boiling at 1 atm (liquid + vapour). (Use $F = C - P + 2$.)

56. Find P, C and F for: Pure CO_2 at its triple point. (Use $F = C - P + 2$.)

57. Find P, C and F for: Sulphur: rhombic + monoclinic + vapour (transition). (Use $F = C - P + 2$.)

58. Find P, C and F for: Hg(l) in equilibrium with Hg vapour. (Use $F = C - P + 2$.)

59. Find P, C and F for: $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ (closed vessel). (Use $F = C - P + 2$.)

60. Find P, C and F for: Solid $I_2 \rightleftharpoons I_2$ vapour. (Use $F = C - P + 2$.)

61. Find P, C and F for: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ with extra CO_2 added. (Use $F = C - P + 2$.)

62. Find P, C and F for: $CaSO_4{\cdot}2H_2O(s) \rightleftharpoons CaSO_4(s) + 2H_2O(g)$. (Use $F = C - P + 2$.)

Complete the F-table

63. Complete the table of $F = 3 - P$ for the water system: single-phase areas (ice, water, vapour), the three curves, the triple point, the critical point and the metastable curve.

64. Complete the table of $F = C - P + 1$ for the Pb–Ag system: liquid melt, liquid+Pb, liquid+Ag, solid Pb+Ag (below eutectic), the eutectic point, the freezing curves, and the pure-metal melting points A and B.

65. At the triple point of water the pressure is 4.58 mmHg. Why can't you change either T or P without losing a phase?

66. At 1 atm water boils at 100 °C. Why is the temperature fixed? Use the phase rule.

67. A system of 3 components can have at most how many phases at equilibrium? Generalise.

68. What is the maximum number of phases for C = 1, 2, 3, 4 (general and condensed)?

Desilverisation (Pattinson) mass balance

Applies the Pb–Ag diagram: the melt gets enriched in silver as pure lead crystallises. Transcript-only topic; the arithmetic is an extension.

69. (Desilverisation of argentiferous lead — transcript) 1000 kg of molten lead contains 0.05 % Ag. It is cooled slowly and pure solid Pb is skimmed off until the melt reaches the eutectic composition (2.6 % Ag). How much Pb is removed and how much Ag-rich melt remains?

70. (Desilverisation of argentiferous lead — transcript) 1000 kg of molten lead contains 0.1 % Ag. It is cooled slowly and pure solid Pb is skimmed off until the melt reaches the eutectic composition (2.6 % Ag). How much Pb is removed and how much Ag-rich melt remains?

71. (Desilverisation of argentiferous lead — transcript) 1000 kg of molten lead contains 0.2 % Ag. It is cooled slowly and pure solid Pb is skimmed off until the melt reaches the eutectic composition (2.6 % Ag). How much Pb is removed and how much Ag-rich melt remains?

72. (Desilverisation of argentiferous lead — transcript) 500 kg of molten lead contains 0.5 % Ag. It is cooled slowly and pure solid Pb is skimmed off until the melt reaches the eutectic composition (2.6 % Ag). How much Pb is removed and how much Ag-rich melt remains?

Clausius–Clapeyron numbers (extension)

Shows what the curves mean quantitatively: very steep OC, gentle OA. Altitude cooking and pressure cookers use curve OA.

73. (Extension) Using $dP/dT = \Delta H / T\Delta V$ with $\Delta H_{fus} = 6010$ J mol^{-1} and $\Delta V = -1.63 \times 10^{-6}$ m^3 mol^{-1} at 273.15 K, find the change in melting point of ice when the pressure rises by 1 atm.

74. (Extension) Using $dP/dT = \Delta H / T\Delta V$ with $\Delta H_{fus} = 6010$ J mol^{-1} and $\Delta V = -1.63 \times 10^{-6}$ m^3 mol^{-1} at 273.15 K, find the change in melting point of ice when the pressure rises by 10 atm.

75. (Extension) Using $dP/dT = \Delta H / T\Delta V$ with $\Delta H_{fus} = 6010$ J mol^{-1} and $\Delta V = -1.63 \times 10^{-6}$ m^3 mol^{-1} at 273.15 K, find the change in melting point of ice when the pressure rises by 100 atm.

76. (Extension) Using $dP/dT = \Delta H / T\Delta V$ with $\Delta H_{fus} = 6010$ J mol^{-1} and $\Delta V = -1.63 \times 10^{-6}$ m^3 mol^{-1} at 273.15 K, find the change in melting point of ice when the pressure rises by 500 atm.

77. (Extension) Using $dP/dT = \Delta H / T\Delta V$ with $\Delta H_{fus} = 6010$ J mol^{-1} and $\Delta V = -1.63 \times 10^{-6}$ m^3 mol^{-1} at 273.15 K, find the change in melting point of ice when the pressure rises by 1000 atm.

78. (Extension) Using $\ln(P_2/P_1) = \dfrac{\Delta H_{vap}}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$ with $\Delta H_{vap} = 40.7$ kJ mol^{-1}, find the boiling point of water at 0.5 atm.

79. (Extension) Using $\ln(P_2/P_1) = \dfrac{\Delta H_{vap}}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$ with $\Delta H_{vap} = 40.7$ kJ mol^{-1}, find the boiling point of water at 0.7 atm.

80. (Extension) Using $\ln(P_2/P_1) = \dfrac{\Delta H_{vap}}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$ with $\Delta H_{vap} = 40.7$ kJ mol^{-1}, find the boiling point of water at 2.0 atm.

81. (Extension) Using $\ln(P_2/P_1) = \dfrac{\Delta H_{vap}}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$ with $\Delta H_{vap} = 40.7$ kJ mol^{-1}, find the boiling point of water at 5.0 atm.

Where are we on the water diagram?

Read the diagram like a map.

82. At 25 °C, 1 atm, which phase(s) of water are present, and what is F?

83. At −10 °C, 1 atm, which phase(s) of water are present, and what is F?

84. At 150 °C, 1 atm, which phase(s) of water are present, and what is F?

85. At 100 °C, 1 atm, which phase(s) of water are present, and what is F?

86. At 0 °C, 1 atm, which phase(s) of water are present, and what is F?

87. At 0.0098 °C, 4.58 mmHg, which phase(s) of water are present, and what is F?

88. At 374 °C, 218 atm, which phase(s) of water are present, and what is F?

89. At −10 °C, 1 mmHg, which phase(s) of water are present, and what is F?

90. At −10 °C, 10 mmHg, which phase(s) of water are present, and what is F?

91. At 25 °C, 760 mmHg, which phase(s) of water are present, and what is F?

Interpret cooling curves

92. A pure metal's cooling curve shows a flat portion at 327 °C. Give F on the flat portion, and the phases.

93. A Pb–Ag melt's cooling curve shows a change of slope at 800 °C and a long halt at 303 °C. Interpret.

94. A melt of unknown composition shows only a halt at 303 °C, with no break above it. What is its composition?

95. Which of 0 %, 2.6 %, 50 % and 100 % Ag in Pb shows (i) a halt only; (ii) a break then a halt; (iii) halt only at 961 °C?

96. How does the duration of the eutectic halt vary with composition?

97. Explain how the Pb–Ag diagram is built from six cooling curves.

Definitions in your own words

98. Define phase with three examples that are not water.

99. Define component. Why are 'zero and negative values' allowed in the definition?

100. Define degree of freedom/variance with an example.

101. What is a condensed system?

102. What is a metastable equilibrium? Give an example.

103. State the conditions under which F = C − P + 2 is valid.

104. What are bivariant, univariant and invariant systems?

Derivation variants and logic

105. Derive the reduced phase rule $F = C - P + 1$ from $F = C - P + 2$.

106. Show that for $C = 2$ the unreduced rule gives $F = 3$ when $P = 1$, and why this needs three dimensions.

107. In deriving the phase rule, why is the number of composition variables $P(C-1)$ and not PC?

108. In the derivation, why are there $C(P-1)$ equations?

109. For a system with $C = 3$ and $P = 2$: number of variables, constraints, F.

110. Why do we say chemical potential 'drives' a system to equilibrium?

Quick-fire checks and spot the error

True or false

111. The phase rule $F = C - P + 2$ applies to the Pb–Ag diagram at constant pressure.

112. Ice present as many pieces is still one phase.

113. NH_4Cl heated in a closed vessel has C = 1.

114. At the triple point of water $F = 2$.

115. The fusion curve of water has a positive slope.

116. Increasing pressure raises the melting point of ice.

117. The critical point is where the liquid–vapour interface disappears.

118. At the eutectic point of Pb–Ag, three phases coexist and $F = 0$.

119. The eutectic mixture has the highest melting point of all compositions.

120. Metastable water at -5 °C has a lower vapour pressure than ice at -5 °C.

121. A component is the same as a chemical species.

122. Gas mixtures form one phase regardless of the number of gases.

Spot the error

123. $CaCO_3 \rightleftharpoons CaO + CO_2$ has C = 3 because there are three species.

124. Water has P = 3 at the triple point so F = 3 − 3 + 2 = 2.

125. Variables = PC + 2.

126. The Pb–Ag eutectic has F = 2 − 3 + 2 = 1.

127. The melting curve of water slopes up because ice is denser than water.

128. Along curve OB, P = 2, so F = 2.

Answers

Set 1: Count P, C and F

10. ***Working.*** single substance, three physical phases.
    Answer: $P = 3$, $C = 1$, $F = 1 - 3 + 2 = \mathbf{0}$ (invariant)

11. ***Working.*** one substance.
    Answer: $P = 2$, $C = 1$, $F = 1 - 2 + 2 = \mathbf{1}$ (univariant)

12. ***Working.*** 3 species − 1 reaction.
    Answer: $P = 3$, $C = 2$, $F = 2 - 3 + 2 = \mathbf{1}$ (univariant)

13. ***Working.*** 3 species − 1 reaction.
    Answer: $P = 3$, $C = 2$, $F = 2 - 3 + 2 = \mathbf{1}$ (univariant)

14. ***Working.*** gas has NH_3 : HCl = 1 : 1, an extra restriction.
    Answer: $P = 2$, $C = 1$, $F = 1 - 2 + 2 = \mathbf{1}$ (univariant)

15. ***Working.*** the 1 : 1 restriction is lost.
    Answer: $P = 2$, $C = 2$, $F = 2 - 2 + 2 = \mathbf{2}$ (bivariant)

16. ***Working.*** NaCl and water; solid, solution, vapour.
    Answer: $P = 3$, $C = 2$, $F = 2 - 3 + 2 = \mathbf{1}$ (univariant)

17. ***Working.*** two components; liquid, liquid, vapour.
    Answer: $P = 3$, $C = 2$, $F = 2 - 3 + 2 = \mathbf{1}$ (univariant)

18. ***Working.*** one component, three phases.
    Answer: $P = 3$, $C = 1$, $F = 1 - 3 + 2 = \mathbf{0}$ (invariant)

19. ***Working.*** single gas phase; composition fixed by $[PCl_3] = [Cl_2]$.
    Answer: $P = 1$, $C = 1$, $F = 1 - 1 + 2 = \mathbf{2}$ (bivariant)

20. ***Working.*** 3 species − 1 reaction.
    Answer: $P = 1$, $C = 2$, $F = 2 - 1 + 2 = \mathbf{3}$ (multivariant)

21. ***Working.*** salt and water as components; hydrate, anhydrate, vapour.
    Answer: $P = 3$, $C = 2$, $F = 2 - 3 + 2 = \mathbf{1}$ (univariant)

Set 2: Number of variables and constraints (derivation check)

22. ***Working.*** Variables = $P(C - 1) + 2$; constraints = $C(P - 1)$.
    Answer: Composition: 1; total variables: 3; constraints: 0; $F = 3 - 0 = \mathbf{3}$

23. ***Working.*** Variables = $P(C - 1) + 2$; constraints = $C(P - 1)$.
    Answer: Composition: 2; total variables: 4; constraints: 2; $F = 4 - 2 = \mathbf{2}$

24. ***Working.*** Variables = $P(C - 1) + 2$; constraints = $C(P - 1)$.
    Answer: Composition: 3; total variables: 5; constraints: 4; $F = 5 - 4 = \mathbf{1}$

25. ***Working.*** Variables $= P(C-1)+2$; constraints $= C(P-1)$.

    Answer: Composition: 4; total variables: 6; constraints: 3; $F = 6-3 = \mathbf{3}$

26. ***Working.*** Variables $= P(C-1)+2$; constraints $= C(P-1)$.

    Answer: Composition: 8; total variables: 10; constraints: 9; $F = 10-9 = \mathbf{1}$

Set 3: Water diagram

27. ***Working.*** $F = 3 - P$.

    Answer: Area $F = 2$ (bivariant); line $F = 1$ (univariant); triple point $F = 0$ (invariant).

28. **Answer:** OA vaporisation (water $\rightleftharpoons$ vapour); OB sublimation (ice $\rightleftharpoons$ vapour); OC fusion (ice $\rightleftharpoons$ water, negative slope); OA′ metastable supercooled water $\rightleftharpoons$ vapour.

29. **Answer:** Triple point 0.0098 °C, 4.58 mmHg; critical point 374 °C, 218 atm.

30. **Answer:** The liquid–vapour interface vanishes — above the critical temperature no pressure can liquefy the vapour.

31. **Answer:** It decreases: ice melts to a smaller volume ($\Delta V < 0$), so by Clausius–Clapeyron $dP/dT < 0$.

32. **Answer:** $P = 2$, $F = 1$; at fixed 1 atm T is fixed at 0 °C (normal melting point).

Set 4: Clausius–Clapeyron slopes (extension)

33. ***Working.*** $dP/dT = \Delta H/(T\Delta V)$.

    Answer: $dP/dT = -1.35 \times 10^7$ Pa K^{-1} ≈ -133 atm K^{-1} — large and negative: very steep curve OC.

34. ***Working.*** Same equation.

    Answer: $dP/dT \approx 3567$ Pa K^{-1} $\approx 35.2\times10^{-3}$ atm K^{-1} — positive and tiny by comparison (large ΔV), so OA is gentle.

35. **Answer:** $\Delta V > 0 \Rightarrow dP/dT > 0$: melting curve leans right (e.g. CO_2).

Set 5: Reduced phase rule and the Pb–Ag system

36. ***Working.*** Condensed system (no gas), pressure fixed.

    Answer: Because $C = 2$ would need 3-D at $F = C - P + 2$; at constant pressure one variable is removed.

37. ***Working.*** $C = 2, P = 3$.

    Answer: $F = 2 - 3 + 1 = 0$ (invariant). Phases: solid Pb, solid Ag and liquid melt (303 °C, 2.6 % Ag).

38. **Answer:** Curves: $P = 2$ so $F = 1$ (univariant); liquid region $P = 1$ so $F = 2$ (bivariant).

39. ***Working.*** 10 % Ag is richer in Ag than the eutectic (2.6 % Ag), so the melt cools until it meets the Ag freezing-point curve (O to the Ag melting point): solid Ag separates first and the melt becomes richer in Pb.

    Answer: Solid Ag separates first on the Ag-side curve; the melt composition slides down that curve to O (303 °C, 2.6 % Ag); there Pb and Ag crystallise together at constant temperature (halt, F = 0); below 303 °C: solid Pb + solid Ag. (A melt with less than 2.6 % Ag would separate Pb first.)

40. **Answer:** A mixture of two components with the lowest freezing point of all compositions; it has a definite composition and a sharp melting point and solidifies at constant temperature (F = 0).

41. **Answer:** At the eutectic $F = 0$: temperature stays fixed while all three phases coexist until solidification is complete.

Set 6: Cooling curves (thermal analysis)

42. **Answer:** Smooth fall, then a **halt** at the freezing point (solid and liquid coexist, $F = C - P + 1 = 1 - 2 + 1 = 0$), then a fall again. (Fig. G24)

43. **Answer:** Fall → **break** (change of slope) when the first solid appears (heat of crystallisation slows cooling, $F = 1$) → fall → **halt** at the eutectic temperature ($F = 0$) → fall.

44. **Answer:** Record cooling curves for several compositions; read the break and halt temperatures; plot them against composition and join the break points (liquidus) and the common halt temperature (eutectic line).

Counting phases, components and degrees of freedom — 18 systems

45. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 1$, $C = 2$, $F = 3$ (trivariant)

46. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 2$, $F = 2$ (bivariant)

47. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 1$, $C = 2$, $F = 3$ (trivariant)

48. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 2$, $F = 1$ (univariant)

49. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 3$, $F = 2$ (bivariant)

50. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 1$, $F = 1$ (univariant)

51. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 2$, $F = 2$ (bivariant)

52. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 2$, $F = 2$ (bivariant)

53. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 2$, $F = 1$ (univariant)

54. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 2$, $F = 2$ (bivariant)

55. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 1$, $F = 1$ (univariant)

56. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 1$, $F = 0$ (invariant)

57. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 1$, $F = 0$ (invariant)

58. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 1$, $F = 1$ (univariant)

59. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 2$, $F = 2$ (bivariant)

60. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 2$, $C = 1$, $F = 1$ (univariant)

61. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 2$, $F = 1$ (univariant)

62. ***Working.*** Count phases first; then components = species − independent reactions − composition restrictions.

    Answer: $P = 3$, $C = 2$, $F = 1$ (univariant)

Complete the F-table

63. ***Working.*** $C = 1$.

    Answer: Areas: P=1, F=2. OA, OB, OC: P=2, F=1. OA′ (metastable, liquid+vapour): P=2, F=1. Triple point O: P=3, F=0. Critical point A: liquid and vapour become identical; treated as invariant (F=0).

64. ***Working.*** $C = 2$, reduced rule.

    Answer: Liquid melt: P=1, F=2. Liquid + one solid: P=2, F=1. Solid Pb + solid Ag: P=2, F=1. Curves AO and BO: P=2, F=1. Eutectic O: P=3 (Pb + Ag + melt), F=0. Pure metal melting point A or B: C=1, P=2 ⇒ F = 1−2+1 = 0.

65. ***Working.*** $F = 0$.

    Answer: F = 0 (invariant): T and P are both fixed; changing either removes at least one phase.

66. ***Working.*** Two phases, $C = 1$.

    Answer: F = 1 − 2 + 2 = 1; fixing the pressure at 1 atm uses up the single degree of freedom, so T is fixed at 100 °C.

67. ***Working.*** $F \geq 0 \Rightarrow P \leq C + 2$.

    Answer: P_max = C + 2 = 5 (general rule); for a condensed system, P_max = C + 1.

68. ***Working.*** $P_{max} = C + 2$ and $C + 1$.

    Answer: General: 3, 4, 5, 6. Condensed: 2, 3, 4, 5 (so Pb–Ag has at most 3 phases at the eutectic).

Desilverisation (Pattinson) mass balance

69. ***Working.*** Ag is conserved: $M\,x_0 = m_{melt} \times 2.6\%$.

    Answer: m_melt = 1000×0.05/2.6 = 19.2 kg (2.6 % Ag); Pb removed = 980.8 kg (98.1 % of the charge). The melt then holds all the 0.50 kg of Ag. [Idealised: pure solid Pb, no Ag in solid.]

70. ***Working.*** Ag is conserved: $M\,x_0 = m_{melt} \times 2.6\%$.

    Answer: m_melt = 1000×0.1/2.6 = 38.5 kg (2.6 % Ag); Pb removed = 961.5 kg (96.2 % of the charge). The melt then holds all the 1.00 kg of Ag. [Idealised: pure solid Pb, no Ag in solid.]

71. ***Working.*** Ag is conserved: $M\,x_0 = m_{melt} \times 2.6\%$.

    Answer: m_melt = 1000×0.2/2.6 = 76.9 kg (2.6 % Ag); Pb removed = 923.1 kg (92.3 % of the charge). The melt then holds all the 2.00 kg of Ag. [Idealised: pure solid Pb, no Ag in solid.]

72. ***Working.*** Ag is conserved: $M\,x_0 = m_{melt} \times 2.6\%$.

    Answer: m_melt = 500×0.5/2.6 = 96.2 kg (2.6 % Ag); Pb removed = 403.8 kg (80.8 % of the charge). The melt then holds all the 2.50 kg of Ag. [Idealised: pure solid Pb, no Ag in solid.]

Clausius–Clapeyron numbers (extension)

73. ***Working.*** $\Delta T \approx T\Delta V\,\Delta P/\Delta H$ (small changes).

    Answer: ΔT = -0.008 K — melting point falls (-0.01 °C); sign negative because ΔV < 0.

74. ***Working.*** $\Delta T \approx T\Delta V\,\Delta P/\Delta H$ (small changes).

    Answer: ΔT = -0.075 K — melting point falls (-0.08 °C); sign negative because ΔV < 0.

75. ***Working.*** $\Delta T \approx T\Delta V\,\Delta P/\Delta H$ (small changes).

    Answer: ΔT = -0.751 K — melting point falls (-0.75 °C); sign negative because ΔV < 0.

76. ***Working.*** $\Delta T \approx T\Delta V\,\Delta P/\Delta H$ (small changes).

    Answer: ΔT = -3.753 K — melting point falls (-3.75 °C); sign negative because ΔV < 0.

77. ***Working.*** $\Delta T \approx T\Delta V\,\Delta P/\Delta H$ (small changes).

    Answer: ΔT = -7.506 K — melting point falls (-7.51 °C); sign negative because ΔV < 0.

78. ***Working.*** $P_1 = 1$ atm, $T_1 = 373.15$ K.

    Answer: T_2 = 354.4 K = 81.3 °C

79. ***Working.*** $P_1 = 1$ atm, $T_1 = 373.15$ K.

    Answer: T_2 = 363.3 K = 90.1 °C

80. ***Working.*** $P_1 = 1$ atm, $T_1 = 373.15$ K.

    Answer: T_2 = 394.0 K = 120.8 °C

81. ***Working.*** $P_1 = 1$ atm, $T_1 = 373.15$ K.

    Answer: T_2 = 425.3 K = 152.2 °C

Where are we on the water diagram?

82. ***Working.*** Locate the point on **Fig. G22**.

    Answer: liquid water; area, F=2.

83. ***Working.*** Locate the point on **Fig. G22**.

    Answer: ice; area, F=2.

84. ***Working.*** Locate the point on **Fig. G22**.

    Answer: water vapour; area, F=2.

85. ***Working.*** Locate the point on **Fig. G22**.

Answer: liquid + vapour; on OA, F=1.

86. ***Working.*** Locate the point on **Fig. G22**.

Answer: ice + liquid; on OC, F=1.

87. ***Working.*** Locate the point on **Fig. G22**.

Answer: ice + liquid + vapour; triple point, F=0.

88. ***Working.*** Locate the point on **Fig. G22**.

Answer: critical point (liquid and vapour indistinguishable); A, F=0.

89. ***Working.*** Locate the point on **Fig. G22**.

Answer: vapour (ice's vapour pressure is 1.95 mmHg); area, F=2.

90. ***Working.*** Locate the point on **Fig. G22**.

Answer: ice; area, F=2.

91. ***Working.*** Locate the point on **Fig. G22**.

Answer: liquid water; area, F=2.

Interpret cooling curves

92. ***Working.*** Reduced rule, $C = 1$.

Answer: Solid + liquid in equilibrium: $F = 1 - 2 + 1 = 0$ — temperature stays constant until all liquid solidifies.

93. **Answer:** Break at 800 °C = start of crystallisation of the first solid (F=1, cooling slows); halt at 303 °C = eutectic crystallisation, three phases, F=0.

94. **Answer:** The eutectic composition, 2.6 % Ag (97.4 % Pb).

95. **Answer:** (i) 2.6 % (halt at 303 °C) and the pure metals (halt at 327 °C for 0 %, 961 °C for 100 %); (ii) 50 %; (iii) 100 % Ag.

96. **Answer:** It is longest at the eutectic composition (all liquid solidifies there) and decreases towards the pure-metal ends (little eutectic liquid).

97. **Answer:** Plot break temperatures against composition for the liquidus curves, the common halt temperature for the eutectic line, and the pure-metal halts for the two ends (see Fig. G24).

Definitions in your own words

98. **Answer:** Homogeneous, physically distinct, mechanically separable part. Examples: a mixture of gases is one phase; two immiscible liquids (oil + water) are two phases; each polymorph (rhombic/monoclinic sulphur) is a separate phase.

99. **Answer:** Smallest number of independently variable constituents that express every phase's composition. Zero/negative coefficients let a constituent be absent from a phase or be consumed (e.g. $CaO = CaCO_3 - CO_2$).

100. **Answer:** Number of intensive variables (T, P, concentration) that can be varied independently without changing the number of phases; for liquid water alone F = 2.

101. **Answer:** A system in which the gas phase is absent, so small pressure changes have little effect (solid–liquid).

102. **Answer:** A state not at the true minimum of Gibbs energy but persisting until disturbed — supercooled water below 0 °C (curve OA′).

103. **Answer:** Heterogeneous system at equilibrium, not influenced by external forces (gravity, electrical, magnetic).

104. **Answer:** F = 2, 1, 0 respectively.

Derivation variants and logic

105. **Answer:** Pressure is held constant (condensed system), removing one variable: $F = C - P + 2 - 1 = C - P + 1$.

106. **Answer:** $F = 2 - 1 + 2 = 3$: T, P and composition must all be specified — a 3-D diagram.

107. **Answer:** In each phase the mole fractions sum to 1, so only $C - 1$ are independent.

108. **Answer:** For each of C components, P chemical potentials must be equal: $P - 1$ independent equations.

109. **Answer:** variables $= 2(2) + 2 = 6$; constraints $= 3$; $F = 3$ $(= 3 - 2 + 2)$.

110. **Answer:** Matter flows from higher to lower μ; at equilibrium μ is equal in all phases and there is no net transfer.

True or false

111. **Answer: False** — at constant pressure use $F = C - P + 1$.

112. **Answer: True** — a phase is not defined by the number of pieces.

113. **Answer: True** — NH_3 : HCl = 1 : 1 in the gas, so one component; C=2 if extra NH_3 or HCl is added.

114. **Answer: False** — F = 1 − 3 + 2 = 0.

115. **Answer: False** — negative: $\Delta V < 0$.

116. **Answer: False** — it lowers it slightly.

117. **Answer: True** — 374 °C, 218 atm.

118. **Answer: True** — solid Pb, solid Ag and melt.

119. **Answer: False** — the lowest.

120. **Answer: False** — its vapour pressure is higher.

121. **Answer: False** — components are the minimum independent set; species − reactions − restrictions.

122. **Answer: True** — gases are completely miscible.

Spot the error

123. **Answer:** Three species minus one reaction gives $C = 2$.

124. **Answer:** $F = C - P + 2$ with $C = 1$: $1 - 3 + 2 = 0$.

125. **Answer:** Variables $= P(C - 1) + 2$ (mole fractions sum to 1).

126. **Answer:** Use the reduced rule: $2 - 3 + 1 = 0$.

127. **Answer:** Ice is less dense ($\Delta V < 0$ on melting), so the slope is negative.

128. **Answer:** $F = 1 - 2 + 2 = 1$.

Chapter 6: Computational Chemistry

Topics T13

A recently added topic: one question in 2025 so far (definition + three applications). Expect short-answer questions.

Formulas to remember

Classical: MM, MD | Quantum: semi-empirical, ab initio, DFT

Past ESA questions with solutions

1. (i) What is computational chemistry? Give any three applications of computational chemistry.

   Jan-25 · UE24 · Set 2 (7 marks)

   Solution.

   i. **Definition:** a set of techniques that solves chemical problems by simulating molecular, biological or materials systems to give reliable, accurate, atomic-level information, using theoretical chemistry built into efficient computer programs.

   ii. **Three applications:** (1) predicting molecular properties — HOMO/LUMO and UV-Vis, IR, NMR spectra; (2) studying reaction mechanisms and kinetics; (3) designing new materials (conductivity, thermal stability); (also drug discovery by simulating drug–target interaction).

   Answer: Definition + any three applications

Practice: exam-style variants

Set 1: Short answers

Practise writing 2–3-mark answers in your own words.

2. Define computational chemistry and name the first theoretical investigation.
3. List four areas in which computational modelling is used.
4. Differentiate classical and quantum methods of computational chemistry.
5. What is molecular mechanics? Why is it fast?
6. How does DFT differ from ab initio methods?
7. Rank MM, semi-empirical, DFT and ab initio by speed and accuracy.
8. List three advantages of computational chemistry. Does it replace experiments?
9. Write the steps of the electronic-structure determination of ethylene.
10. Describe the bonding and HOMO/LUMO of ethylene.

11. Give two scopes and two advantages of AI-enhanced computational chemistry.

New question types (not yet asked)

Which method would you use?

Scenario questions: choose and justify (speed vs accuracy).

12. Optimise the geometry of a 400-atom steroid derivative in seconds.
13. Study how a protein folds over nanoseconds.
14. Quick qualitative geometry and excited states of a mid-size organic molecule at modest cost.
15. Obtain the electron distribution of a 20-atom molecule with good accuracy without the full cost of a wave-function calculation.
16. Compute the most accurate energy of a very small molecule such as H_2 or ethylene.
17. Predict HOMO/LUMO and the UV-Vis spectrum of a new dye.
18. Screen a million candidate drug molecules for binding to a target.
19. Estimate how electrolyte molecules interact in a battery.

Compare the methods

20. Tabulate MM, MD, semi-empirical, ab initio and DFT: theory, handles electrons?, speed, accuracy.
21. What is a force field?
22. What is a wave function? What does $|\psi|^2$ give?
23. What is a basis set?
24. Give two limitations of computational chemistry.
25. Name the 1998 Nobel laureates linked with computational chemistry.

Ethylene and the electronic-structure workflow

26. How many σ and π bonds does ethylene (C_2H_4) have? State the hybridisation of carbon.
27. Draw the π and π^* orbitals of ethylene; which is HOMO and which LUMO?
28. List the 5 computational steps to determine the electronic structure of ethylene.
29. How does the HOMO–LUMO gap connect to UV-Vis spectroscopy?
30. Why is ethylene's π bond the site of reactivity?

31. Why is the ethane rotational barrier (≈3 kcal/mol) a typical molecular-mechanics problem?

AI-enhanced computational chemistry

32. Describe the AI-enhanced workflow in four steps.
33. List four scopes of AI-enhanced computational chemistry.
34. List three advantages of AI-enhanced computational chemistry.
35. Why can't an AI model work without quantum calculations first?
36. Give an engineering example of AI-assisted screening.

Quick-fire checks and spot the error

True or false

37. Molecular mechanics treats atoms as spheres and bonds as springs.
38. DFT calculates the full many-electron wave function.
39. Ab initio methods are usually faster than semi-empirical ones.
40. Computational chemistry replaces laboratory experiments.
41. The first theoretical investigation was by Heitler and London in 1927.
42. In ethylene HOMO = π and LUMO = π^*.
43. Quantum methods ignore electrons.
44. AI-enhanced models can reduce prediction time to milliseconds once trained.
45. Semi-empirical methods can calculate transition states and excited states.
46. Molecular dynamics is based on the laws of motion.

Spot the error

47. Ab initio means 'based on experimental parameters'.
48. DFT is faster than semi-empirical methods.
49. Ethylene's HOMO is π^*.

Answers

Set 1: Short answers

2. **Answer:** Techniques to simulate chemical systems and compute properties at the atomic level. First theoretical work: Heitler and London, 1927.

3. **Answer:** Molecular properties/spectra, reaction mechanism and kinetics, material design, drug discovery.

4. **Answer:** Classical (molecular mechanics, molecular dynamics): Newtonian, atoms as spheres and bonds as springs, electrons ignored, very fast. Quantum (semi-empirical, ab initio, DFT): based on the Schrödinger equation, treat electrons explicitly, slower but more accurate.

5. **Answer:** Model of a molecule as balls (atoms) joined by springs (bonds), energy from classical force fields; large molecules such as steroids can be optimised in seconds.

6. **Answer:** DFT derives the electron density directly instead of a conventional many-electron wave function; faster than ab initio but slower than semi-empirical.

7. **Answer:** Speed: MM $>$ semi-empirical $>$ DFT $>$ ab initio. Accuracy (generally): ab initio $>$ DFT $>$ semi-empirical $>$ MM.

8. **Answer:** Easier, cheaper and safer than experiments. It does not replace experiments; it helps chemists design and develop desired products.

9. **Answer:** 1 Define geometry of C_2H_4; 2 select a basis set; 3 apply the method to get orbital energies and density; 4 solve the equations for the wave function / electron density; 5 visualise the orbitals (HOMO = π, LUMO = π^*).

10. **Answer:** Each carbon is sp^2: 2 C–H σ bonds + 1 C–C σ bond; one unhybridised p orbital per carbon forms the π bond. HOMO = π, LUMO = π^*.

11. **Answer:** Scopes: accelerated molecular discovery; property prediction; reaction-pathway mapping; structure generation. Advantages: massive speed-up (hours/days $\rightarrow$ milliseconds once trained); large-scale screening; pattern recognition of structure–property relations.

Which method would you use?

12. **Answer:** Molecular mechanics — balls and springs, very fast.

13. **Answer:** Molecular dynamics — applies laws of motion to follow atom movements.

14. **Answer:** Semi-empirical — a middle road; can handle transition states and excited states.

15. **Answer:** DFT — derives electron density directly; faster than ab initio, slower than semi-empirical.

16. **Answer:** Ab initio — based on the Schrödinger equation and the wave function; the most demanding.

17. **Answer:** A quantum method (DFT or semi-empirical).

18. **Answer:** Fast classical methods (MM/MD) plus AI/ML surrogate models trained on quantum data.

19. **Answer:** MD for dynamics, DFT for electronic structure; AI to speed up screening.

Compare the methods

20. **Answer:** MM: classical, no electrons, fastest, lowest. MD: classical dynamics, no electrons, fast. Semi-empirical: quantum (approximate Schrödinger), yes, fast, medium. DFT: quantum (density), yes, medium, high. Ab initio: quantum (wave function), yes, slowest, highest.

21. **Answer:** The set of equations/parameters that gives the potential energy of a molecule as the sum of attractive and repulsive atom–atom interactions (bond stretching, bending, non-bonded).

22. **Answer:** A mathematical function from the Schrödinger equation describing electrons; $|\psi|^2$ gives the electron probability density.

23. **Answer:** The set of mathematical functions used to build molecular orbitals in a calculation (step 2 of the ethylene workflow).

24. **Answer:** Cost grows quickly with molecular size and accuracy; results depend on the chosen method/basis set and are not a replacement for experiment.

25. **Answer:** Walter Kohn and John Pople (DFT and computational methods). [Added for clarity; 2013: Karplus, Levitt, Warshel.]

Ethylene and the electronic-structure workflow

26. **Answer:** 5 σ (4 C–H + 1 C–C) and 1 π; each carbon is sp^2.

27. **Answer:** π: in-phase overlap of the two p orbitals (bonding, HOMO). π*: out-of-phase overlap with a node between the carbons (antibonding, LUMO).

28. **Answer:** 1 Define geometry. 2 Select basis set. 3 Apply the method (orbital energies and density). 4 Solve equations (wave function / density). 5 Visualise (orbital diagrams, density maps, HOMO and LUMO).

29. **Answer:** Absorption of light promotes an electron HOMO → LUMO; a smaller gap absorbs at longer wavelength (see Ch. 4 N1).

30. **Answer:** Its π electrons are higher in energy and more exposed than the σ framework; the HOMO (π) donates and the LUMO (π*) accepts electrons.

31. **Answer:** It is described by torsional terms in a force field: staggered conformers lowest, eclipsed highest (Fig. G25).

AI-enhanced computational chemistry

32. **Answer:** Quantum simulations generate data → build a dataset → train an ML model → predict properties in milliseconds.

33. **Answer:** Accelerated molecular discovery; property prediction; reaction-pathway mapping; structure generation.

34. **Answer:** Massive speed-up (hours/days → milliseconds after training); broad screening of vast chemical spaces; pattern recognition of subtle structure–property relationships.

35. **Answer:** It is trained on data produced by quantum simulations (or experiments); the model interpolates, it does not replace the underlying physics.

36. **Answer:** Battery-electrolyte or polymer screening: predict stability and conductivity of thousands of candidates before synthesis.

True or false

37. **Answer: True** — classical force field.

38. **Answer: False** — it derives the electron density directly.

39. **Answer: False** — slower but more accurate.

40. **Answer: False** — it complements them.

41. **Answer: True** — slide 3.

42. **Answer: True** — slide 16.

43. **Answer: False** — classical methods ignore electrons.

44. **Answer: True** — slide 20.

45. **Answer: True** — slide 10.

46. **Answer: True** — slide 9.

Spot the error

47. **Answer:** Ab initio ('from the beginning') uses the Schrödinger equation with no empirical parameters (that is what semi-empirical methods add).

48. **Answer:** DFT is faster than ab initio but slower than semi-empirical.

49. **Answer:** HOMO is π (bonding); LUMO is π*.

Chapter 7: Mock Question 1 Papers

Topics All Unit 1

Three full-length mock 'Question 1' papers in the exam format, with new molecules and new numbers (nothing repeated from the solved past questions). Allow about 30 minutes per paper; write full answers before checking.

Formulas to remember

$$\varepsilon_J = BJ(J+1),\ I = \mu r_0^2$$

$$\bar{\nu} = \frac{1}{2\pi c}\sqrt{k/\mu},\ \text{ZPE} = \tfrac{1}{2}\bar{\nu}$$

$$\bar{\omega}_e(1-2x_e),\ 2\bar{\omega}_e(1-3x_e)$$

$$F = C - P + 2,\ F = C - P + 1$$

Mock papers: attempt cold, then check

Mock paper 1

Total 22 marks · about 30 minutes · constants: $h = 6.626\times10^{-34}$ J s, $c = 3\times10^{10}$ cm s^{-1}, 1 amu = 1.66×10^{-27} kg, $\pi = 3.14$.

1. The first line in the rotational spectrum of CO is at 3.86 cm^{-1} (C = 12.0, O = 15.995 amu). (i) Derive $I = \mu r_0^2$. (ii) Calculate I and the bond length. (iii) Draw the energy levels to $J = 4$ and the spectrum. (6 marks)

2. HF ($k = 966$ N m^{-1}; H = 1.008, F = 18.998 amu) is treated as an SHO. (i) Find $\bar{\nu}$ and the zero-point energy (cm^{-1} and J). (ii) Find ε_2 and ε_5. (iii) Why is only one line seen? (6 marks)

3. HCl: fundamental 2885.3 cm^{-1} and first overtone 5665.0 cm^{-1}. Find x_e, $\bar{\omega}_e$ and the second overtone; state the selection rule. (4 marks)

4. Derive the phase rule. For $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$ find P, C, F. (4 marks)

5. Differentiate Stokes and anti-Stokes lines and state which is stronger. (2 marks)

Mock paper 2

Total 22 marks · about 30 minutes · constants: $h = 6.626\times10^{-34}$ J s, $c = 3\times10^{10}$ cm s^{-1}, 1 amu = 1.66×10^{-27} kg, $\pi = 3.14$.

1. The first line in the rotational spectrum of HF is at 41.9 cm^{-1} (H = 1.008, F = 18.998 amu). (i) Derive $I = \mu r_0^2$. (ii) Calculate I and the bond length. (iii) Draw the energy levels to $J = 4$ and the spectrum. (6 marks)

2. NO ($k = 1595$ N m^{-1}; N = 14.0, O = 16.0 amu) is treated as an SHO. (i) Find $\bar{\nu}$ and the zero-point energy (cm^{-1} and J). (ii) Find ε_2 and ε_5. (iii) Why is only one line seen? (6 marks)

3. State the Franck–Condon principle. Draw the vibrational–electronic transitions and spectra when (i) $r_e' = r_e''$ and (ii) $r_e' \gg r_e''$. What does a continuum imply? (4 marks)

4. Draw the water phase diagram with triple and critical points and give F in each region. Why is $F = C - P + 1$ used for Pb–Ag? (4 marks)

5. State the Born–Oppenheimer approximation and the order of energies. (2 marks)

Mock paper 3

Total 22 marks · about 30 minutes · constants: $h = 6.626\times10^{-34}$ J s, $c = 3\times10^{10}$ cm s^{-1}, 1 amu = 1.66×10^{-27} kg, $\pi = 3.14$.

1. The first line in the rotational spectrum of HBr is at 17.0 cm^{-1} (H = 1.008, Br = 79.9 amu). (i) Derive $I = \mu r_0^2$. (ii) Calculate I and the bond length. (iii) Draw the energy levels to $J = 4$ and the spectrum. (6 marks)

2. CO ($k = 1857$ N m^{-1}; C = 12.0, O = 15.995 amu) is treated as an SHO. (i) Find $\bar{\nu}$ and the zero-point energy (cm^{-1} and J). (ii) Find ε_2 and ε_5. (iii) Why is only one line seen? (6 marks)

3. HF: fundamental 3958.5 cm^{-1} and first overtone 7737.2 cm^{-1}. Find x_e, $\bar{\omega}_e$ and the second overtone; state the selection rule. (4 marks)

4. Derive the phase rule. For $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$, with excess NH_3 added find P, C, F. (4 marks)

5. What is computational chemistry? Name two quantum methods and one classical method. (2 marks)

Answers

Mock paper 1

1. ***Working.***
   i. (i) Derivation Q1-D.
   ii. (ii) $B = 1.93$ cm^{-1}; $I = 1.45 \times 10^{-46}$ kg m^2; $\mu = 6.8562$ amu; $r_0 = 1.129$ Å.
   iii. (iii) Levels 0, 2B, 6B, 12B, 20B = 0, 3.86, 11.58, 23.16, 38.60 cm^{-1}; lines every 3.86 cm^{-1}.

   Answer: $I = 1.45 \times 10^{-46}$ kg m^2; $r_0 = 1.129$ Å

2. ***Working.***
   i. $\mu = 0.9572$ amu; $\bar{\nu} = 4139$ cm^{-1}.
   ii. ZPE $= 2069$ cm^{-1} $= 4.11 \times 10^{-20}$ J; $\varepsilon_2 = 10346$, $\varepsilon_5 = 22762$ cm^{-1}.
   iii. (iii) Equal spacing and Δv $= \pm 1$.

   Answer: $\bar{\nu}$ = 4139 cm^{-1}, ZPE = 2069 cm^{-1}

3. ***Working.***
   i. $\bar{\omega}_e = 3f - o = 2990.9$; $x_e = 0.01765$; 2nd overtone $= 3\bar{\omega}_e(1 - 4x_e) = 8339.1$ cm^{-1}.
   ii. Δv $= \pm 1, \pm 2, \pm 3$...

   Answer: x_e = 0.01765, $\bar{\omega}_e$ = 2990.9 cm^{-1}, 2nd overtone 8339.1 cm^{-1}

4. ***Working.***
   i. Derivation Q5-D.
   ii. $P = 3$, $C = 2$, $F = 2 - 3 + 2 = 1$.

   Answer: F = C − P + 2; P = 3, C = 2, F = 1

5. **Answer:** Stokes: lower frequency (molecule gains energy), stronger; anti-Stokes: higher frequency, weaker (fewer molecules in excited levels).

Mock paper 2

1. ***Working.***
   i. (i) Derivation Q1-D.
   ii. (ii) $B = 20.95$ cm^{-1}; $I = 1.34 \times 10^{-47}$ kg m^2; $\mu = 0.9572$ amu; $r_0 = 0.917$ Å.
   iii. (iii) Levels 0, 2B, 6B, 12B, 20B = 0, 41.90, 125.70, 251.40, 419.00 cm^{-1}; lines every 41.90 cm^{-1}.

   Answer: $I = 1.34 \times 10^{-47}$ kg m^2; $r_0 = 0.917$ Å

2. ***Working.***
   i. $\mu = 7.4667$ amu; $\bar{\nu} = 1904$ cm^{-1}.
   ii. ZPE $= 952$ cm^{-1} $= 1.89 \times 10^{-20}$ J; $\varepsilon_2 = 4760$, $\varepsilon_5 = 10472$ cm^{-1}.
   iii. (iii) Equal spacing and Δv $= \pm 1$.

   Answer: $\bar{\nu}$ = 1904 cm^{-1}, ZPE = 952 cm^{-1}

3. ***Working.*** Statement; case (a) and case (c) of Fig. G21.

   Answer: (0,0) strongest for (i); continuum (dissociation) for (ii)

4. ***Working.*** Fig. G22; areas F=2, lines F=1, triple point F=0; condensed system at constant pressure.

   Answer: F = 2, 1, 0; reduced rule for condensed Pb–Ag

5. **Answer:** $E = E_{rot} + E_{vib} + E_{elec}$, independent; $E_{rot} \ll E_{vib} \ll E_{elec}$.

Mock paper 3

1. ***Working.***
   i. (i) Derivation Q1-D.
   ii. (ii) $B = 8.5$ cm^{-1}; $I = 3.29 \times 10^{-47}$ kg m^2; $\mu = 0.9954$ amu; $r_0 = 1.412$ Å.
   iii. (iii) Levels 0, 2B, 6B, 12B, 20B = 0, 17.00, 51.00, 102.00, 170.00 cm^{-1}; lines every 17.00 cm^{-1}.

   Answer: $I = 3.29 \times 10^{-47}$ kg m^2; $r_0 = 1.412$ Å

2. ***Working.***
   i. $\mu = 6.8562$ amu; $\bar{\nu} = 2144$ cm^{-1}.
   ii. ZPE $= 1072$ cm^{-1} $= 2.13 \times 10^{-20}$ J; $\varepsilon_2 = 5360$, $\varepsilon_5 = 11792$ cm^{-1}.
   iii. (iii) Equal spacing and Δv = $\pm$1.

   Answer: ν^{-}= 2144 cm^{-1}, ZPE = 1072 cm^{-1}

3. ***Working.***
   i. $\bar{\omega}_e = 3f - o = 4138.3$; $x_e = 0.02172$; 2nd overtone $= 3\bar{\omega}_e(1 - 4x_e) = 11336.1$ cm^{-1}.
   ii. Δv = $\pm$1, $\pm$2, $\pm$3...

   Answer: x_e = 0.02172, $\bar{\omega}_e$ = 4138.3 cm^{-1}, 2nd overtone 11336.1 cm^{-1}

4. ***Working.***
   i. Derivation Q5-D.
   ii. $P = 2$, $C = 2$, $F = 2 - 2 + 2 = 2$.

   Answer: F = C − P + 2; P = 2, C = 2, F = 2

5. **Answer:** Simulating chemical systems at the atomic level; quantum: ab initio, DFT (semi-empirical); classical: molecular mechanics/dynamics.